

Landslide Frequency and Magnitude in the Southern Alps, New Zealand

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Whāia te iti kahurangi ki te tūohu koe me he maunga teitei

Seek the treasure you value most dearly: if you bow your head, let it be to a lofty mountain

Abstract

Mountain regions are shaped by complicated interactions between tectonics and climatic forces. Understanding the orientation of the processes influencing landslide magnitude and frequency is an important part of geomorphological research and hazard mitigation. The Southern Alps are formed by dextral transpression at the Alpine Fault, where oblique convergence between the Pacific and Australian plate result in horizontal and vertical strike-slip displacement. Consequently, the orogen receives significant uplift rates, along with abundant daily and annual rainfall. This study uses Google Earth Pro and historical aerial imagery to identify and map landslides over a 70-year period. Volume-area scaling relationships establish landslide-derived erosion rates in the range 0.09–0.51 mm/yr, with proximal activity concentrated within 5–10 km of the Alpine Fault. Landslides display a power-law magnitude-frequency distribution in the range 10^0 – 10^{-5} km⁻² yr⁻¹. Power-law regressions yield a lower scaling coefficient compared to previous data, reflecting the shift to shallower landslides and the accuracy of Google Earth Pro implementation. Strong positive correlations are found between landslide-derived erosion rates and Late Quaternary throw rates ($R^2 = 0.974$, $r = 0.987$), indicating the first-order control of tectonics on mass wasting. Analysis of rainfall data reveals that intense, short-duration events are significant triggers for landslides, and a decrease in median landslide area since the 1960s further supports the transition from a system dominated by large, infrequent landsliding events to one characterised by smaller, more frequent events. The data reveals the critical influence of Southern Alps tectonics in setting the stage for climatically driven landsliding and the implications for understanding sediment transport and connectivity in long-term landscape evolution.

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Chapter 1: Introduction

Landslides are dynamic geomorphological processes driven by mechanical forces and environmental triggers, and they exert a profound influence on the evolution of montane landscapes. In New Zealand's Southern Alps, landslides and other forms of gravitational mass wasting significantly contribute to erosion (Allen and Owens, 2011; Hovius et al., 1997; Korup et al., 2004). Understanding their occurrence, magnitude, and frequency is crucial for assessing their impact on erosion, sediment transport, and overall hillslope morphology.

The Western Southern Alps of New Zealand represent a unique tectonically active orogen driven by high uplift rates, frequent rainfall, and high landslide susceptibility. This study focuses on five catchments—Hokitika, Kokatahi, Styx, Arahura, and Taipo—situated west of Arthur's Pass National Park, a region with relatively low uplift and high precipitation. Through employing a combined mapping approach targeting recent landslides (within the last 20 years) using Google Earth Pro, and historical landslides from 1960 using aerial imagery, this research establishes new minimum estimate landslide-derived erosion rates. In doing so, long-term assessments of the trends embedded in landscape evolution are made available, with emphasis on the influence of Alpine Fault throw rates, median annual rainfall, and rainfall intensity. The inclusion of these metrics reflects the understanding that common first-order controls on the rates and patterns of mass wasting include tectonics (Roda-Boluda et al., 2023), storms and earthquakes (Densmore and Hovius, 2000).

Recognition of landslide inventories has increased in recent years due to their importance in quantitative hazard and risk assessment (Korup, 2005). A new landslide dataset is presented of $n=2757$ landslides in previously unmapped mountainous terrain of the Western Southern Alps. The key objectives are to examine the relationship with landslide frequency and magnitude with proximity to the Alpine Fault, assess the role of rainfall as a triggering mechanism,

evaluate the evolution of landslide patterns through time, and consider to what extent this study's landslide-derived erosion rates contribute to overall catchment erosion, highlighting the complexities of catchment sediment connectivity and transport. Through the integration of a range of analytical methods, including landslide density calculations, frequency normalisation, volume-area scaling, and magnitude-frequency distribution analysis, this research provides a comprehensive understanding of erosional drivers within a newly studied area of the Southern Alps.

Chapter 2: Background

In steep terrains, which is common throughout the Southern Alps, mass wasting is deemed to be the dominant erosional process on hillslopes and the main source of sediment to rivers (Burbank et al., 1996; Hovius et al., 1997; Korup, 2004, 2005). Mass wasting events, specifically landslides, demonstrate the equilibrium between gravitational and resisting forces, such as bedrock strength (Clarke and Burbank, 2010). Landslide initiation can be clearly understood through fracture mechanics (Martel, 2004), where slopes transition to catastrophic failure as a result of critical threshold exceedance. Structural discontinuities in rock, such as fractures, jointing, bedding, and strength are known to predispose hillslopes to failure, however, hydrometeorological influence is emerging as a triggering mechanism that initiates active instability (McColl, 2015). Here, the mechanics of landslides are discussed, exploring their initiation and propagation. The physical principles governing critical thresholds are stated, as well as to what extent hydrometeorological interactions can necessitate surface failure, in the context of the Southern Alps.

2.1 Landslide Mechanics

The ability of a hillslope to resist shearing stress, or the shear strength, is controlled by its angle of internal friction, substrate cohesion (Campforts et al., 2022), and effective normal stress. The Coulomb equation mathematically acknowledges such a relationship:

$$(1) \quad SS = C + \sigma' \tan \phi$$

Where SS is shear strength, C is cohesion, σ' is effective normal stress, and ϕ is the angle of internal friction (Bierman and Montgomery, 2014).

Frictional strength of rock and soil arises from the potential shearing resistance between mineral grains in the proximal area of a rupture site, where frictional strength is directly proportional to the effective normal stress that maintains grain surface contact (Bierman and Montgomery, 2014). The angle of internal friction (ϕ) states that with increased effective normal stress comes increased shear strength; a co-dependent relationship where an increase in the normal stress component drives a proportional change at a rate set by ϕ . the friction angle typically operates within a range of 10° to 40° , and within this lies the variability of cohesion values that span many orders of magnitude between the two interfaces of soil and rock (Bierman and Montgomery, 2014).

Landslides are classified both by depth (Larsen et al., 2010) and material type (Cruden and Couture, 2011). Due to the large strength discontinuity between soil and bedrock, shallow planar landslides detach and slide along the soil-bedrock interface (Bierman and Montgomery, 2014). In the Southern Alps, shallow planar surfaces are typical in steep, soil-mantled slopes, where regolith formation is pronounced. Additionally, deep-seated bedrock failures are operating within a different regime, and excavate deeper into the subsurface including coarser grains and surficial material. These processes are predominant along discrete planes of weakness such as faults, bedding planes, or rotational surfaces (Kojima et al., 2015; Xu et al., 2014; Margielewski, 2006). Bedding plane orientation is a large proponent in slope stability and maintains shear strength to a degree such that shear stress doesn't overcome it. For example, bedding that dips back into the slope promotes slope stability, whereas bedding that follows a slope will generally promote instability (Bierman and Montgomery, 2014). Maintaining a clear understanding of this framework is imperative when examining landslide dynamism and their respective failure mechanisms, especially in tectonically active and erosion-susceptible alpine regions.

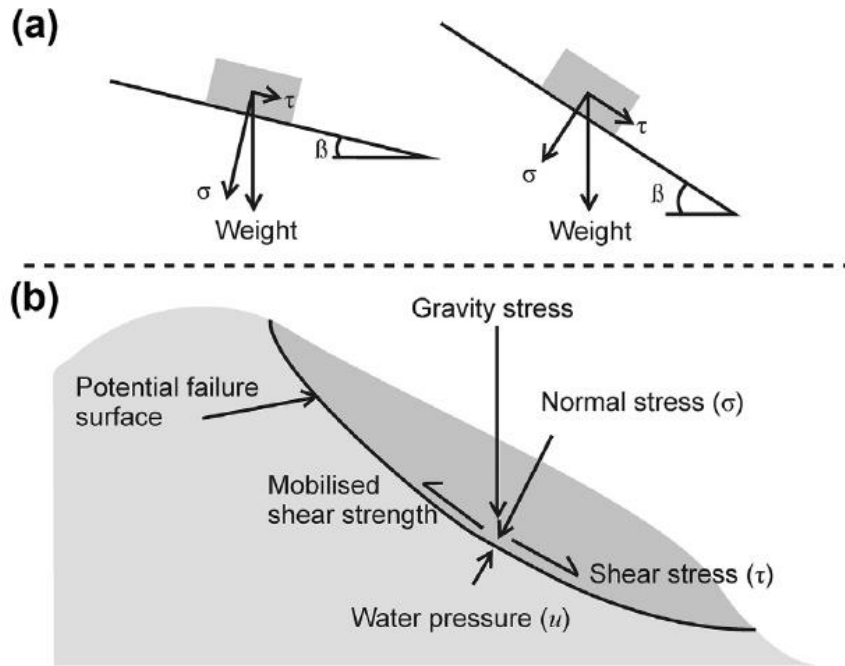


Figure 1. Slope angle (β) influence of the magnitude of shear stress (τ) and normal stress (σ). (b) stresses acting along a potential failure surface. From McColl (2015), adapted after Selby (1993).

2.2 Hydrometeorological Drivers of Failure

In areas dominated by steep regolith-covered slopes, rainfall serves a key trigger for landsliding (Brand, 1981; Chen et al., 2020; Crozier, 1999). The relationship between rainfall and slope failure is complex and changes as the regolith is removed from the hillslope. Once regolith is stripped, the triggering threshold for landslides also changes, making the landscape more stable. However, the sensitivity to different storm types varies at each phase of regolith stripping (Brooks et al., 2002). For example, undisturbed regolith and redeposited regolith, after landsliding, are more susceptible to failure during high-intensity storms, while partially stripped regolith is the least susceptible (Brooks et al., 2002). On the other hand, redeposited regolith tends to have higher porewater pressures and reduced shear strength, making it more unstable (Brooks et al., 2002), though it is possible for increased cohesion within the material to counteract strength reductions promoting longer term stability. In areas of storm-triggered landslides, steep slopes are largely concentrated at the lowermost part of the hillslope, forming

an "inner gorge" like morphology (Densmore and Hovius, 2000). This is a result of storm conditions driving an increase in soil saturations at slope bases, causing a form of undercutting failure. In contrast, earthquake-triggered landslides can appear to be more uniform across a slope gradient due to ground-shaking (Densmore and Hovius, 2000).

Different hydrometeorological conditions can produce movement styles, and by determining scar, runout and landslide morphology, it is possible to understand slope stability and failure mechanisms associated with an event. Hydrometeorological triggers can manifest as either a single, high-magnitude event producing almost instantaneous reactions, or develop cumulatively, destabilising a slope over a multitude of events (McColl, 2015). For instance, precipitation events drive shifts in groundwater exacerbating temporary exceedances in failure thresholds that often result in the growth of fractures within pore spaces (McColl, 2015; Clarke and Burbank, 2011). Rainfall specifically influences slope stability by increasing pore water pressure and surface material weight, reducing effective stress during infiltration, and amplifying the downslope motion of gravity. It also raises the groundwater table, alters soil structure strength, and loosens surface material through rainwash (Zhao et al., 2022). Through all of this, these factors ultimately reduce shear strength, causing a decrease in the effective normal stress needed for slope failure when precipitation directly interacts with slope surfaces (Smith et al., 2023).

Antecedent conditions are inextricably linked with the groundwater table, and can further increase landsliding susceptibility under low frequency, high magnitude rainstorms (Smith et al., 2023). Material excavation at the base of slopes, linked with increased groundwater table, promotes slope steepening beyond their critical shear stress and also influences over-steepening through the increased negative pore pressure within the soil matrix (McColl, 2015). Take shallow landslides, for example. Such events are typically occurring in response to smaller

scale rainfall cycles, where larger bedrock sliding could be attribute to the long-term influence of both tectonics and rainfall patterns (Sidle and Boogard, 2016).

Chapter 3: Physical Setting

The study covers 1,360 km² of the Western Southern Alps, immediately west of Arthur's Pass National Park, extending from S-N, West of the Main Divide primarily adjacent to the Alpine Fault (Fig. 2). Formed by dextral transpression, the Southern Alps were, and still undergo activity due to the Alpine Fault. The Alpine Fault is a major fault running along the western edge of the South Island 800-km-long (Korup, 2005). It is a right-lateral, oblique-reverse fault (up-to-the-east), where it exhibits both horizontal and vertical displacement bounding west of the transpressive orogen (Cox and Findley, 1995) at a rate of uplift 10 ± 2 mm/yr (Norris and Cooper, 2001).

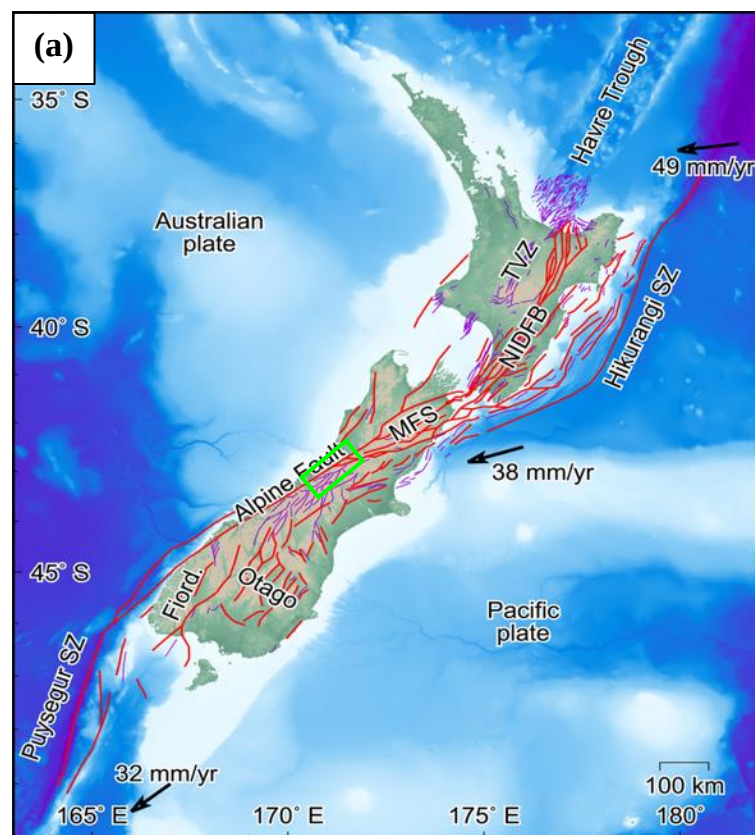


Figure 2. (a) an inset showing the relative location of the study area within the broader 'New Zealand' (green line), also included are the most pronounced fault systems on both islands, placed within a geodynamic context (adapted from Hirschenberg and Sutherland, 2022).

The Southern Alps represents a classic example of a two-sided orogen, where the rate of tectonic material accumulation on the drier, eastern side is roughly balanced by the rate of

erosion and sediment transport on the wetter, western side (Little et al., 2005). This balance contributes to a state of approximate equilibrium in terms of the Southern Alps' topography, thermal structure, and exhumation (Little et al., 2005). The rock in the Southern Alps is low-grade metamorphosed greywacke comprising the Torlesse group (Clarke and Burbank, 2011), with the metamorphic grade decreasing towards the east. This metamorphic material is detached from the upper portion of the westward subducting Pacific plate and is accreted onto the orogenic wedge of the Southern Alps, then thrust up to the surface along the steep, east-dipping Alpine Fault (Koons, 1989, 1990; Beaumont et al., 1996). Additionally, Late Quaternary vertical slip rates are highest within the central Southern Alps at ~12 mm/yr and decrease towards both SW and NE (Langridge et al., 2010). Fracture densities in the Southern Alps exhibit notable variation. In the western Southern Alps, extensive fracturing of the bedrock is associated with deeper exhumation and significant transpression along the Alpine Fault ramp. In contrast, the eastern side of the range features significantly lower fracture densities, reflecting shallower burial and exhumation pathways (Clarke and Burbank, 2010; Roda-Boluda et al., 2023). Various studies on landslides in the region indicate that high-intensity rainstorms, fluvial incision, and seismic ground-shaking are the most likely triggering mechanisms (Korup, 2004, 2005; Kritikos et al., 2015; Rosser et al., 2020). The Southern Alps

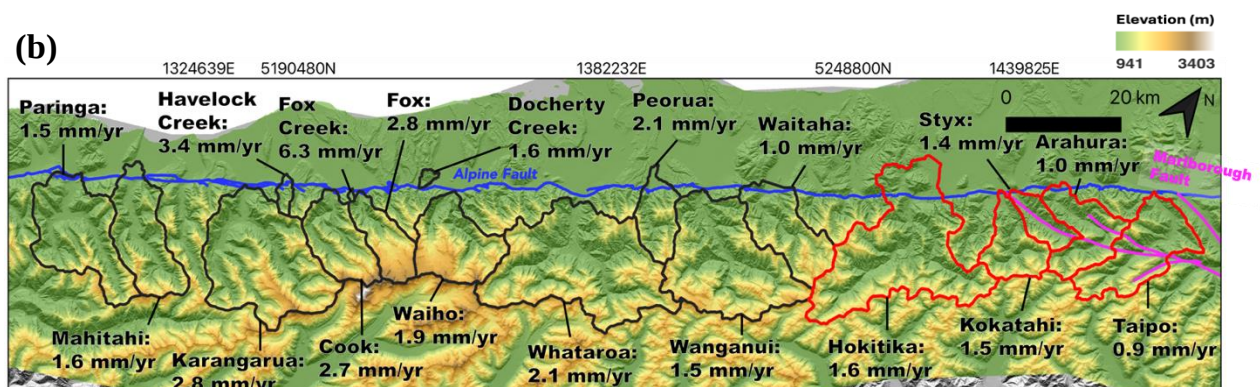


Figure 2. (b) The overarching catchment selection (red line) compared against other central Southern Alps catchments, each named accordingly coupled with their respective annual exhumation rates. The Alpine Fault can be seen running through the catchments (blue line), complimented by the Marlborough Fault system (pink line). Exhumation data from Roda-Boluda et al., (2023).

of New Zealand are surrounded by ocean and are located between the subtropical anticyclones and the travelling depressions of the Southern Ocean exposing them to extreme westerly winds (Sturman and Wanner, 2001), with such systems interacting with the orogen creating unique atmospheric conditions. The interaction between the westerly winds and the Southern Alps creates large-scale variability in average annual rainfall between the west and east, where the western flanks of the Southern Alps receive incredibly high amounts of annual precipitation (>8 mm/yr), but an orographic gradient sharply reduces precipitation from west to east (Clark and Burbank, 2011), meaning that around 600 mm is received on the eastern coastal plains and in the mountain basins.

Chapter 4: Methodology

Google Earth Pro (GEP) was utilised for mapping landslides occurring within the last 20 years. Recent failures were identified following Cruden and Varnes' (1996) criteria for landslide classification, emphasising geometric features and movement patterns. Recognising landslide typology is critical for compiling inventories, with very high-resolution imagery aiding in identifying movement types (Murillo-García et al., 2015). The analysis often relied on the 3D environment in GEP to distinguish between rotational and translational landslides, characterised by concave and planar rupture surfaces, respectively (Murillo-García et al., 2015). Additionally, understanding regional geology informed interpretations of landslide susceptibility based on variations in rock type.

This study included only active landslides and reactivated movements, as defined by Cruden and Varnes (1996). Active landslides encompass both first-time failures and reactivations facilitated by pre-existing unstable shear surfaces. Inactive landslides, identifiable by vegetation regrowth on scars without recent reactivations, were excluded. Vegetation disturbances were critical to identifying active failures due to their distinct spectral signatures (Asner et al., 2000). Determining the dates of landslides varied in complexity, as some failures could be accurately dated, while others were constrained by periods of poor-quality imagery. Mapped landslides were dated to their initial occurrence, assuming that as long as they showed no signs of inactivity, they contributed to sediment budgets. Recurring wasting events were remapped to account for newly detached and deposited material.

Efforts were made to avoid amalgamation, where multiple landslides are erroneously combined into a single polygon. While mapped polygons ideally represent individual landslides, amalgamation often occurs in regions with high landslide density and shared runout zones,

especially when image resolution or mapping focus is limited and when the primary goal of the mapper is not to map landslide extent precisely, but rather to rapidly evaluate the area affected by slope failure (Marc and Hovius, 2015). Automated detection methods (Guzzetti et al., 2012) can help, but poor resolution and dense terrain disturbance can complicate differentiation between individual events, leading to merged polygons.

In this study's catchments, amalgamation was minimal, with large scree zones and frost-shattered mountain tops being more prominent. However, in areas of high landslide convergence, detailed examination of scarps, runout channels, and slope detachment was employed to delineate individual events. In cases of extreme amalgamation, mapping was omitted. For historical aerial imagery, challenges such as image parallax (caused by vertical camera shifts) and variations in solar zenith angles introduced uncertainties in distinguishing landslides. While amalgamation was more likely in older imagery, it was less of an issue with recent high-resolution data. This study acknowledges that avoiding all amalgamation is inherently challenging and beyond the scope of the current analysis but asserts that the likelihood of amalgamation is lower in recent datasets compared to 1960s imagery.

4.1 Historical Imagery

Historical landslides were identified using black-and-white aerial imagery captured by the Wild RC-8 camera. The imagery, formatted at 18x18 cm, was taken at altitudes ranging from 16,500 to 25,000 feet with a nominal focal length of 114 mm (Fig. 4). These images were freely accessed through the Retrolens Historical Image Resource (<https://www.retrolens.nz>). Using the Retrolens interface, the study area was located, and imagery was filtered to dates between 1960 and 1969. Careful consideration was given to selecting the most suitable years to ensure adequate image quantity and quality.

In some cases, complete catchment coverage was unavailable due to gaps in the imagery, limiting the extent to which certain areas could be georeferenced and mapped.

4.2 Georeferencing

Georeferencing is the transformation of raster imagery to align it with a spatial reference system, whereby it may represent a specific geographical area (Yao Xiaobai, 2020). Here historical aerial imagery is sourced from Retrolens (Fig. 3). This imagery lacked embedded geospatial metadata, which necessitated the need for georeferencing.

Georeferencing was carried out in Quantum Geographical Information System (QGIS) (Fig. 4), utilising the built in georeferencing toolkit. Key feature identification was included into the process, allowing for a best estimate placement of ground control points (GCP).

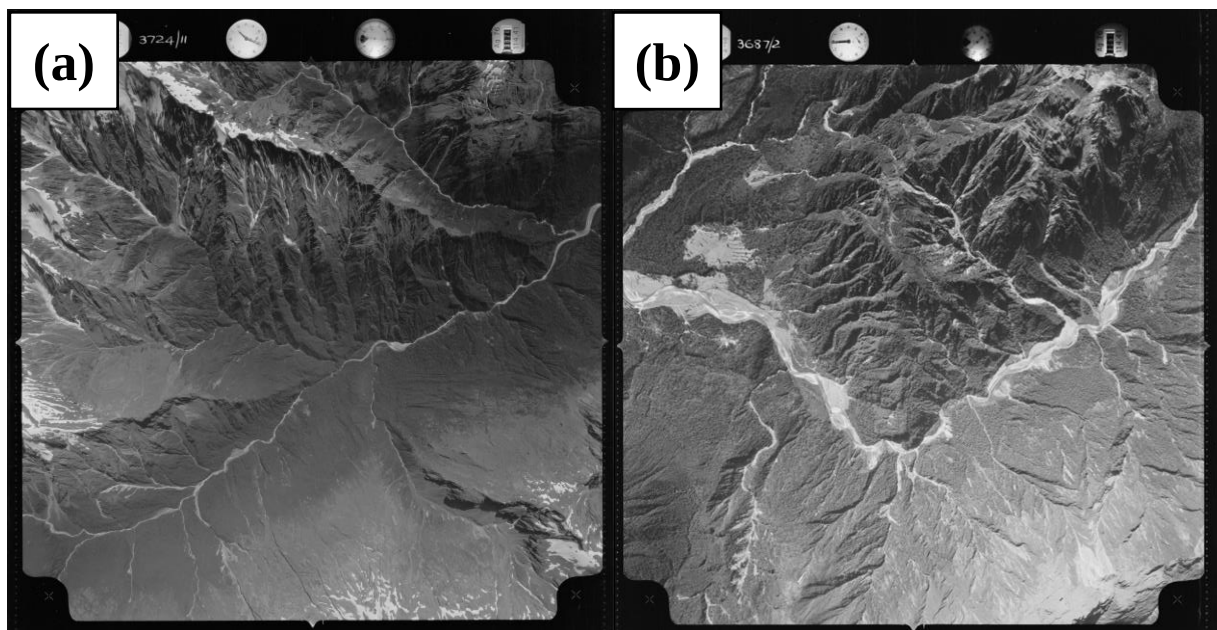


Figure 3. Two examples of aerial imagery (a and b) derived from the Wild RC-8 camera and downloaded from <https://retrolens.nz>

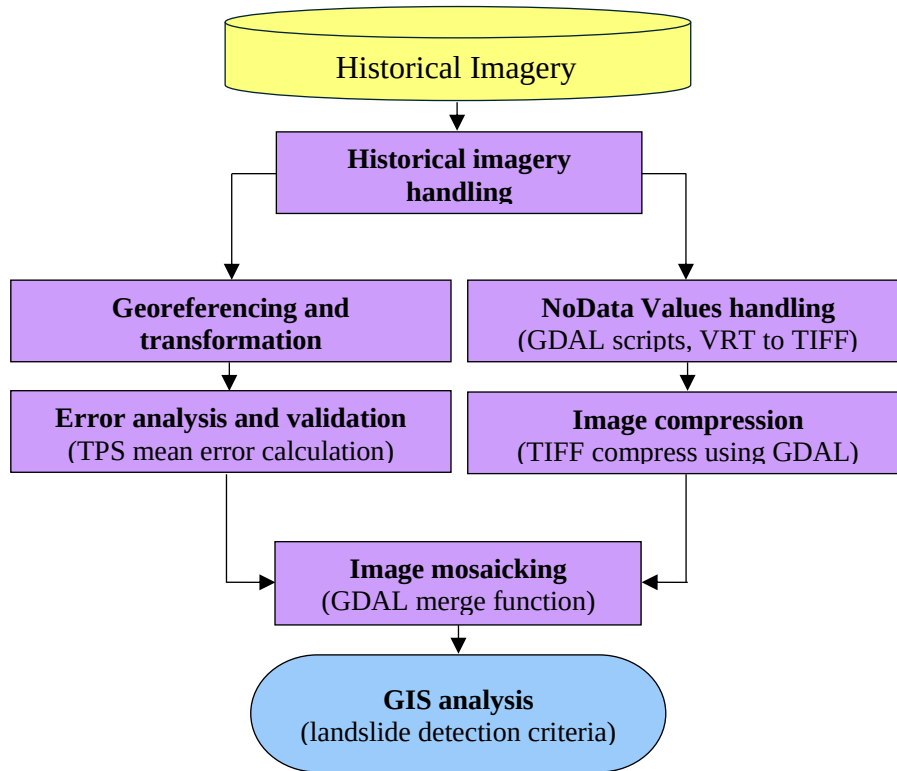


Figure 4. Algorithm for photogrammetric historical imagery processing and analysis

Table 1. Average $\bar{E}_{TPS}$ per catchment after georeferencing using TPS transform and Cubic 4x4 resampling.

Catchment	$\bar{E}_{TPS}$
Hokitika	$6.60526e^{-10}$
Kokatahi	$1.2937e^{-10}$
Styx	$2.84526e^{-10}$
Arahura	$1.93599e^{-9}$
Taipo	$4.50248e^{-9}$

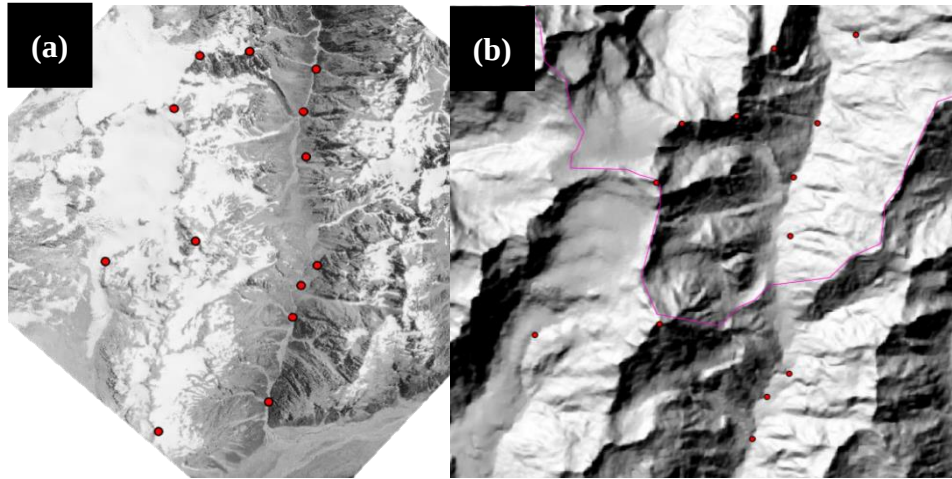


Figure 5. (a) GCPs (the red markers) placed on the historical image (left), while using the hillshade map (b; right) as a reference point for topographic distinction. The historical image displays a reasonable example of poor image quality due to snow cover, making it difficult to confidently place GCPs.

Key features included river junctions, prominent topographic landmarks such as rock outcrops, mountain ridgelines, and peaks. Additionally, the hillshade map would be used to compensate in areas of poor resolution, cloud cover, or abundant snow coverage, as it aided in the interpretation of topographic features (Fig. 5).

Priority was given to the GCP placement on stable and unchanging features to ensure that accuracy is as high as possible. These GCPs would be evenly distributed across the orthophoto, aiming to target each corner, and the edges, and central aspects. This would decrease the level of distortion that can arise from uneven GCP placement. The number of GCPs placed in a given image depended on the given characteristics of the image and its target location. More specifically, the transformation algorithm selected in most cases require a minimum amount of GCPs, and the level of complexity and distortion per image would be a deciding factor. On average, 90 to 100 GCPs were assigned per image. It was also essential to maintain a transformation mean error value ($\overline{E}_{TPS}$) as close to zero as possible, though a threshold of ($\overline{E}_{TPS}$) ≤ 5 would suffice. See the respective values for this in Table 1, and this study's georeferenced images in Appendix I.

The process for image transformation follows two aspects: the resampling procedure and the transformation algorithm. This study applied Cubic (4x4) kernel resampling and Thin Plate Spline (TPS) transformation. TPS is an interpolation-based method that utilises polynomials to maintain continuity and smoothness between all neighbouring polynomials (Sempio et al., 2019; Ha et al., 2018). This is recognised as the best fit approach due to the generation of smooth and more flexible surfaces, that in the context of the Southern Alps, suit the rough and complex topography. Higher order polynomial transformations such as polynomial degree 2 and 3 (PD2 and PD3) which fit quadratic and cubic surfaces, TPS offers higher precision in image overfitting with data.

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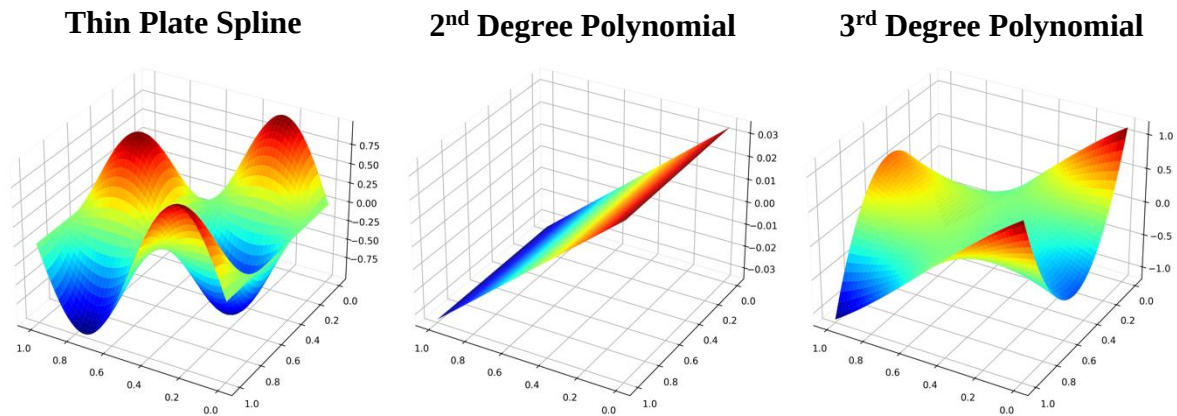


Figure 6. Comparison of surface fitting techniques illustrates three separate methods of fitting surfaces to GCPs when georeferencing: thin plate spline, 2nd degree polynomial, and 3rd degree polynomial.

4.3 Image Mosaic

Georeferencing involves aligning images with a specific coordinate system, resulting in a transformation of their spatial data structure (Wylie et al., 2019). After completing the georeferencing process, areas outside the original image boundaries are typically filled with NoData values, displayed as a black void.

The images were processed through command line scripting (CLI) in order to minimise memory intensive computation. NoData values were reclassified, converting the rasters into virtual raster tiles (VRT) using GDAL and setting NoData values to 0. Using VRTs allows for the processing of large raster spatial data arrays, with an emphasis on increased efficiency (McCormack and Hogg, 1997). Another command would translate the VRTs back into a readable format. The images were then compressed using the GDAL translate function with the ‘DEFLATE’ method as to reduce their overall file size. Then the processed catchment images were merged using the ‘merge.py’ function from GDAL. See Appendix I for the CLI commands.

4.3.1 Mosaic Reliability

The final mosaics varied in quality, with some displaying noticeable misalignments, primarily at image seamlines and in areas with complex topography. Despite applying the same $\overline{E}_{\text{TPS}}$, targeting values ≤ 5 , these discrepancies are believed to arise from differences in image resolution, lighting conditions, terrain relief, and potential human errors. The impact of relief displacement—where terrain appears to “lean away” when photographed from above (Lillesand et al., 2015)—is considered a significant contributor to the misalignment observed at image seamlines (Fig. 7).

Another challenge encountered was the absence of imagery in certain areas of the catchments. In the context of this study, these gaps often correspond to regions of high frost cracking and scree slopes (but not always). Further details regarding each catchment’s imagery, the $\overline{E}_{\text{TPS}}$ test results, and their respective weighted averages, as well as the imagery mosaics, can be found in Appendix I.

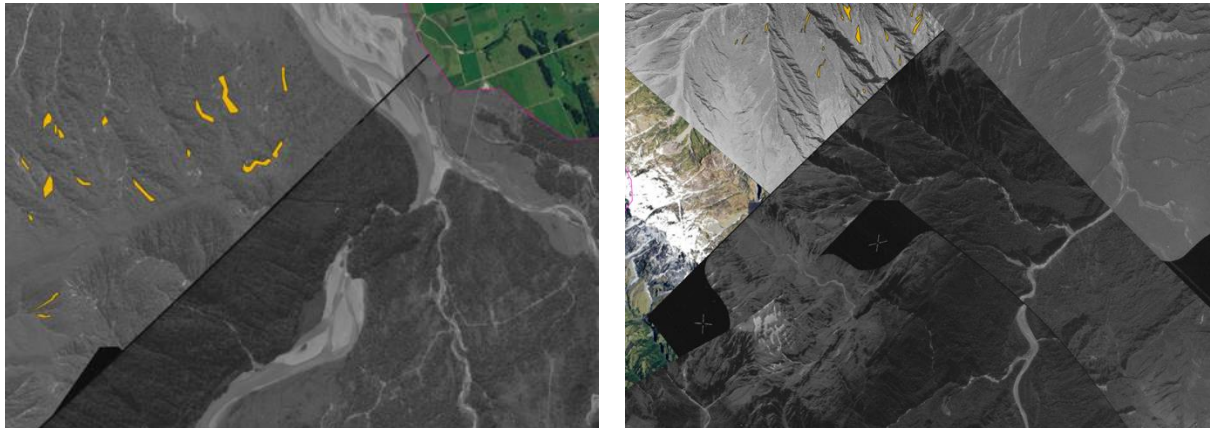


Figure 7. (left) Good mosaic seamline correlation, (right) complex mosaic seamline structure, with low accuracy correlation

4.4 Climate Data

The following climate data are included in this study’s landslide frequency and magnitude analyses: (i) mean annual rainfall, (ii) mean annual rainfall intensity, and (iii) mean annual

temperature. Data (i) and (iii) are sourced from the National Institute of Water and Atmospheric Research (NIWA, 2012), while data (ii) is derived by dividing median annual rainfall by median rain days raster (after Roda-Boluda et al., 2023).

Although New Zealand's climate network is relatively dense, with 680 rainfall stations (200 of which are automated), 93% of these are located below 500 m above sea level (asl) (Tait et al., 2012). This results in a lack of high-elevation data, necessitating interpolation for mountainous regions. The absence of rain gauges in higher elevations means that data must be estimated based on nearby stations, especially since the rain-gauge network is concentrated in valleys (Duncan, 1992). To address this, NIWA developed the Virtual Climate Station (VCS) dataset in 2005, which interpolates daily rainfall observations onto a grid covering New Zealand. This interpolation uses a TPS method, incorporating latitude, longitude, and 1951-1980 mean annual rainfall as covariates (Tait et al., 2012). Higher elevation stations receive a greater weighting in the interpolation as a result of the lower number stations at these altitudes, hence relative contributions need to be enhanced (Tait and Turner, 2005).

VCS data is considered to be a reasonable estimate of rainfall, however, its inherent uncertainty is high, particularly in areas with complex, mountainous terrain where station density is low and rainfall patterns are spatially heterogeneous (Tait and Turner, 2005; Tait et al., 2012). Therefore, it is important to acknowledge that rainfall estimates in remote locations may not accurately reflect the true distribution of rainfall, as low station density often leads to deviations from actual spatial rainfall patterns (Wagner et al., 2012).

4.5 Landslide Analysis

Landslide density (km^{-2} or m^{-2}) is calculated using the following formula:

$$(2) \quad \text{density} = \frac{\text{number of landslides}}{\text{total area of band}}$$

To enhance the representation of landslide occurrences, landslide frequency was normalised. This would offer a clearer comparison of landslide occurrences across different catchments, spatially and temporally. The frequency of landslides occurring within specific logarithmic area bands was also calculated. These logarithmic bands were automatically generated in QGIS based on the area values derived from each catchment's landslide shapefile.

As the intervals varied between catchments, the units for each plot might differ slightly. However, when these data sets were combined and plotted, a script was used to select bands that best fit the values within each array. The choice to use logarithmic bands was deliberate, given the broad range of area values that often spanned multiple orders of magnitude. These bands proved effective in clearly illustrating trends in landslide area.

4.5.1 Distance from Fault

Distance from bands ranging from 0 to 25 km were provided (Roda-Boluda in prep), stepping up in increments of 5 km highlight the distance from the Alpine Fault. Landslide data was clipped to these band ranges and analyses were carried out. Subsequently, the normalised frequency of landslides within each band was also employed.

Normalising the frequency of landslides is essential, as it allows for a fair comparison of landslide occurrences across different distances from the Alpine Fault. By adjusting for the varying catchment areas and the frequency of events within each distance band, the analysis can reveal spatial trends that might otherwise be obscured. The data was then visualised using the matplotlib library.

4.5.2 Volume–Area Scaling

The Larsen et al. (2010) model estimates landslide volume in cubic meters (m³) by utilising Python libraries to calculate the area of uploaded vector shapefiles, with error estimates produced via standard Gaussian error propagation. The volume–area (V–A) relationship is employed to define landslide volume and erosion estimates, based on scaling relationships derived from a limited number of field measurements, where predicted volume V of a given landslide area A depends on a scaling exponent γ and intercept α such that $V = \alpha A^\gamma$ (Larsen et al., 2010).

For soil landslides, where $A < 3 \cdot 10^3 \text{ m}^2$, volume V is estimated using:

$$(3) \quad V = 0.612 \cdot A^{1.193}$$

For mixed soil–bedrock landslides, where $3 \cdot 10^3 \geq A \leq 1 \cdot 10^5 \text{ m}^2$, the scaling equation is given as:

$$(4) \quad V = 0.14 \cdot A^{1.36}$$

with $\gamma_{mixed} = 1.36 \pm 0.01$, $\log(\alpha) = -0.86 \pm 0.005$, with $\sigma_\alpha = 0.065$ and $\sigma_\gamma = 0.01$.

For bedrock landslides, where $A \geq 1 \cdot 10^5 \text{ m}^2$, the scaling equation is given as:

$$(5) \quad V = 0.025 \cdot A^{1.49}$$

with $\gamma_{rock} = 1.49 \pm 0.03$, $\log(\alpha) = -1.60 \pm 0.19$, with $\sigma_\alpha = 0.065$ and $\sigma_\gamma = 0.03$.

The distinct values of γ and α reflect differences in scaling behavior due to the materials involved (Larsen et al., 2010). Mixed-soil landslides typically show moderate scaling,

indicative of combined soil and bedrock failures, while bedrock landslides are characterised by higher γ values, reflecting deeper failures (see Appendix II for derived values).

The model incorporates two key error propagation calculations for different components of the volume estimation process. For power-law relationships of the form $V = A^\gamma$, the error propagation formula applied is:

$$(6) \quad \sigma_f = |f| \cdot \sqrt{\gamma \cdot \left(\frac{\sigma_A}{A}\right)^2 + (\ln(A) \cdot \sigma_\gamma)^2}$$

Where σ_f is uncertainty in the final result, f is the absolute value where $f = A^\gamma$, σ_A is area uncertainty, and σ_γ is exponent uncertainty.

For the complete formula of $V = \alpha A^\gamma$, the error propagation formula is:

$$(7) \quad \sigma_V = |V| \cdot \sqrt{\left(\frac{\sigma_\alpha}{\alpha}\right)^2 + \left(\frac{\sigma_f}{f}\right)^2}$$

In this case, σ_V is the uncertainty in volume, V is the absolute value of calculated volume, σ_α is the coefficient uncertainty, and σ_f is power law uncertainty. The error propagation methods applied in this model assume normal distribution of errors, independence between variables, uniform uncertainty across area ranges, and exclude systematic errors.

For the purposes of this study, the model was adapted to account for areas smaller than $3 \cdot 10^3 \text{ m}^2$, using scaling parameters derived from Roda-Boluda (in prep), given as $V = 0.612 \cdot A^{1.193}$. This adaption meant that there was no error propagation and estimation in place to account for the newly defined area threshold (e.g. 3000 m^2). Such adaptations remain untested and undocumented, likely due to the inherent variability in estimates for smaller volumes (Guzzetti et al., 2009). Although an attempt was made to develop a simplified approach for error

estimation in this instance, the high variability in estimates proved to be more complex and unreliable than initially anticipated. This variability may be attributed to the diverse factors influencing hillslope and soil mechanics, such as antecedent soil moisture, compaction, and composition, which contribute to the significant uncertainty in error estimation for smaller landslides. Furthermore, this paper suggests that power-law functions for volume estimation and error propagation may not fully capture the complexities inherent in smaller landslides. Detailed results of this data can be found in Appendix III.

4.5.3 Magnitude–Frequency Distribution

The mapped landslides follow a magnitude–frequency distribution (Hovius et al., 1997), that can be satisfied by a power-law relation:

$$(8) \quad N_C = k(A)^{-\beta}$$

Where N_C is the number of landslides per magnitude interval of area A and k is a scaling coefficient that defines the rate of landsliding per area per year (Hovius et al., 1997). The exponent β defines the slope of the power-law trend in log-log space, and regressions are fit to the most linear and robust selection of the data distribution (Clarke and Burbank, 2010). Landslide areas are converted to km^2 , and each scar area is used in empirical calculations to derive appropriate correction factors, which adhere to predefined parameters (Hovius et al., 1997). For $A < 2E^{-3} \text{ km}^2$, the factor of 2.1 is applied, for A between $2E^{-3}$ and $2E^{-2} \text{ km}^2$, the factor is 2.5, for $A > 2E^{-2} \text{ km}^2$, the factor is 2.9.

Chapter: 5 Results

5.1 Catchment-Scale Landslide-Derived Erosion Rates

Landslide-derived catchment-averaged erosion rates are established based on estimated landslide volumes (Chapter 4.5.2). The volume of landslides within each catchment were normalised to the respective catchment area, yielding a landslide-derived erosion rate (Emberson et al., 2016). The results of these calculations (Fig. 8) reveal significant spatial variability across the study area. The highest erosion rates were observed in Styx (0.51 ± 0.21 mm/yr) and Kokatahi (0.49 ± 0.21 mm/yr), both of which are characterised by the highest mean slopes (Table 2) and the greatest rates of both annual and daily rainfall. Arahura maintains a moderate erosion rate (0.41 ± 0.15 mm/yr), consistent with its high daily and annual rainfall levels. On the other hand, Hokitika shows an overall lower erosion rate (0.33 ± 0.15 mm/yr), despite having the highest number of mapped landslides per catchment, and being the largest

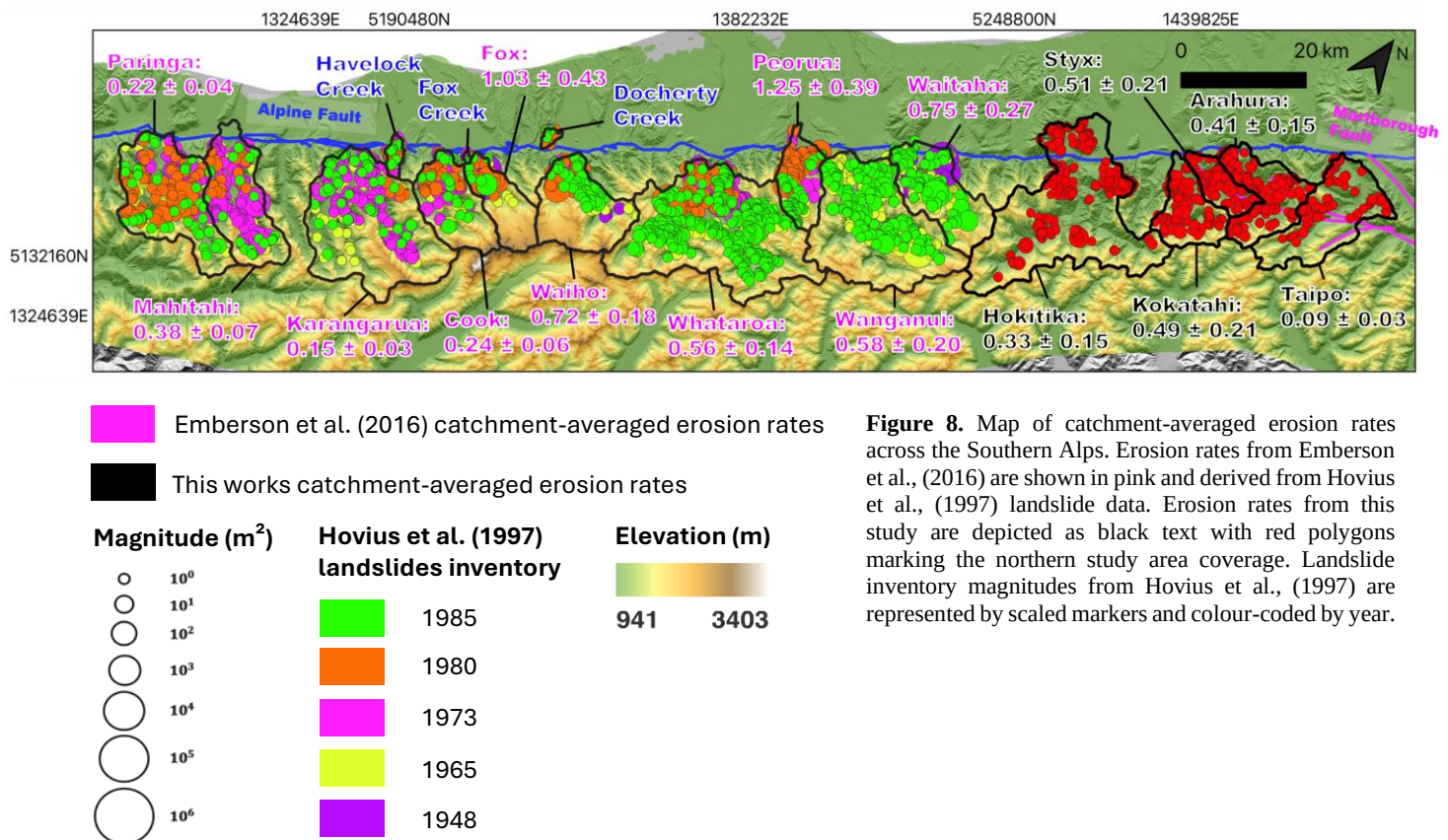


Figure 8. Map of catchment-averaged erosion rates across the Southern Alps. Erosion rates from Emberson et al., (2016) are shown in pink and derived from Hovius et al., (1997) landslide data. Erosion rates from this study are depicted as black text with red polygons marking the northern study area coverage. Landslide inventory magnitudes from Hovius et al., (1997) are represented by scaled markers and colour-coded by year.

catchment in the study area. It is noted that this results from lower levels of rainfall and the fact that 3.4% of its area remains glaciated, while the other catchments are all less than 1% glaciated indicating that there is less available hillslopes to undergo landsliding. The lowest erosion rate (0.09 ± 0.03 mm/yr) established is for the Taipo catchment, where it is assumed to be a resultant from the lower number of mapped landslides and its smaller catchment area.

It is suggested that in smaller catchments, erosional processes are more concentrated making the processes appear more localised. This would create an effect whereby the individual landslide events appear to induce a disproportionately larger impact on catchment-averaged erosion rates, compared to elsewhere. It is therefore possible for this effect to work in reverse, where larger catchments distribute sediment over a larger areas, suppressing the true effect of localised erosion rates. These events may also be linked to shorter transport pathways, given the proximity of the smaller catchments. Styx is exemplary of this. Also, the data supports the idea of higher erosion rates concentrating in the central Southern Alps and progressively decreases further north (Roda-Boluda et al., 2023). A significant observation appears within the data, such that the erosion rates derived in this study are significantly lower than the suspended sediment load-based erosion rates (SSL erosion rates) (NIWA, 2012) and ^{10}Be catchment-averaged denudation rates (Roda-Boluda et al., 2023), highlighting the underrepresentation of landslide contributions to catchment-wide sediment budgets and mass wasting. This is further discussed in Chapter 6.

Table 2. Catchment metrics for Hokitika, Kokatahi, Styx, Arahura, and Taipo. ^{10}Be catchment averaged erosion rates are from Roda-Boluda et al., (2023), and suspended sediment load erosion rates are after NIWA (2012).

Metrics	Hokitika	Kokatahi	Styx	Arahura	Taipo
^{10}Be catchment averaged denudation rates $\pm 1\sigma$ (mm/yr)	5.871 ± 1.323	5.159 ± 1.513	2.651 ± 0.397	1.622 ± 0.200	2.639 ± 0.406
Suspended sediment load-based erosion rates (mm/yr)	2.29	5.35	5.08	0.36	2.04
This work's landslide-derived catchment-averaged erosion rates (mm/yr)	0.33 ± 0.15	0.49 ± 0.21	0.51 ± 0.21	0.41 ± 0.15	0.09 ± 0.03

5.2 Landslide Area

In the absence of a single definitive measure for landslide magnitude, landslide area is often used as a discernible proxy, particularly for complex failures (e.g., Guzzetti et al., 2005). This study investigated trends in landslide area over a 60-year period (1960–2024) across five catchments in the region: Hokitika, Kokatahi, Styx, Arahura, and Taipo. Analysis of landslide frequency in log-transformed area (km²) revealed distinct patterns, some more subtle than others (Fig. 9).

Hokitika is the southernmost catchment in the study area, and it exhibits a decrease in peak recent landslide areas compared to the 1960s; however, both datasets show a similar drop-off in area from 10³–10⁵ km². Despite the temporal variation in peak area values, landslide area frequencies appear not to surpass the established ranges. A shift is observed, where there is a higher frequency of smaller landslides through time, owing to a broader distribution within the catchment. Additionally, Kokatahi illustrate minimal variation in landslide area distribution, apart from a slight increase in frequency for lower magnitude events, distantly resembling the patterns of Hokitika. Styx is recognised as the catchment with a significant reduction in lower magnitude landslides, where most of the events occurring in this catchment operate within the 10³–10³ km² range. Arahura is the only catchment showing a marked shift in trends, with landslide areas spanning from small to large, including the proximal peak, increasing by an order of magnitude since the 1960s. Such patterns require further undertaking, however, for simplicity a baseline hypothesis will suffice.

While a scope of factors influence these trends, including non-linear triggering mechanisms, landscape evolution, and sampling biases, it is important to consider the influence of rainfall in dictating magnitude and frequency. Arahura is one of three catchments greatly affected by both rainfall metrics, highlighting their committed influence, and even though historical

precipitation data was not considered in this analysis, it is more than probable that rainfall experienced in the 1960s was, to an extent, more influential in its contributions to larger, more frequent landsliding. The exhaustion of hillslope material could therefore be reflected in the fact that overall magnitude and frequency of landslides is lower in the recent dataset, resulting from less prone-to-fail hillslopes. It is stated that there is a shift toward smaller, more frequent landslides in Arahura, yet sampling errors in the data collection process are not ruled out. Refer

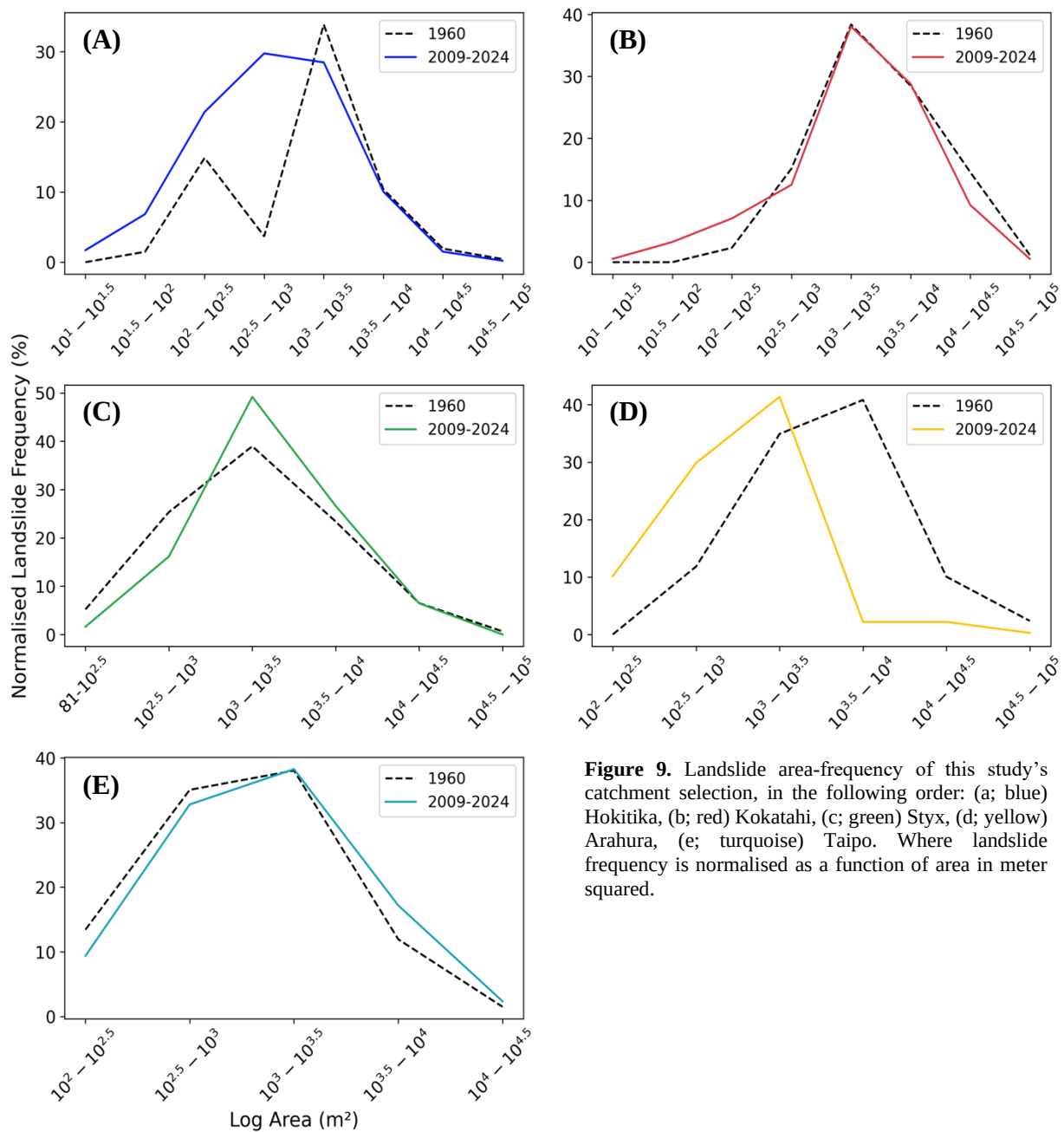


Figure 9. Landslide area-frequency of this study's catchment selection, in the following order: (a; blue) Hokitika, (b; red) Kokatahi, (c; green) Styx, (d; yellow) Arahura, (e; turquoise) Taipo. Where landslide frequency is normalised as a function of area in meter squared.

to Appendix IV for further information pertaining to landslide density as a function of elevation for both datasets.

5.3 Climate

In this study, two types of climate data were analysed: (i) Median Annual Rainfall (MAR) and (ii) Median Rainfall Intensity (MRI). Median Annual Air Temperature was excluded, as both annual and daily rainfall directly influence hillslope processes such as surface runoff, infiltration, and slope stability, which controls hillslope diffusion, including creep (Fernandes and Dietrich, 1997). Temperature is considered to have an indirect effect on slopes, for example, through frost cracking via freeze-thaw cycles, which are more pronounced within specific temperature ranges. Landslide-derived catchment-averaged erosion rates were cross-checked against MAR and MRI (Fig. 10), with similar catchment patterns observed in both plots.

All of the catchments in this study area experience significant amounts of rainfall; though the strongest influence on erosion can be seen in Styx, Kokatahi, and Arahura, with very little influence in Hokitika and Taipo. It is reasonable to state that rainfall does not always contribute to higher rates of erosion. Kokatahi and Styx are two catchments receiving ~9000 mm/yr of rainfall, and maintaining a common rate of 0.5 mm/yr, compared to Arahura, which receives the highest amounts of rainfall (~9500 mm/yr) and maintains a lower rate of 0.4 mm/yr. These trends highlight that factors beyond rainfall are contributing to erosion rates. This is also observable in Taipo, another catchment receiving substantial amounts of rainfall (7000 mm/yr) but occupies the lowest erosion rate of the study (~0.1 mm/yr). It is unclear what drives such an observation, but it is assumed that sediment retention per catchment is potentially playing a role.

Hokitika has the lowest rainfall (~6500 mm/yr) and exhibits a higher erosion rate (~0.3 mm/yr) than Taipo, reinforcing once more the idea that rainfall is a part of the minority that influence the processes shaping sediment routing and hillslope diffusion. Regarding MRI, higher rainfall intensity appears to correlate with higher erosion rates, though the relationship is not entirely linear. Arahura, Kokatahi, and Styx are all experiencing similar rainfall intensities (~33–34 mm/day) but only slight variations in erosion rates are observed. Despite Taipo having a relatively high rainfall intensity (~32 mm/day), it shows minimal erosion. However, the similar patterns of daily rainfall and erosion in Styx, Kokatahi, and Arahura suggest that rainfall intensities above a certain threshold, such as ≥ 33 mm/day, may have a predictable effect on erosion rates, provided other catchment factors, such as mean slope gradient and sediment supply, are favorable.

5.4 Distance from Alpine Fault

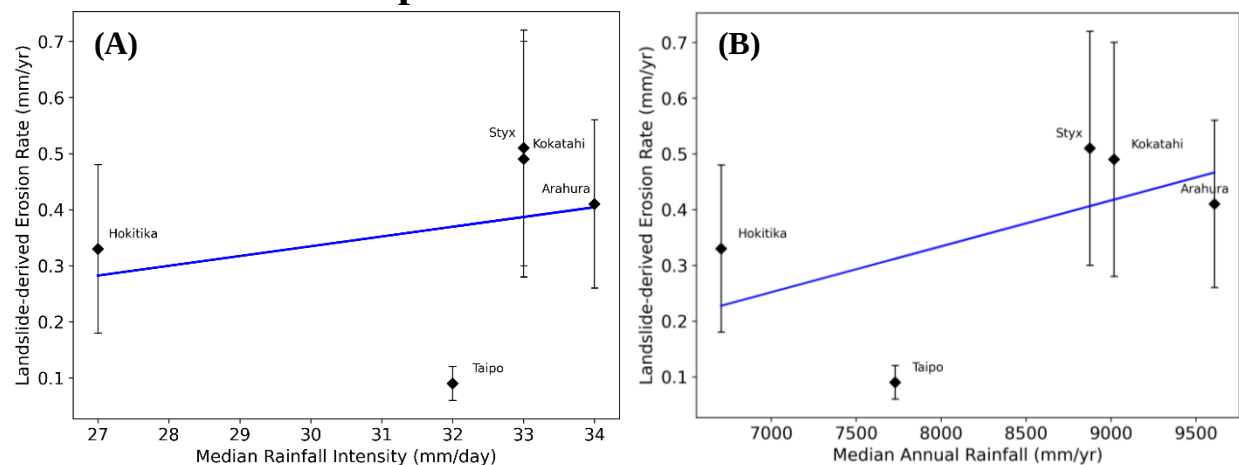


Figure 10. This study's landslide-derived erosion rates as a function of daily and annual rainfall with a blue regression line indicating small overall correlation. (a) Median Annual Rainfall, and (b) Median Rainfall Intensity.

The relationship between normalised landslide frequency and distance from the Alpine Fault was analysed for both the 1960s and recent landslide datasets. Understanding these relationships provides insights into the tectono-geomorphological controls on landslide frequency and magnitude. Each catchment displays its own unique relationship with landslide

frequency at specific distances from the Alpine Fault (Fig. 11b–f). However, the focus here will be on the combined results (Fig. 11g).

Starting with the 1960s data, the 0–5 km band represents 15% of landslide frequency, showing moderate values. The 5–10 km band accounts for 42% of total landslides, indicating a significant increase in frequency and highlighting the dominant role of fault-related influence, particularly the vertical throw at the fault. At 10–15 km, landslide frequency decreases to 35%, suggesting a diminishing effect of fault activity. Beyond this, the 15–20 km and 20–25 km bands account for only 0–10% of landslide frequency, reflecting the reduced influence of the fault on landslide events at greater distances.

The recent data reveals a slight increase in landslide frequency compared to the 1960s, possibly reflecting both the growing significance of the Alpine Fault in conditioning landslide processes and improved detection techniques. The 5–10 km zone remains the area with the highest frequency, reinforcing the notion that faulting and seismic activity play a critical role in landslide dynamics within this range. This pattern is consistent with the 1960s data, showing a decrease in frequency with increasing distance from the Alpine Fault. Most catchments exhibit a peak in landslide frequency at 5–10 km, except Kokatahi, which shows a peak at 10–15 km, due to its geographical positioning relative to the fault (Fig. 11a). The general trend of frequency drop-off around 15 km suggests that most catchments fall outside the more fault-affected areas.

In terms of median landslide area (A_{median}) at the 5–10 km peak, a noticeable decrease in landslide size is observed from the 1960s to recent years. In Hokitika, A_{median} decreases sharply from 1732 m² to 386 m², indicating a shift toward smaller landslides. Kokatahi shows a minor

reduction, from 1750 m² to 1598 m². In Styx, however, A_{median} increases from 1611 m² to 1958 m², suggesting a trend toward larger landslides in this catchment.

Substantial decreases in A_{median} are also noted in Arahura and Taipo, where values drop from 2953 m² to 841 m² and 2247 m² to 964 m², respectively. These trends, especially in Hokitika, Arahura, Taipo, and the overall dataset, suggest a significant shift toward smaller landslides

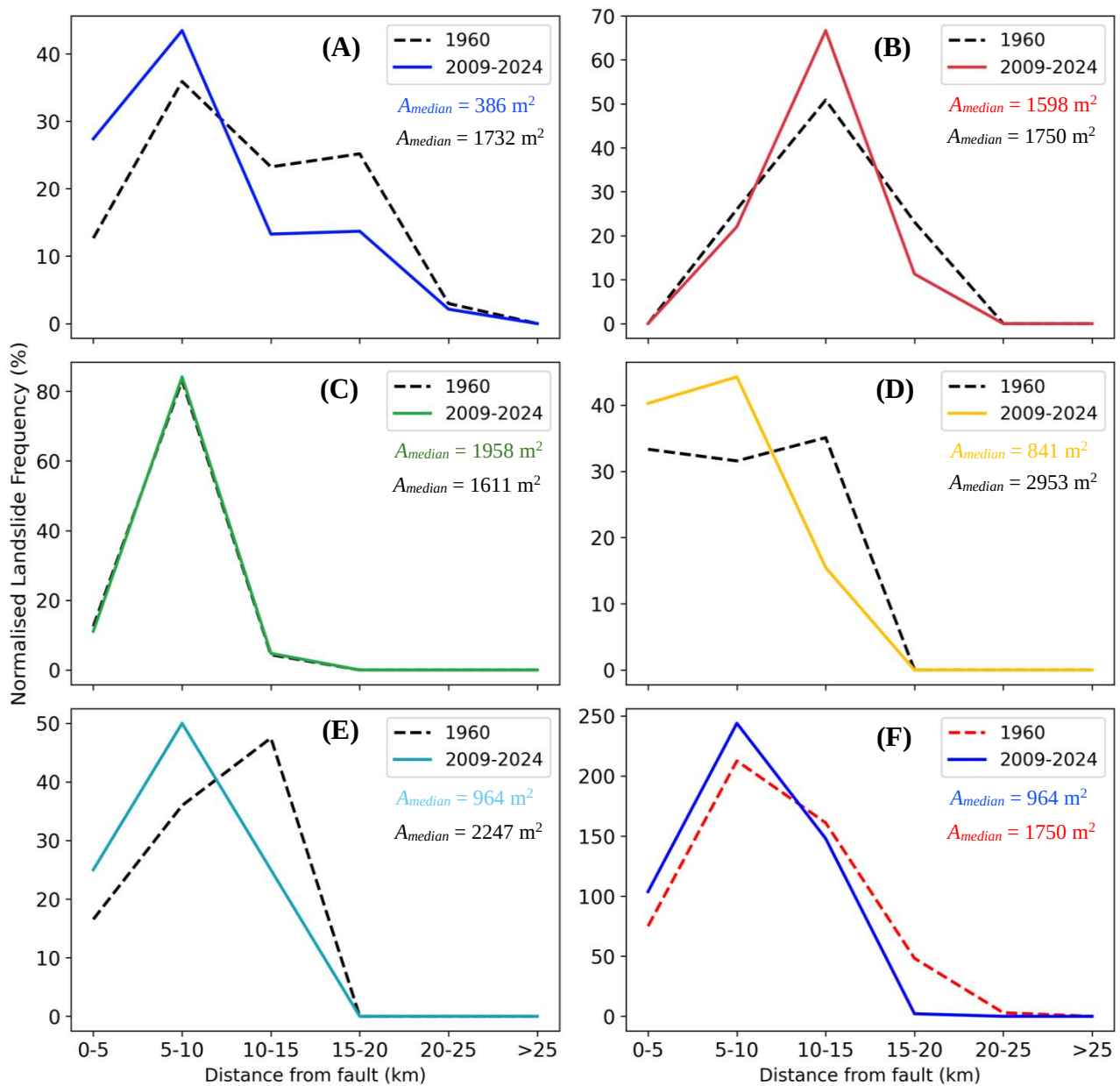


Figure 11. A series of plots showcasing historical and recent trends, where normalised landslide frequency is plotted over a distance of 0–25km (a) A spatial map visually illustrating the reference distance bands that were used in order to calculate relative densities, where blue band is 0-5, green band is 5-10, yellow band is 10-15, pink band is 15-20, and purple band is 20-25km (b; 1) Hokitika, (c; 2) Kokatahi, (d; 3) Styx, (e; 4) Arahura, (f; 5) Taipo, (g) entire catchment study area. Each graph contains the median area across both recent and historical dataset at 5-10km distance, where colours corresponds to each line.

over time. Styx is the only catchment where A_{median} has increased, reflecting local influences that may be driving this contrasting pattern. For a detailed list of median area values at each respective distance interval for both datasets, refer to Appendix V.

5.5 Magnitude–Frequency Distribution

To assess the rates at which landslides contribute to landscape evolution, both the size of individual landslide events and the frequency of events of different sizes can be analysed (e.g., Hovius et al., 1997). This analysis allows for a more comprehensive understanding of the mechanisms governing slope failure and, subsequently, sediment transport. These distributions are often expressed as power-law relationships, which illustrate the balance between larger magnitude, less frequent events and smaller, more frequent events that cumulatively drive erosion.

Logarithmic landslide magnitude-frequency distributions, normalised by area and time, were calculated (e.g., Clarke and Burbank, 2010; Hovius et al., 1997). The expanded dataset allowed for adjustments in the logarithmic binning to accommodate a wider area range. This involved extending the original 10^0 – 10^{-3} range to 10^0 – 10^{-5} , as indicated in Figure 12. It is important to note that the binning strategy itself was not altered; rather, the minimum and maximum values were expanded to better capture the study’s values. This analysis incorporates both newly compiled data and previously published inventories by Hovius et al. (1997) and Emberson et al. (2016). The newly mapped landslide inventory spans several decades of geomorphological activity and emphasises the improvements gained from higher-resolution, imagery-driven analyses.

A key objective of this analysis was to assess the influence of improved detection techniques on observed landslide distributions, especially in capturing smaller landslide events. However,

it is essential to note that the findings of this study provide only a minimum estimate, as historical landslides were mapped only for the decade of the 1960s. Including data from the 1970s through the 1990s would be crucial for obtaining more reliable first-order estimates.

The analyses were conducted in three ways: (i) using legacy datasets (e.g., Hovius et al., 1997; Emberson et al., 2016), (ii) combining all datasets, and (iii) using this study's new dataset independently. Power-law regressions were calculated for all datasets between 10^0 – 10^{-3} km², yielding k values of 1.28, 4.7, and 0.98 respectively. According to Ohmori and Hirano (1988) and Hovius et al. (1997), when the slope exponent (β) is ≤ 1.5 , the largest landslide events contribute the most to overall denudation. The β value for this study is 1.19, for combined data it is 1.31, and for the legacy datasets, it is 1.26.

The roll-over observed in this study's data, compared to the other two (e.g., green and purple lines), suggests that the mapping approach used here captured a greater number of smaller landslides. This is likely due to the use of higher-resolution imagery with Google Earth Pro (GEP), compared to older datasets like Hovius et al. (1997), which were limited by the quality of physical orthophotos. Not only did this approach improve detection precision, but it also enhanced accuracy. Amalgamation in landslide mapping is a known issue (Marc and Hovius, 2015), and earlier datasets might have been affected by this problem. The use of GEP in this study accounted for amalgamation to a greater degree than previous methods.

Combining the three datasets and the individual data (Fig. 12) resulted in a lower overall scaling coefficient. As is well-known, the scaling coefficient indicates whether sediment flux is driven by high-magnitude events or a series of smaller events (Clarke and Burbank, 2010). The increased number of shallow landslides observed suggests a reduction in the frequency of larger events, as indicated by the k values. This decrease in the scaling exponent reflects the

smoothing effect of this dataset, producing a more accurate balance between landslide density and magnitude.

The detection of smaller landslides in this study's data can be explained by the following factors: (i) Hovius et al. (1997) covered a larger study area, which includes regions of the Southern Alps with much higher uplift rates than the catchments in this study, potentially leading to differences in landslide frequency and size, (ii) the use of Google Earth Pro (GEP) in this study allowed for the identification of previously undetected shallow landslides, providing greater precision and accuracy in mapping, and (iii) the Hovius et al. (1997) dataset likely contains higher levels of landslide amalgamation, leading to an underrepresentation of smaller landslides in their dataset.

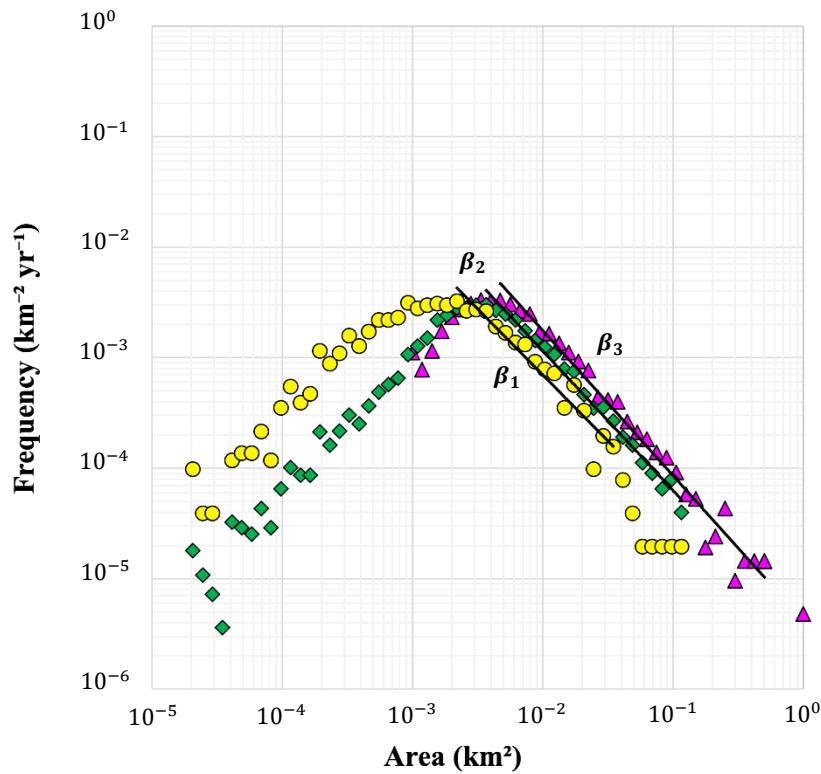


Figure 12. Magnitude-frequency density (mapped landslides per km²) as a function of landslide area A . The yellow curve represents this study's dataset alone ($n=2755$), the green curve combines this study's data with Hovius et al., (1997) and Emberson et al., (2016) ($n=11207$), and the purple curve represents only the datasets from Hovius et al., (1997) and Emberson et al., (2016) ($n=8493$). The power-law exponents are $\beta_1=1.19$, $\beta_2=1.31$, and $\beta_3=1.26$ for the yellow, green, and purple datasets, respectively.

Chapter 6: Discussion

Here, the aforementioned results are analysed and compared with related research. The comparative factors involved in this analysis include Alpine Fault throw rates at the catchment outlet, exhumation rates derived from low temperature thermochronology, and mean slope metrics. Additionally, annual and daily rainfall data are incorporated to examine the combined effects of climate and tectonics.

6.1 Landslide Magnitude: The Pre-conditioning Factor

Geospatial analysis reveals that the majority of landsliding predominates within 5 to 10 km of the Alpine Fault, therefore, this study finds 5 to 10 km distance of the Alpine Fault to be the proximal hillslope failure zone. Norris and Cooper (2001) highlight the study area being subject to high long-term uplift rates, where faulting weakens the overall bedrock strength involved in slope stability (Roda-Boluda et al., 2023). It is argued that long-term uplift rates over the last 50,000 years have contributed to the steepening of slopes, making them more susceptible to failure. Additionally, that rainfall acts as an immediate trigger for landsliding, especially when steep hillslopes are present (Robinson et al., 2016). See Chapter 6.3.

Landslide-derived erosion rates are examined against Late Quaternary throw rates, which represent the vertical component of fault slip rates (Roda-Boluda et al., 2023). A strong positive correlation is found (Fig. 13), suggesting that higher throw rates correspond with higher annual erosion rates. This relationship highlights that high Late Quaternary uplift rates have contributed to catchment slope steepening, particularly nearer to the Alpine Fault, reinforcing the geomorphological conditions at the catchment outlet where landslide frequency is concentrated, as previously indicated in Chapter 5.3.

The tectono-geomorphological processes at play here drive hillslopes toward their critical thresholds of stability (Montgomery and Dietrich, 1994). Initially, the landscape response to this may be slow, with gradual erosion and steepening. Over time, the combined effects of uplift, erosion, and stress propagate through the landscape, leaving an imprint, particularly around the proximal zone near the fault. The correlation between throw rates and erosion (Fig. 14) is indicative of a progression to active hillslope failures, reflecting a lag time response to tectonics. This accords with the idea that uplift is a first-order control on hillslope processes (Roda-Boluda et al., 2023; Korup, 2006, 2008; Clarke and Burbank, 2011).

The high landslide density at 5 to 10 km from the Alpine Fault suggests that this zone represents an active failure area, potentially indicative of a landscape still adjusting to Quaternary uplift. The decrease in landslide median area since the 1960s (Fig. 11) suggests a transition toward smaller landslides, which may be a reflection of the ongoing adjustment of the catchments to Quaternary tectonics. The observed larger landslides in the 1960s may be representative of such adjustment cycles, where the geomorphological response aided in the production of larger landslides (Fig. 11).

Hovius et al. (1997) states that active landslide scars have a lifespan of ~25 years, as determined by the cyclic nature of vegetation regrowth. It is suggested here that some of the imagery used to capture landslides may include failure deposits from up to 25 years prior. The influence of events prior to the 1960s may well be reflected within the historical data and thus cannot be entirely excluded. The inclusion of additional historical data, such as mapped landslides from the 1950s and 1940s, would offer better speculation on this matter than currently exists.

The lack of access to historical mean slope data is recognised, which directly limits the analysis of hillslope evolution, but the current mean slope data (Fig. 13c, d) show an overall weak signal with erosion. It might be that this arises from the suggested former adjustment period, supported by the observed decrease in median landslide area since the 1960s (Fig. 11; Appendix V). This observation points to the idea that threshold slopes, which are maintained by bedrock landsliding, could have reached a steady state where further steepening is limited by the strength of the bedrock at the mountain scale (Schmidt and Montgomery, 1995). However, it is increasingly difficult to quantitatively account for this at such great scales.

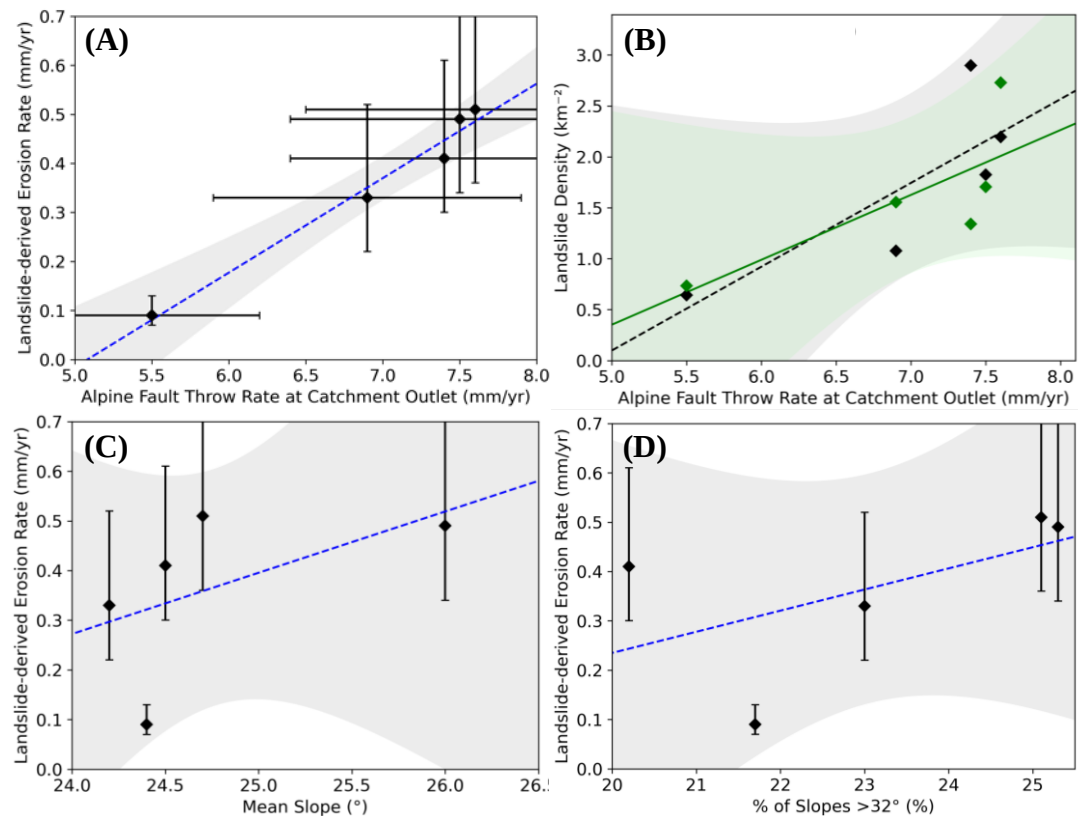


Figure 13. (a) Landslide-derived erosion rates as a function of Alpine Fault throw rate, showcasing a strong positive correlation, yielding $R^2=0.974$ and pearsons correlation (r) of 0.987. (b) landslide density per km² as a function of Alpine Fault throw rate, showcasing a positive relationship for both datasets, where green data is 1960s dataset yielding $R^2=0.585$ and $r=0.765$, and black data is recent dataset yielding $R^2=0.641$ and $r=0.800$. (c) the relationship between catchment mean slopes and this works landslide-derived erosion rates with $R^2=0.270$ and $r=0.369$, and (d) showcases the relationship between the percentage of slopes greater than 32° and its influence on landslide-derived erosion rates with $R^2=0.305$ and $r=0.334$. Plots a, c, and d have associated error bars and all four plots include calculated 95% confidence bands showcased by grey feature, and also green in plot b.

Processes such as channel incision, which removes mass from the base of slopes, may be operating at a rate similar to the supply of material, effectively maintaining valley slopes at a threshold gradient controlled by large-scale rock strength (Binnie et al., 2007). The higher magnitude landslides of the 1960s, or perhaps even earlier, could be the overarching contributor to higher erosion rates. The sharp decrease in median landslide area since this period (Fig. 11e, f) supports this idea, though the exact timing of this change remains uncertain due to the lack of data before or after the 1960s.

Without data before or after this decade it would be unreasonable to assume otherwise. However, based on presented results, this study assumes a greater magnitude of historical landsliding that would decouple erosion from slope gradients (Fig. 13c) in some catchments. The catchment processes operate in a heterogenous manner, given that two catchments with more than 25% of their slopes greater than 32° (Fig. 13d) are producing the highest recorded erosion rates. The shift from high magnitude landsliding events in the 1960s, to much lower magnitude and higher frequency landsliding reaffirms the transition to an overall more stable configuration, dominated by a multitude of smaller landslides (Fig. 11 and 12). It is indeed a reflection of the cumulative impact of fault related processes which drives variability in slope stability. The positive correlations observed between throw rates, exhumation, and erosion further suggest tectonics drive the landscape towards instability (Clarke and Burbank, 2011), resulting in contributions to sediment flux (Fig. 14).

The 60% decrease in landslide magnitude in the Arahura and Taipo catchments (Fig. 12) are thought to be indicative of a historical phase in which deeper excavation dominated landslide events, leading to larger, more significant landslides during the 1960s. Since then, the size of landslides has diminished, which might explain the observed weak correlation between exhumation and erosion in these catchments. In such cases, deeper landslides from the past

could have contributed more significantly to the sediment flux, while current landslides, which happen to be smaller, contribute less; a result of catchment adjustment to a steady-state configuration.

As for the very low erosion rate observed in Taipo, two possible explanations are suggested: (i) the temporal gap between the historical and recent data is not fully capturing the sediment fluxes within the catchment, or (ii) Taipo may be unique with respect to its sediment retention capabilities, possibly through landslide dams that trap sediment in the valley bottoms, preventing downstream transport (Korup, 2004). This is further elaborated on Chapter 6.3.

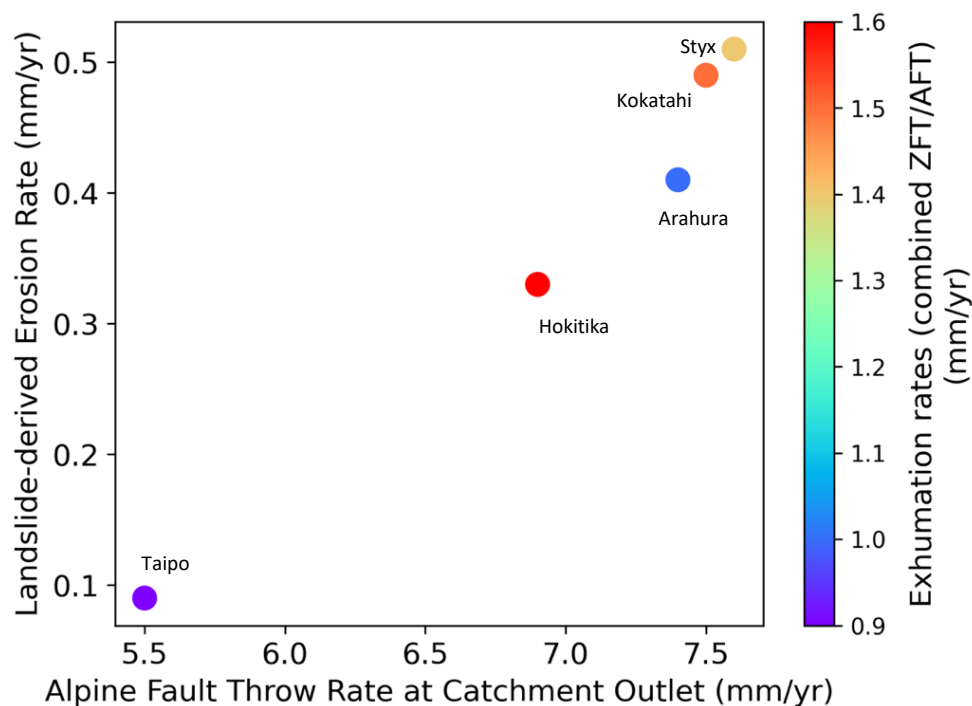


Figure 14. Landslide-derived erosion rates as a function of throw rate at the Alpine Fault, with exhumation rates (mm/yr) represented by the colourbar. The scatter plot shows the potential correlation between erosion, throw, and exhumation for individual catchments.

6.2 Landslide Frequency: The Triggering Mechanism

Landslides triggered by rainfall and storms are significant concern across New Zealand (Glade, 1998). The mechanism that leads to slope failures is well understood and relates to the increase

in negative pore-water pressures (Brand, 1981) when water begins infiltrating unsaturated soil. The loss of negative pore-water pressures decrease the shear strength of the soil below the mobilised shear strength along the potential slip surface (Tsaparas et al., 2001), and can be significantly facilitated through site-specific hydrological conditions such as high antecedent soil saturation. Additionally, in deforested regions the loss of tree root reinforcement and the associated altered hydrological regimes increases slope failure susceptibility during rainfall events (Glade, 1998, 2003). This is achieved through the reduced mechanical cohesion of slope surfaces, where the given intensity of rainfall events will largely determine the rate and magnitude at which regional thresholds are exceeded owing to increased widespread triggering of landslides.

In the Taipo catchment, the role of rainfall in triggering landslides appears to be less significant, despite the substantial rainfall received on both a daily and annual basis (Fig. 11). The data suggests rainfall alone is not a major factor driving landslide frequency or magnitude in Taipo. Naturally, a conceptual understanding of regional rainfall thresholds can be valuable for understanding landslide susceptibility, but this study's results highlight the need for cautious interpretation (Smith et al., 2023) for the reason that such thresholds can ignore variations in climate and landscape factors that influence susceptibility (Marc et al., 2019).

The relationship between rainfall and landslides is not straightforward, and is influenced by various factors including rainfall intensity, duration, soil properties (e.g., antecedent conditions), land cover, and topography (Glade, 1998; Smith et al., 2023; Tsaparas et al., 2001). While this study does not assess the role of land cover, it acknowledges its significance and would integrate it in the future, if possible.

Rainfall is compared against the relationships between throw rates and erosion rates, as well as landslide density (Fig. 16) to determine how exactly the three operate with one another. It is accepted that rainfall intensity is critical in determining the magnitude of landslide events (Glade, 1998, 2003; Smith et al., 2023), and the data from this study supports this. Figure 15a and 15b demonstrate that rainfall significantly conditions both landslide magnitude and frequency, with clear patterns emerging, particularly toward the higher end of correlations where the highest levels of daily and annual rainfall are sufficiently aligned with the highest rates of erosion and Alpine Fault throw. However, two outliers in the data reaffirm the heterogenous nature of catchment processes, complicating a straightforward interpretation. Using Styx, Kokatahi, and Arahura as an example, similar median rainfall intensities are inline with the higher rates of erosion, reiterating that intense, short-duration rainfall events are more influential in triggering landslides than longer term rainfall patterns (Crozier, 1999; Kritikos et al., 2015; Brook, 2019). Such findings highlight the importance of considering rainfall intensity and duration as key factors in landslide susceptibility.

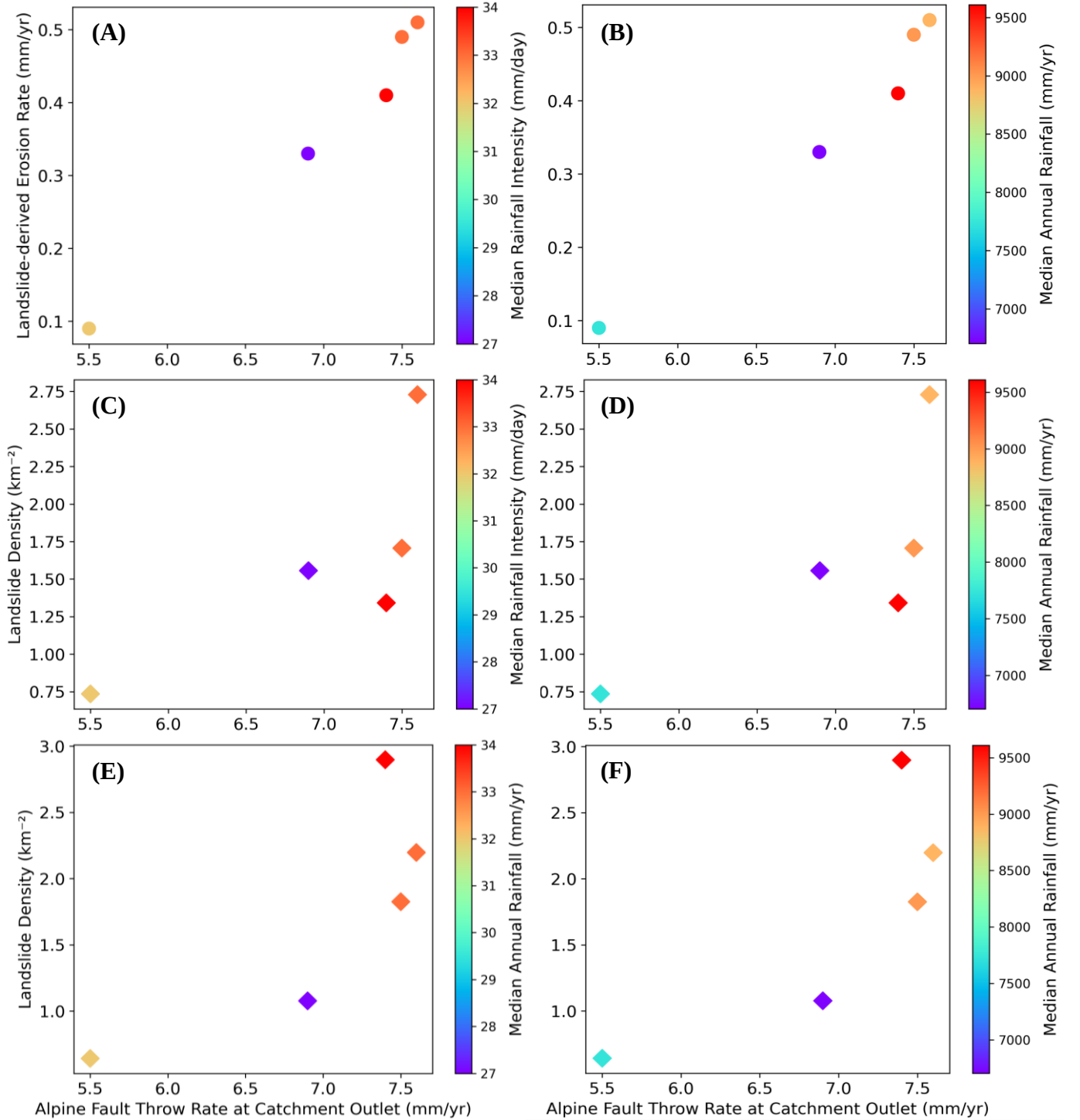


Figure 15. A series of plots showing the relative contribution of median rainfall intensity and median annual rainfall against both landslide-derived erosion rates (**a** and **b**), and landslide density for recent and historical datasets using Alpine Fault throw rate. Where **c** and **d** is this study's historical landslide dataset, and **e** and **f** is the recent landslide dataset. For clarity, graphs **a**, **c**, and **e** are all median annual rainfall, and graphs **b**, **d**, **f** are median rainfall intensity.

6.3 Contribution of Landslides to Catchment-Averaged Erosion Rates

In active orogens, landslides play a key role in driving rates of erosion, especially when these orogens are subject to rapid uplift and influenced by glacial and periglacial processes (Burbank, 2002; Hovius et al., 1997). In the case of the Western Southern Alps, research has shown that landslides contribute significantly to the region's denudation, with a landslide-driven erosion rate of approximately 9 ± 4 mm/yr (Hovius et al., 1997). This rate aligns closely with the region's long-term rock uplift rate, reinforcing the link between tectonic forces and surface processes.

6.3.1 *Sediment Connectivity*

It is previously stated that the landslide data of this work alone do not account for all erosion and suspended-sediment load (SSL) erosion rates, as not all landslide material directly enters the fluvial system. Other sources of sediment, such as surface runoff, contribute to total sediment flux (Huggel et al., 2021). Landslide material must be transported into rivers for it to contribute to SSL erosion rates.

Battista et al. (2020) demonstrated that sediment mobilisation on hillslopes, particularly in transport-limited environments, can occur through overland flow. This would be where sediment is transported to river channels, though only if the hillslopes are connected to the river network. For disconnected landslides to contribute to sediment flux, overland flow must have sufficient bed shear stress to transport the sediment off the slope. This transient functional connectivity to the river network is essential for the delivery of landslide material (Battista et al., 2019, 2020; Chen et al., 2020).

Sediment generated by landslides often remains immobile on hillslopes, despite the cumulative daily and annual effect of rainfall. Rainfall facilitates the mobilisation of sediment via surface runoff, transporting accumulated material downslope into river channels. However, these relationships are not straightforward. For example, sediment yields in the Western Southern Alps can exceed 70,000 t km² per year (Korup, 2004), and rivers draining the Western flanks of the Southern Alps have annual sediment yields approximately 10 times higher than global averages (Griffiths, 1979). This is subjective to the catchment, based on the angle of its slopes (e.g., if they are steeper there is a greater overall transport efficiency) and rainfall patterns, where the latter relates mostly to Arahura which receives 9,500 mm/year (Fig. 10). This could suggest that while sediment supply may appear abundant, catchment-specific factors ultimately control the rate at which sediment flux from catchment to catchment (Beel et al., 2011).

Despite Arahura receiving the highest annual rainfall, it is shown that it doesn't necessarily align with higher rates of erosion, especially when compared to other catchments. However, the erosion rate in Arahura is nearly 14% higher than the National Institute of Water and Atmospheric Research (NIWA, 2012) suspended-sediment load (SSL) erosion rates, suggesting that sediment generation is greater than sediment flux. As previously discussed, this is likely tied to the ineffective transport and removal of sediment, where sediment retention in Arahura is of a greater efficiency than other catchments. The dual role of landslides—both supplying and retaining sediment—is crucial, as large landslides can dam river channels, causing forced upstream alluviation that traps sediment and reduces downstream transport (Korup, 2004).

To confirm these findings and better understand the dynamics of sediment flux and storage in Arahura, further analysis using longitudinal stream profiles could identify knickpoints that help

explain sediment storage and flux. By developing and simulating these profiles, possibly through the HyLands Landscape Evolution Model (Campforts et al., 2020), it would be possible to assess the formation of knickpoints, epigenetic gorges, and sediment accumulation zones (Campforts et al., 2022). The large net sediment delivery rates of landslides can be catastrophic (Korup, 2004), and heavily influence fluvio-geomorphological processes. For instance, the accumulation of landslide debris in valleys can not only directly block river channels, creating temporary or permanent dams (Korup, 2004), but also act as a tools cover effect and shield the underlying bedrock from further rates of incision and abrasion, promoting the formation of epigenetic gorges (Campforts et al., 2022). Separately, the method of data collection, including temporal considerations, date ranges, and calculation methods will influence the interpretation of results.

6.3.2 Data Completeness and Uncertainty Evaluation

Understanding erosion rate estimates in Arahura requires evaluating the methodologies used and acknowledging their inherent limitations. Here, the two proxies, SSL erosion rates and ^{10}Be denudation rates, are examined. Special attention is appointed to considering their respective contributions in interpreting this study's catchment processes.

SSL erosion rates are derived from measurements spanning the period from 1960 to the 1990s, reflecting short-term sediment transport dynamics (Hicks et al., 2011). These rates are influenced by high-frequency climatic variability, which may not capture long-term sediment dynamics, especially since they exclude the effects of net sediment retention on floodplains and estuaries (Hicks et al., 2011). While SSL rates provide a snapshot of sediment transport during this period, they may not adequately represent the full range of sediment delivery processes or the long-term evolution of the catchment's erosion dynamics.

On the other hand, ^{10}Be denudation rates, which integrate over approximately 240 years (Roda-Boluda et al., 2023), offer a longer-term average. These rates won't capture recent shifts in sediment dynamics, which could be influenced by recent anthropogenic influence or shifts in climate variability. The study does not consider denudation and erosion rates as directly comparable due to their differing processes, where erosion involves the physical transport of sediment, while denudation includes both erosion and weathering, complicating comparisons between the two proxies.

A significant source of uncertainty arises from the temporal biases inherent in each proxy, though it is suggested that the SSL rates are most comparable with the established erosion rates of this study. Additionally, the volume-area scaling method for estimation of landslide-derived erosion rates introduces further uncertainty, particularly in the case of large-magnitude landslides, which is otherwise expected. However, no uncertainty has been calculated for smaller, more frequent soil landslides due to very little existing literature covering the topic. Additionally, catchment-specific factors such as slope, channel density and connectivity complicate cross-method analysis.

Chapter 7: Closing Remarks

This study integrates the use of Google Earth Pro (GEP) for mapping recent landslides (spanning the past 20 years) with historical aerial imagery to identify landslides from the 1960s. This approach is coupled with an analysis of climate data and Alpine Fault throw rate to examine the interplay between landslide-derived erosion rates, tectonic activity, and climatic influences.

The findings reveal that the vertical displacement component of the Alpine Fault significantly influences landslide frequency and magnitude, with landslide activity being primarily concentrated within a 5 to 10 km range of the fault. Strong positive correlations between landslide-derived erosion rates and Late Quaternary throw rates ($R^2 = 0.974$, $r = 0.987$) underscores this relationship. Higher throw rates are closely associated with increased erosion rates, highlighting a strong connection between tectonic processes and slope instability as a preconditioning factor.

Rainfall emerges as a key factor in triggering landslides within the selected catchments, with intense, short-duration events playing a particularly prominent role. The influence of rainfall on this study's erosion rates align closely with existing research on landslide triggers in New Zealand, where high rainfall intensities are known to increase negative pore pressures and reduce shear strength (Brooks, 2002). The strong correlation between median rainfall intensity and erosion rates further solidifies rainfall as an immediate triggering mechanism for landsliding in the Southern Alps (Robinson et al., 2016). However, it is recognised that rainfall is not the sole determinant of landslide patterns; other hydrometeorological factors, including antecedent conditions, soil mechanical properties, and vegetation cover, may also significantly influence landslide activity (McColl, 2015).

Significant decreases in median landslide area are observed since the 1960s, indicating a shift toward smaller, more frequent landslides. A trend that is attributed to a gradual, ongoing adjustment to tectonics, with an assumption that some, but not all, catchments are approaching a more stable configuration. The larger magnitude landslides documented in the 1960s likely represent an initial phase of catchment response to uplift, as suggested by the strong correlation between erosion rates and throw rates, but it is noted that the landslide events within this decade could be associated with earlier decades due to the vegetation regrowth cycle of approximately 25 years (Hovius et al., 1997). These interpretations are supported by magnitude-frequency distribution analysis, which shows a roll-over in the distribution of landslides in this study compared to legacy datasets (e.g., Hovius et al., 1997; Emberson et al., 2016). Additionally, the increased frequency of smaller landslides may reflect both the enhanced detection capabilities granted by Google Earth Pro and the ongoing geomorphological response to uplift.

Landslides in the study area contribute to catchment-averaged erosion in the Western Southern Alps, although the complexity of understanding sediment routing systems and connectivity within catchments is also emphasised. While landslides are commonly recognised for delivering substantial volumes of sediment, not all of the sediment derived from landslides in this study seems to enter the fluvial system directly. Instead, it may remain immobilised on hillslopes, or stored in the form of landslide dams which increase sediment retention within the catchment, thereby limiting the amount of sediment reaching river channels.

The multi-faceted relationship between uplift, climate, and slope stability is discernible, and reveals broad implications for sediment transport processes. It is also demonstrated that bulk erosion was once driven by high-magnitude, low-frequency events (Hovius et al., 1997) but has now shifted to a phase dominated by low-magnitude, high-frequency events. Future research is recommended to extend the temporal coverage of the dataset, particularly

addressing the gap between 1970 and 1990, to refine minimum estimates. The development of model simulations in targeting longitudinal river profiles for the identification of knickpoints, dams, and sediment accumulation would also be an asset. Furthermore, a detailed investigation into the influence of vegetation cover on landslide susceptibility would offer better understanding to the mechanical properties that promote the stability of slopes, particularly for catchments north of the Central Southern Alps, where uplift associated with the Alpine Fault gradually diminishes. Finally, GEP proved to be a valuable asset to this study, though its limitations should be acknowledged. Challenges related to image quality—particularly shadowing in areas of high relief and 3D-mode distortions—were evident and impacted the accuracy of landslide detection during mapping efforts.

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Appendix I — Georeferenced Imagery

This study's georeferenced imagery mosaics and associated mean error values, as well as the Command Line commands used in data processing.



Figure 16. Imagery mosaic of Hokitika (left)

Catchment	$\bar{E}_{\text{TPS}}$
HOK2	8.28923e^{-10}
HOK4	4.76724e^{-10}
HOK6	1.02513e^{-9}
HOK8	2.8167e^{-10}
HOK9	5.43272e^{-10}
HOK10	1.03171e^{-9}
HOK11	4.62533e^{-10}
Average	6.60526e^{-10}

Table 3. Mean error values for each historical image in Hokitika (above)

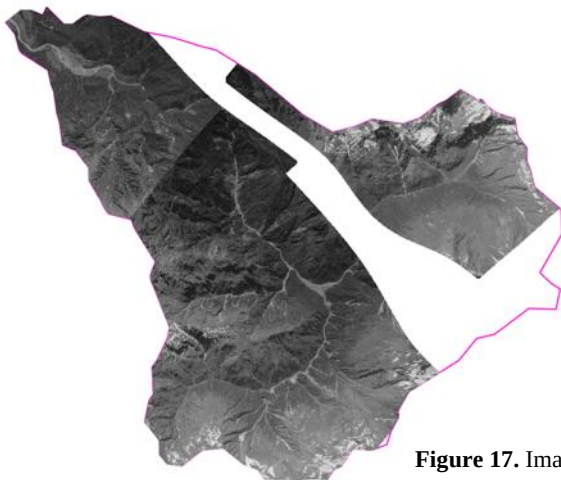


Figure 17. Imagery mosaic of Kokatahi (left)

Catchment	$\bar{E}_{\text{TPS}}$
KOK1	3.81298e^{-10}
KOK2	3.88129e^{-10}
KOK3	1.4377e^{-10}
Average	1.2937e^{-10}

Table 4. Mean error values for each historical image in Kokatahi (above)

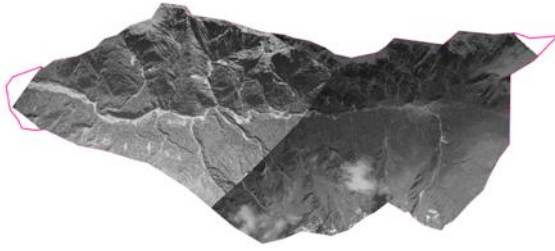


Figure 18. Imagery mosaic of Styx (above)

Catchment	$\overline{E}_{\text{TPS}}$
STYX1	1.81581e^{-10}
STYX2	3.87471e^{-10}
Average	2.84526e^{-10}

Table 5. Mean error values for each historical image in Styx (above)

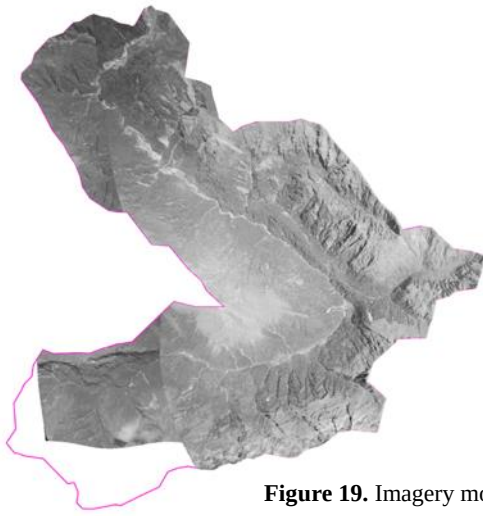


Figure 19. Imagery mosaic of Arahura (left)

Catchment	$\overline{E}_{\text{TPS}}$
ARA1	5.91841e^{-10}
ARA2	2.00727e^{-9}
ARA3	3.20887e^{-9}
Average	1.93599e^{-9}

Table 6. Mean error values for each historical image in Arahura (above)



Figure 20. Imagery mosaic of Taipo (left)

Catchment	$\overline{E}_{\text{TPS}}$
TAI1	4.50248e^{-9}

Table 7. Mean error values for each historical image in Taipo (above)

CLI Commands

NoData Value removal:

```
gdalbuildvrt -srcnodata 0 output.vrt "input.tif"
```

where, gdalbuildvrt [access GDAL to create VRT] → -srcnodata [src function to define no data value '0'] → <write the output VRT file name> → <insert input TIF file>

Then,

```
gdal_translate -a_nodata 0 input.vrt output.tif
```

where, gdal_translate [convert raster data between formats] → -a_nodata [specify NoData value constraint] → <insert input VRT file> → <write output TIF file>

File Compression:

```
gdal_translate -co COMPRESS=DEFLATE input.tif output.tif
```

where, gdal_translate [convert raster data between formats] → -co [creates option to configure properties] → COMPRESS=DEFLATE [application of compression type] → <insert input TIF> → <write output TIF>

Mosaic Creation:

```
gdal_merge.py -o output.tif "<input raster>" "<input raster>"
```

where, gdal_merge.py [merge multiple raster datasets] → -o [specify output file name] → <write output TIF> → <insert series of TIFs>

Appendix II — Volume–Area Scaling Values

V-A values associated with recent landslides, showing soil area and volume, mixed soil-bedrock area and volume, and the associated errors for mixed values.

Table 8. Hokitika volume–area scaling values for soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)
342.094028	645.634895	3849.63822	3784.01498	455.6548127
99.2441131	147.509569	3318.25244	3091.85112	371.7719558
1178.43711	2823.70919	4813.99352	5128.48679	618.918003
483.412536	975.310022	8529.5277	11164.5118	1355.343737
238.849041	420.584013	9502.29453	12930.8925	1571.591465
987.17192	2285.92525	4896.97057	5249.0795	633.579996
539.732751	1112.34805	6795.76323	8196.44945	992.6540323
551.198149	1140.59527	3803.04449	3721.86375	448.1175379
1175.50087	2815.31768	4491.22827	4666.57591	562.7835271
131.896549	207.104677	7816.53113	9914.7298	1202.511381
84.8447166	122.349109	16229.0083	26778.0442	3273.804341
64.8118079	88.7267975	3037.2375	2741.27664	329.3393792
22.2238368	24.746087	3144.25629	2873.46662	345.3346974
29.7346705	35.0230801	6092.48758	7064.84254	854.6422278
49.700981	64.6420054	4752.15297	5039.09701	608.0515161
44.2782919	56.319283	3777.23178	3687.54987	443.9566032
101.283521	151.132965	3097.27235	2815.22907	338.2871023
41.9193038	52.7583734	3301.75029	3070.95819	369.2420586
98.6788259	146.507758	3027.14804	2728.89948	327.8420082
130.428188	204.357027	8669.02066	11413.557	1385.816427
97.2608699	143.999716	3251.87882	3008.04604	361.6249087
210.232183	361.186549	5034.89895	5451.16328	658.155886
93.5253071	137.426311	3120.14044	2843.53509	341.7124078
57.761087	77.3361244	3497.15874	3320.73813	399.4959948
65.7956267	90.3359209	12536.7076	18849.9982	2297.911706
206.02982	352.590016	4534.59122	4727.95825	570.2406709
280.843343	510.233693	4096.59284	4117.90922	496.1633662
612.715274	1294.05001	3391.41718	3184.93236	383.0446213
926.292886	2118.76201	3482.3396	3301.61543	397.1791718
242.346584	427.941739	4787.42171	5090.02659	614.2424805
350.409604	664.401479	48455.6789	118536.071	14680.66123
100.454302	149.657983	8785.95819	11623.448	1411.502641
427.597094	842.511568	4383.56056	4515.09086	544.3834299
162.748792	266.128954	3466.8305	3281.63373	394.7583871
621.249481	1315.58165	5104.76171	5554.28811	670.699965
75.4117068	106.300639	29438.7373	60188.2674	7409.413666

445.050327	883.697258	7009.6379	8549.24481	1035.714322
586.508239	1228.29432	6645.96021	7951.70539	962.7908102
381.664898	735.695856	19819.2308	35141.4599	4306.087864
297.508312	546.557494	3774.00629	3683.26801	443.4374022
178.310287	296.759661	10495.8695	14803.582	1801.127272
98.6235919	146.409931	3439.39578	3246.36606	390.4859826
259.152441	463.578088	4139.06739	4176.08333	503.2238245
832.20708	1864.60777	7572.70674	9496.49421	1151.40143
1034.7443	2417.94959	3472.24418	3288.60499	395.6029429
891.153108	2023.2267	9061.24139	12121.5186	1472.471389
1026.85669	2395.97707	4591.17615	4808.37509	580.0114122
495.068727	1003.43064	4074.11833	4087.21523	492.4384201
60.1679077	81.1958362	3219.12202	2966.91208	356.6452316
1193.10212	2865.68083	4106.88912	4131.9914	497.8724166
60.3374211	81.4688165	3470.24876	3286.035	395.2915922
165.932139	272.350717	29742.0721	61033.2676	7514.35225
1211.9006	2919.62823	3558.78718	3400.57613	409.1699235
484.281789	977.402625	4230.8013	4302.45697	518.5641488
358.814529	683.457251	5684.72682	6429.64232	777.2498872
495.068727	1003.43064	6906.95989	8379.38199	1014.979923
639.774715	1362.51634	13339.7576	20510.8167	2502.092614
1182.39226	2835.01908	4312.0099	4415.15768	532.2477244
494.319412	1001.61903	4919.37876	5281.77279	637.5554004
1048.31141	2455.81906	5674.65526	6414.15508	775.3636826
922.099735	2107.32468			
893.601411	2029.85974			
673.90592	1449.67401			
215.445821	371.897916			
687.727272	1485.2139			
375.769605	722.1592			
89.6546508	130.668468			
79.4150387	113.066879			
262.938973	471.67015			
116.537534	178.667331			
105.955949	159.487247			
726.172156	1584.78976			
231.089155	404.334168			
156.734896	254.439247			
839.159029	1883.20517			
410.573844	802.651863			
100.557657	149.8417			
687.099289	1483.5961			
271.348832	489.722743			
404.264371	787.958464			

364.374976	696.111541			
287.52175	524.741692			
524.353265	1074.63957			
120.57429	186.075115			
646.131168	1378.68169			
1194.54623	2869.81933			
754.277942	1658.2362			
359.45423	684.911149			
306.216699	565.696986			
1084.03564	2555.98552			
496.04311	1005.78717			
206.898278	354.36382			
802.699116	1786.00592			
1158.68739	2767.34429			
960.429669	2212.24307			
294.9458	540.945966			
472.569486	949.268289			
781.43957	1729.71931			
365.023583	697.59006			
386.350141	746.482868			
1103.74177	2611.51379			
358.051287	681.72323			
137.409707	217.473427			
405.758792	791.434675			
526.624278	1080.19452			
600.517986	1263.37699			
126.031657	196.165956			
196.936331	334.104367			
375.160105	720.762004			
116.728263	179.016234			
1051.71294	2465.32852			
379.984797	731.833901			
676.09063	1455.28244			
133.184169	209.518985			
341.208121	643.640729			
125.365231	194.929115			
593.943065	1246.89243			
520.961843	1066.35272			
375.823769	722.283385			
736.951039	1612.89358			
292.508771	535.617949			
365.578484	698.855376			
190.018046	320.150125			
1190.294	2857.63618			

857.610541	1932.70924			
1023.1441	2385.64617			
1174.20657	2811.61997			
599.019175	1259.61611			
163.250822	267.10861			
488.430518	987.400067			
265.560048	477.284762			
563.1226	1170.09399			
148.466025	238.507768			
238.089397	418.988699			
49.6257298	64.5252597			
75.885255	107.097467			
431.717593	852.206261			
989.683272	2292.86468			
1086.96741	2564.23446			
252.478107	449.370261			
291.068167	532.472412			
514.684788	1051.04239			
545.212099	1125.83318			
786.198325	1742.29317			
662.61825	1420.75322			
242.690358	428.666041			
598.380282	1258.01353			
78.6515659	111.771304			
69.4413566	96.3389549			
20.6595733	22.6825141			
54.1695311	71.6343341			
22.6326159	25.2900672			
132.857339	208.905745			
880.276094	1993.80086			
198.067539	336.395126			
532.829627	1095.39648			
148.266337	238.12511			
108.027619	163.21439			
741.681325	1625.25202			
407.562361	795.633291			
428.828648	845.407281			
865.630625	1954.29099			
103.745667	155.526212			
128.994711	201.680404			
337.31031	634.878695			
287.249598	524.149194			
327.846618	613.68651			
191.086233	322.298359			

422.096147	829.597066			
225.321088	392.323223			
113.167006	172.519892			
103.437737	154.975656			
954.107214	2194.88039			
1116.65148	2647.99509			
1031.65209	2409.33174			
961.410263	2214.93796			
1107.35992	2621.72998			
285.959382	521.341757			
916.116821	2091.02293			
1043.5837	2442.61194			
325.703464	608.903568			
532.395043	1094.33071			
184.245898	308.582313			
972.919274	2246.60672			
683.944892	1475.47416			
1025.28462	2391.60163			
102.060874	152.517805			
43.9379169	55.803174			
28.7420838	33.632852			
129.483536	202.592508			
177.280194	294.715557			
144.729158	231.363461			
474.487436	953.86631			
229.47906	400.975554			
621.946753	1317.34339			
1083.73717	2555.14597			
70.076699	97.391436			
540.512095	1114.26447			
216.955405	375.008747			
334.844065	629.344804			
999.230906	2319.27791			
294.445266	539.850965			
171.726109	283.733857			
327.238509	612.32876			
1051.04521	2463.46133			
367.17472	702.497268			
535.250705	1101.33697			
133.01046	209.193013			
579.210959	1210.08448			
138.811414	220.122613			
431.639873	852.023236			
318.446024	592.752192			

627.818283	1332.19358			
260.719903	466.925106			
368.342244	705.162967			
231.447302	405.081868			
478.66765	963.900213			
192.741226	325.631298			
580.222271	1212.6055			
171.856019	283.989946			
1180.0573	2828.3413			
800.443219	1780.01943			
1158.19563	2765.94317			
262.40998	470.538299			
305.172523	563.396465			
211.483515	363.752771			
302.615772	557.769872			
468.915995	940.519519			
672.912664	1447.12536			
224.992203	391.640153			
669.962013	1439.55839			
47.8422393	61.7684268			
128.263971	200.318153			
691.823349	1495.77309			
165.950946	272.387543			
1043.51066	2442.40799			
419.208608	822.830992			
1234.68262	2985.22409			
1019.41193	2375.26805			
57.5983183	77.0762041			
729.175516	1592.61239			
151.17016	243.699393			
399.373498	776.599055			
547.42637	1131.29013			
881.189524	1996.2693			
21.1493536	23.3254945			
323.005283	602.890591			
692.383035	1497.21683			
347.966036	658.877825			
653.690376	1397.94578			
1093.24653	2581.91609			
413.936654	810.500985			
1221.33495	2946.76378			
343.805768	649.490824			
466.666543	935.13944			
615.576439	1301.26226			

315.660078	586.570849			
559.633682	1161.45051			
693.346201	1499.70189			
668.15699	1434.93257			
406.268206	792.620199			
1002.34459	2327.90237			
409.392958	799.898497			
789.608353	1751.31239			
388.50382	751.449858			
319.994536	596.192487			
228.667717	399.284834			
205.658574	351.832196			
473.343024	951.122307			
421.72039	828.716087			
223.827	389.221663			
1236.97918	2991.84958			
668.132363	1434.86947			
294.735813	540.48654			
1116.55019	2647.70852			
235.258888	413.053064			
111.868419	170.16078			
496.705656	1007.39004			
1015.87845	2365.44921			
1112.73349	2636.91465			
128.515381	200.786665			
1202.74006	2893.3192			
197.646427	335.542055			
477.631881	961.412441			
146.164805	234.104034			
192.899937	325.951212			
578.686313	1208.77696			
218.699672	378.608397			
739.251343	1618.90149			
126.11	196.31144			
1075.63055	2532.36053			
96.4108648	142.499623			
360.741139	687.837516			
225.462177	392.616313			
257.093405	459.187343			
279.180731	506.632157			
121.447934	187.684691			
145.137142	232.141749			
138.09679	218.771345			
441.460455	875.200078			

74.3738965	104.557724			
816.58839	1822.93524			
561.133491	1165.16488			
455.446285	908.378735			
131.547062	206.450166			
199.303317	338.900534			
21.5763004	23.8883393			
31.9529259	38.1621299			
22.057001	24.5246237			
56.6978686	75.6408734			
60.3823883	81.5412555			
45.2465375	57.7916035			
734.966931	1607.71442			
322.033591	600.727514			
149.656269	240.79067			
574.246295	1197.72076			
739.226667	1618.83703			
186.653964	313.399878			

Table 9. Kokatahi volume–area scaling values for recent soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume	V-A Error (Mixed)
551.97945	1142.52432	4357.4818	4478.59871	539.9516408
2973.7027	8519.13296	3777.51428	3687.92495	444.0020842
625.736396	1326.92502	8244.76246	10660.6587	1293.710787
807.825089	1799.62081	15183.8191	24460.1969	2988.189772
1348.24151	3315.61641	17680.4557	30086.5822	3681.871207
1425.56764	3543.71594	7497.95004	9369.22387	1135.852191
1849.89866	4835.69511	11384.4813	16533.6134	2013.399005
2341.77393	6406.46662	5052.25726	5476.73819	661.2666367
1047.61472	2453.8721	7976.40728	10191.5377	1236.348254
2229.15564	6040.63938	12577.6752	18933.8211	2308.213256
1875.93956	4917.01465	11503.3506	16768.8347	2042.275418
1212.27814	2920.71334	3676.4068	3554.33044	427.8051611
2640.97488	7394.62572	4051.24799	4056.04314	488.6556782
897.349566	2040.02119	4643.96958	4883.72608	589.1678221
1266.47342	3077.14914	4715.04721	4985.66141	601.5564542
211.75178	364.30331	8188.92092	10562.5807	1281.716364
203.361077	347.148223	4632.45687	4867.26781	587.1677695
1018.72783	2373.36657	4606.05785	4829.58404	582.5885411
1117.57217	2650.59996	13879.6235	21647.8821	2641.967134
1822.2419	4749.57131	6461.30683	7652.74876	926.3229168
679.521057	1464.09583	5726.60065	6494.13841	785.1053169

697.375472	1510.10504	16004.6734	26275.8897	3211.907621
620.2266	1312.99791	5487.24612	6127.78893	740.4936135
537.653018	1107.23655	3917.73431	3875.33572	466.7313115
1007.70245	2342.75501	4428.54381	4578.21986	552.0507968
1444.22316	3599.11022	13005.9255	19815.9034	2416.641852
504.28541	1025.75672	3588.80975	3439.65078	413.905219
1107.96523	2623.43977	5882.74027	6736.12466	814.584094
2871.08522	8169.59259	4138.15386	4174.82987	503.0716868
2926.80778	8359.10353	12507.4917	18790.2805	2290.572842
2763.31014	7805.07305	7174.68497	8824.16493	1069.279763
2183.03667	5891.84391	3808.10005	3728.59416	448.9337089
1531.76341	3860.86938	4144.50647	4183.54839	504.129898
1107.02876	2620.79467	3324.77797	3100.12325	372.773653
545.31832	1126.09486	3474.49835	3291.50885	395.9547453
2810.53813	7964.47722	5027.53468	5440.3227	656.8373502
1724.71026	4447.88742	14432.4913	22828.9499	2787.322515
2531.26828	7029.65267	3259.98218	3018.24484	362.859656
1162.4365	2778.02995	4331.20352	4441.90682	535.495882
1613.94146	4109.24032	6934.53049	8424.90405	1020.536262
431.338497	851.313576	17043.4345	28621.9644	3501.176966
1750.57399	4527.57573	7894.64502	10049.7233	1219.011957
1069.77805	2515.93135	8690.84141	11452.6461	1390.599811
2066.47621	5518.50324	3611.86293	3469.73473	417.5512529
1753.46137	4536.48616	10595.1303	14994.3046	1824.518612
2237.13742	6066.45203	3596.95612	3450.27371	415.1926407
1016.16772	2366.25279	3035.82474	2739.54265	329.1296011
1611.7448	4102.56889	14360.8875	22675.0525	2768.378382
1294.45963	3158.44279	5569.42629	6252.93602	755.7308709
1245.80588	3017.33632	3335.44814	3113.66196	374.4131406
2623.23315	7335.40056	5834.6566	6661.35483	805.474699
1052.89121	2468.62394	4958.71528	5339.29399	644.5502791
2583.86813	7204.2694	3424.17802	3226.84701	388.1215546
1262.6503	3066.07057	3064.72358	2775.06998	333.4279111
1733.78619	4475.82503	6266.95288	7341.3935	888.354984
1000.30497	2322.25233	6098.14155	7073.76064	855.7292174
1423.4234	3537.35792	51503.2176	128788.324	15962.45066
1026.42072	2394.76355	5638.51605	6358.66458	768.6057469
2191.08881	5917.77952	6915.11788	8392.84493	1016.623159
1168.57547	2795.5415	4851.20422	5182.47431	625.4816181
1341.72975	3296.52085	21477.0301	39198.2525	4807.643718
1439.19252	3584.15896	5048.6577	5471.43217	660.62124
2236.90639	6065.70464	20567.5538	36958.1438	4530.629887
637.441453	1356.59028	3656.17538	3527.75574	424.5838162
776.796534	1717.46544	7100.13999	8699.70966	1054.083733

2411.66022	6635.20818	5192.85816	5685.05336	686.6088925
391.712737	758.860383	20072.0272	35752.4528	4381.595155
1029.2395	2402.61146	3402.68302	3199.32965	384.7884431
2155.50989	5803.32103	4437.27782	4590.50393	553.5428623
664.22718	1424.86978	6923.47993	8406.65054	1018.308242
314.360592	583.691196	8022.91906	10272.4453	1246.239819
2837.56213	8055.92215	22267.6858	41173.698	5052.048157
2395.13565	6581.00543	5827.96204	6650.96238	804.2086261
1577.63992	3999.21572			
2796.06024	7915.55599			
2053.42709	5476.95548			
722.674741	1575.68816			
849.431357	1910.73945			
2081.17146	5565.35282			
1026.54383	2395.1062			
2536.23781	7046.12038			
321.089221	598.626466			
1476.81477	3696.21626			
1902.29781	4999.54729			
1589.44118	4034.93054			
393.721715	763.505797			
2701.07517	7595.8201			
994.573603	2306.38751			
1083.48667	2554.4414			
2454.45156	6775.90173			
763.464946	1682.35956			
1598.15387	4061.33113			
74.6440688	105.011006			
221.58555	384.576156			
188.680996	317.464463			
469.993712	943.098894			
84.9933392	122.604835			
26.3785638	30.3601588			
68.4699669	94.7333865			
55.1699168	73.2153767			
127.404564	198.717955			
129.834588	203.24795			
125.606052	195.375915			
46.289763	59.3847559			
125.608862	195.38113			
61.9486482	84.0708426			
219.56818	380.402813			
100.431982	149.618314			
100.363486	149.496586			

110.997939	168.582355			
728.975022	1592.08999			

Table 10. Styx volume–area scaling values for recent soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)
2683.58877	7537.19184	10985.4177	15750.4259	1917.278478
2192.5902	5922.61745	11342.4325	16450.6172	2003.211013
591.957623	1241.92145	10678.5046	15155.0005	1844.229266
2022.41158	5378.40855	5125.15402	5584.48556	674.3735291
1046.02229	2449.42284	3529.13739	3362.10296	404.5079279
993.640197	2303.80545	3605.21066	3461.04654	416.4982597
2184.85436	5897.697	5243.07586	5759.95263	695.7224775
988.977561	2290.9143	3687.97985	3569.55577	429.6508367
1291.61897	3150.17572	4043.63961	4045.68701	487.3990084
1436.19514	3575.2554	3577.61831	3425.07121	412.1383328
1617.28838	4119.4086	5292.14264	5833.3852	704.6585217
2442.84359	6737.68878	13374.1532	20582.7745	2510.942442
1844.60231	4819.18282	3525.0809	3356.84834	403.8712298
2003.53371	5318.5694	5480.12693	6116.97914	739.1775869
1387.60566	3431.42755	29834.9982	61292.7547	7546.580049
2657.63906	7450.32381	5631.52542	6347.94544	767.3003666
872.674706	1973.27822	4549.55703	4749.19228	572.8205042
806.555982	1796.24843	4641.67574	4880.4457	588.7691772
2089.32541	5591.37581	3800.38451	3718.32386	447.6882729
1910.64573	5025.73243	5535.73192	6201.54398	749.4733433
363.209014	693.454975	6377.6111	7518.24864	909.9198437
2745.01065	7743.44922	8332.10396	10814.5418	1312.531734
557.625405	1156.47989	3512.70648	3340.83242	401.9306423
317.034969	589.62009	17254.4155	29104.8993	3560.749193
1201.37918	2889.41406	11868.0919	17496.0409	2131.571108
1298.4972	3170.19922	5067.79206	5499.65325	664.0539603
740.397342	1621.89595	7460.23111	9305.18178	1128.028517
465.62024	932.638673	5552.1373	6226.55212	752.5182987
1194.23467	2868.92639	4210.1744	4273.95433	515.1039395
374.100669	718.334434	4232.3704	4304.62723	518.8276248
704.068391	1527.41107	24196.2247	46097.6203	5661.694782
1974.67806	5227.31303	7550.62282	9458.84993	1146.802064
1888.27076	4955.59829	4132.814	4167.50499	502.1826423
1798.31598	4675.26858	9935.94321	13739.9811	1670.729123
2407.82309	6622.61548	3030.13235	2732.5589	328.2847147
1448.29868	3611.23023	4887.50681	5235.28814	631.903068
1207.28191	2906.35857	5938.96465	6823.83262	825.2708178

973.986821	2249.54791	3042.40053	2747.61606	330.106334
1521.88805	3831.19266	4300.54767	4399.20379	530.310512
252.290207	448.971311	6328.21594	7439.1671	900.2765069
952.090092	2189.34564	6660.17054	7974.83737	965.613013
1609.72612	4096.43952	3299.87749	3068.58946	368.9552401
1781.05445	4621.78052	3233.57034	2985.0369	358.8393578
1200.2115	2886.064	3124.82908	2849.34792	342.4158491
1168.94226	2796.58833			
1877.41063	4921.61497			
1839.38755	4802.93381			
1383.65827	3419.78522			
578.729608	1208.88485			
1008.55701	2345.12536			
843.079353	1893.7057			
2014.5034	5353.32799			
707.371602	1535.96399			
1684.19769	4323.52876			
2138.57672	5748.97407			
1499.76932	3764.85809			
2514.30722	6973.49517			
2859.7703	8131.19706			
2854.27397	8112.55662			
2371.60753	6503.95469			
1304.86223	3188.74699			
2060.09911	5498.19252			
1633.47992	4168.65721			
1855.71123	4853.82734			
1668.45885	4275.37105			
1121.38676	2661.39687			
2770.86269	7830.52938			
1991.5607	5280.6736			
1421.65215	3532.10727			
2167.71512	5842.54485			
1914.59533	5038.12894			
1152.55647	2749.88444			
1766.04833	4575.36246			
1924.51148	5069.27424			
1126.10947	2674.77397			
210.693925	362.133144			
336.129797	632.228819			
996.966432	2313.00889			
711.852325	1547.57813			
1341.49896	3295.84438			

Table 11. Arahura volume–area scaling values for recent soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)
914.610291	2086.92129	3020.32678	2720.53996	326.830714
2526.52663	7013.94589	5856.78554	6695.73777	809.663556
2185.75625	5900.60148	5638.3394	6358.39364	768.572752
633.503737	1346.59871	3126.81638	2851.81266	342.714124
540.771545	1114.90259	7849.5179	9971.67734	1209.47199
505.703273	1029.19832	8681.0281	11435.0624	1388.44806
2469.67588	6826.07248	9631.65132	13170.8797	1600.99176
211.578676	363.948048	3915.78112	3872.70837	466.412605
2884.62915	8215.59039	3303.45418	3073.11371	369.503061
1795.55291	4666.69999	5032.10932	5447.05613	657.656333
2015.98975	5358.04045	3967.43903	3942.35487	474.861512
2374.89794	6514.72141	3860.02723	3797.90993	457.340029
602.804851	1269.11878	6606.09845	7886.91238	954.886139
564.296607	1173.00482	4175.15069	4225.67304	509.243024
2821.40761	8001.23749	9565.51666	13048.0384	1585.94218
1088.01556	2567.18463	3689.84128	3572.00626	429.947901
1842.50711	4812.65318	34612.8532	75014.3972	9252.47031
2446.31351	6749.10795	7108.93665	8714.3716	1055.87387
2567.41444	7149.57335	6348.34502	7471.36697	904.202913
1354.78386	3334.81963	12242.4525	18250.8385	2224.28866
1945.09276	5134.01603	9268.93127	12500.9235	1518.92852
2292.95366	6247.45288	7957.27587	10158.3077	1232.28582
1081.18408	2547.96638	3094.11469	2811.32642	337.814864
1654.09911	4231.50955	3369.66731	3157.1856	379.684064
1661.69081	4254.68913	3140.70251	2869.05061	344.800258
1274.61513	3100.76355	3563.46821	3406.66076	409.907267
1807.30371	4703.15803	4400.04255	4538.19456	547.189405
628.461087	1333.82098	9699.3176	13296.8805	1616.42966
1507.35787	3787.59517	3956.86995	3928.07865	473.12955
1054.00808	2471.74829	3146.19615	2875.87788	345.626518
2562.0094	7131.62041	4168.14995	4216.03972	508.073688
1278.54681	3112.17756	4274.50266	4363.00955	525.915805
972.019648	2244.12865	15488.6145	25130.3683	3070.74744
1390.16175	3438.96983	3051.32634	2758.58477	331.433379
1256.61568	3048.59669	3436.91999	3243.18836	390.101047
991.508398	2297.91005	4170.71791	4219.57266	508.50253
1422.97849	3536.03893	7467.54778	9317.5955	1129.545
1777.34032	4610.28465	3994.98902	3979.63243	479.384181
1288.42276	3140.87809	13342.2608	20516.0514	2502.73641
643.686621	1372.46121	4134.60304	4169.95871	502.480457
372.571422	714.832686	8490.18352	11094.532	1346.78216

741.016736	1623.51478	7846.38889	9966.2718	1208.81126
1031.24595	2408.20022	3080.71837	2794.78547	335.81339
731.96963	1599.8956	8835.12314	11711.9957	1422.34017
1018.13009	2371.7053	11621.6344	17003.7677	2071.11992
1542.84026	3894.20071	6287.35466	7373.91596	892.320301
700.301842	1517.6679	5136.96311	5601.99254	676.503352
1872.43172	4906.04771	4673.02781	4925.33229	594.224156
1658.39172	4244.61357	4059.48654	4067.26494	490.017421
2020.72049	5373.04369	3980.60332	3960.15573	477.021146
251.054344	446.348763	7415.08978	9228.69051	1118.68458
288.559398	527.001733	3124.61309	2849.08008	342.383436
319.560784	595.228506	3310.61872	3082.18162	370.601072
739.424865	1619.35484	7311.55659	9053.88866	1097.33374
304.642011	562.228226	5866.65247	6711.08364	811.533192
570.875577	1189.33825	12936.14	19671.4403	2398.88103
1546.98178	3906.67483	18982.8828	33140.1374	4058.84549
350.086175	663.669944	5271.56637	5802.56109	700.907411
889.883646	2019.7888	3095.34242	2812.84364	337.998454
669.704546	1438.89842	14354.5643	22661.4753	2766.70714
1028.77376	2401.31446	4756.84702	5045.8676	608.874515
1004.39983	2333.59795	3299.64637	3068.29718	368.91985
1337.44059	3283.95275	4793.10054	5098.23972	615.24091
1280.40917	3117.58648	18791.2056	32685.8719	4002.74411
2223.41101	6022.07259	3941.45158	3907.27687	470.606007
1806.31443	4700.08693	3085.5806	2800.78606	336.539458
1139.58121	2712.99218	4574.00671	4783.93646	577.041952
1068.7736	2513.11338	3903.21119	3855.81108	464.362951
2295.84043	6256.83742	3230.08688	2980.66437	358.310026
2484.99462	6876.61465	3958.93837	3930.8715	473.468369
2047.29921	5457.46219	3348.53597	3130.28958	376.42676
632.335868	1343.63766	5670.69479	6408.06769	774.622303
2171.6659	5855.25057	4718.17698	4990.16273	602.103566
2348.69824	6429.07214	3487.91708	3308.80921	398.050726
364.44593	696.273259	5480.89486	6118.14492	739.319512
1101.54345	2605.30978	9821.97367	13526.084	1644.51536
2300.02147	6270.4335	3379.68704	3169.95999	381.23121
1847.1669	4827.17723	4385.15097	4517.31886	544.654018
346.41367	655.372604	3090.69093	2807.09651	337.303032
592.070674	1242.20441			
1514.71097	3809.64791			
816.146015	1821.75716			
1081.55534	2549.01022			
2014.02803	5351.82097			
548.320932	1133.49594			

849.387924	1910.6229			
827.984647	1853.3268			
223.401744	388.339607			
876.216164	1982.83537			
319.830099	595.827008			
1092.55448	2579.96635			
2003.19918	5317.50999			
846.328819	1902.4165			
1194.00705	2868.27404			
320.157011	596.55364			
1420.71451	3529.32826			
1127.7985	2679.5608			
556.417401	1153.49167			
607.505627	1280.93454			
1574.3932	3989.39902			
398.326046	774.16975			
2942.90448	8413.9783			
231.654676	405.514903			
353.835039	672.157147			
1985.42315	5261.26466			
706.685142	1534.18592			
1902.87375	5001.35316			
2007.50124	5331.1367			
1755.87606	4543.94008			
371.73798	712.925396			
770.591584	1701.11144			
647.31576	1381.69767			
1431.68129	3561.85402			
2431.77085	6701.27039			
2027.77818	5395.43934			
727.736806	1588.86431			
701.443214	1520.61929			
289.105612	528.192041			
1079.52138	2543.29244			
1790.12777	4649.88346			
816.062301	1821.53423			
988.648575	2290.00517			
702.409559	1523.11882			
1399.11788	3465.41787			
1533.7373	3866.80563			
1155.24967	2757.55203			
1155.61452	2758.59102			
1439.34332	3584.60699			
523.171057	1071.74969			

428.027419	843.523194			
502.987162	1022.6071			
476.367053	958.375914			
526.497357	1079.88394			
358.277768	682.237702			
261.478092	468.545469			
1091.83706	2577.94541			
2350.45971	6434.82479			
1236.74158	2991.164			
2681.47017	7530.09361			
1061.40529	2492.45743			
152.927983	247.083853			
246.932248	437.619585			
416.277473	815.971958			
428.180125	843.882231			
2373.31165	6509.53047			
2229.72634	6042.4844			
649.21857	1386.54448			
577.20307	1205.08167			
254.979594	454.686851			
1997.84746	5300.56637			
1316.19704	3221.81996			
1755.37142	4542.38214			
800.235214	1779.46761			
1745.92189	4513.22537			
445.905789	885.724082			
476.653641	959.063802			
1597.66474	4059.84827			
2057.28139	5489.22211			
2269.44503	6171.11424			
798.94336	1776.04105			
1900.4818	4993.85391			
1405.96665	3485.6648			
403.997099	787.337017			
187.220864	314.535767			
702.533105	1523.43843			
669.637543	1438.72668			
2321.01282	6338.7661			
144.234773	230.420918			
1797.11061	4671.53029			
796.530618	1769.64427			
237.436772	417.618916			
651.889608	1393.35273			
2103.01601	5635.11285			

1413.26708	3507.26791			
610.073978	1287.39775			
2268.66338	6168.57863			
952.690335	2190.9924			
2077.21952	5552.74743			
1575.24231	3991.966			
1761.97574	4562.77794			
901.921675	2052.42752			
2627.9025	7350.98025			
2329.37195	6366.01064			
1592.65017	4044.65097			
2516.43466	6980.53505			
1245.7691	3017.23005			
229.85119	401.751403			
199.761729	339.830678			
447.317042	889.069368			
607.967778	1282.09714			
222.938234	387.378577			
893.225152	2028.84014			
1024.17935	2388.5262			
1475.31097	3691.72655			
1430.22946	3557.54534			
2772.88575	7837.35049			
1123.53105	2667.46925			
2104.86645	5641.02864			
2593.33523	7235.77082			
1156.55607	2761.27261			
671.132479	1442.55929			
1780.20644	4619.15536			
755.821346	1662.28495			
504.002555	1025.07037			
370.668581	710.479338			
766.390026	1690.05207			
306.020099	565.263721			
172.945142	286.138374			
361.896388	690.466211			
273.064826	493.419689			
1758.51419	4552.08596			
1364.76506	3364.15109			
1029.80603	2404.18926			
1191.03252	2859.75152			
1398.53717	3463.702			
1151.09951	2745.73788			
1328.63431	3258.17301			

235.70296	413.983384			
147.246118	236.171637			
334.671575	628.958055			
236.71164	416.097801			
830.731262	1860.66361			
272.409197	492.006668			
204.307772	349.077047			
830.866824	1861.02584			
647.345939	1381.77452			
1546.0282	3903.80209			
392.738587	761.23191			
764.409948	1684.84415			
557.812477	1156.94276			
2260.06292	6140.69058			
1889.54306	4959.58201			
874.203311	1977.40247			
1788.42386	4644.60381			
215.301192	371.600096			
732.420528	1601.07143			
766.11002	1689.31545			
607.289225	1280.39021			
895.888691	2036.0597			
334.931048	629.539847			
137.39396	217.443693			
198.978842	338.242404			
108.617179	164.277601			
818.642816	1828.40798			
2232.08637	6050.11516			
2014.74831	5354.10443			
797.598286	1772.47447			
283.632597	516.284991			
268.071332	482.674233			
79.8166411	113.749345			
607.059941	1279.81351			
1391.55699	3443.08793			
355.977803	677.016048			
835.431681	1873.2303			
535.291819	1101.4379			
551.768727	1142.00399			
467.311797	936.682203			
355.040401	674.889712			
278.037804	504.158756			
280.286706	509.027452			
207.797781	356.202549			

1591.55191	4041.32378			
1308.8277	3200.31124			
2550.75529	7094.26319			
2597.65004	7250.13558			
1561.31328	3949.89051			
908.836841	2071.21474			
453.08781	902.769751			
2013.17771	5349.12545			
845.255757	1899.53924			
1083.8501	2555.46364			
1688.17651	4335.71694			
2847.92887	8091.04626			
847.148615	1904.61514			
2776.16874	7848.42174			
1903.19533	5002.36153			
867.639706	1959.70341			
179.321579	298.768673			
210.268836	361.261675			
375.577666	721.719159			
728.957102	1592.0433			
544.45649	1123.97201			
1772.81047	4596.27026			
969.589545	2237.437			
961.569771	2215.37637			
611.555092	1291.12734			

Table 12. Taipo volume–area scaling values for recent soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)
1328.93882	3259.06389	5447.98306	6068.2348	733.243491
2076.8675	5551.62482	3348.0948	3129.72872	376.358838
455.216585	907.832209	7053.30692	8621.76027	1044.567
1191.47994	2861.03318	5870.55608	6717.15744	812.273192
2006.2435	5327.15224	3145.90978	2875.52188	345.583435
348.778597	660.713781	4333.88272	4445.64408	535.94971
949.530542	2182.32579	3851.93359	3787.0838	456.027003
1752.65976	4534.01214	3029.12849	2731.32781	328.13578
117.402603	180.250695	3259.1325	3017.17501	362.730132
1748.63703	4521.59989	3319.58225	3093.53639	371.976028
136.317835	215.413426	7918.63486	10091.2785	1224.09171
744.175412	1631.77425	5368.05972	5947.48479	718.545142
2710.90375	7628.8055	4842.03165	5169.15229	623.861923
1029.24913	2402.63826	3905.92994	3859.46414	464.806064

1030.75475	2406.83185	5629.93954	6345.51439	767.004316
261.887344	469.420481	23248.1915	43658.7237	5359.64895
427.298551	841.809857	7308.34351	9048.47799	1096.67292
759.057457	1670.77928	5261.9867	5788.22512	699.162863
215.15893	371.307189	6223.19196	7271.76287	879.865719
268.132938	482.806569	3393.59278	3187.71135	383.381212
1132.52415	2692.96096	4510.88329	4694.37222	566.160317
334.504254	628.582932	11059.3493	15894.7603	1934.9896
1376.32881	3398.1849	8002.44428	10236.8084	1241.88286
827.244428	1851.35032	11407.3648	16578.8273	2018.94932
1989.09104	5272.86234	4422.4754	4569.68999	551.014748
2448.4104	6756.01012	4294.8317	4391.25364	529.34518
1495.48479	3752.03041	5323.87662	5881.0086	710.454319
858.079642	1933.97051			
590.534644	1238.3607			
1413.58153	3508.19891			
436.568411	863.642142			
344.096079	650.145159			
825.939354	1847.86644			
618.197901	1307.87597			
661.330647	1417.46019			
680.011085	1465.35551			
347.522381	657.875749			
1274.60074	3100.72178			
654.577037	1400.2082			
1369.82543	3379.03769			
1272.14436	3093.5942			
2281.34111	6209.72492			
718.434249	1564.6642			
543.691891	1122.08919			
866.922078	1957.76986			
1071.47893	2520.70429			
600.580057	1263.53278			
1605.53504	4083.71881			
459.451584	917.917073			
967.64767	2232.09209			
1583.66003	4017.42831			
823.761653	1842.05544			
216.400613	373.864989			
168.082136	276.565906			
862.530373	1945.94373			
509.907919	1039.41526			
1251.04037	3032.46719			
2128.59135	5716.96491			

978.086974	2260.85			
2248.5517	6103.39609			
155.700935	252.438069			
287.577029	524.862052			
419.078788	822.527009			
2133.82774	5733.74712			
2242.84402	6084.91778			
1049.41701	2458.90928			
2645.44193	7409.54966			
1992.78528	5284.5475			
1837.19891	4796.11672			
924.667153	2114.32644			
2454.86724	6777.27078			
532.069203	1093.53174			
1406.85994	3488.30703			
569.153367	1185.05904			
1394.09066	3450.56812			
1571.71344	3981.29951			
1016.36431	2366.79893			
419.114596	822.610854			
601.475735	1265.78117			
1283.13144	3125.49565			
875.305894	1980.37815			
1620.7918	4130.05664			
1414.98365	3512.35067			
1695.45273	4358.02027			
266.772555	479.885704			
548.456559	1133.83043			
2904.23101	8282.23586			
1042.93817	2440.80952			
325.137943	607.642487			
217.709512	376.564318			
1241.21045	3004.06283			
712.298759	1548.73607			
2288.77183	6233.86229			
1274.3422	3099.97148			
670.276547	1440.36471			
610.65409	1288.85832			
528.809057	1085.5429			
1436.66419	3576.64846			
588.573421	1233.45581			
2766.58658	7816.11483			

V-A Metrics for mapped 1960s landslides

V-A values associated with historical landslides, showing soil area and volume, mixed soil-bedrock area and volume, bedrock area and volume and the associated errors for both mixed and bedrock landslide values.

Table 13. Hokitika volume–area scaling values for historical soil, mixed, and bedrock landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)	Bedrock Area (m ²)	Bedrock Volume (m ³)	V-A Error (Bedrock)
2489.36885	6891.05789	22697.9061	42259.3139	5186.407419	141072.04	241932.377	111274.064
2177.4332	5873.80623	5296.78974	5840.35273	705.5064498	118159.062	185781.671	85318.0028
2251.14192	6111.78478	22936.4452	42864.4512	5261.314783	115876.003	180458.486	82859.5903
2400.07243	6597.19118	31046.7722	64702.9584	7970.236411			
2583.37152	7202.61755	5376.30275	5959.9088	720.0573494			
2778.1268	7855.02613	3443.52497	3251.66773	391.1282158			
1600.18977	4067.5042	3865.55556	3805.30937	458.2374729			
2641.92405	7397.79639	12661.9573	19106.5778	2329.445719			
2241.87661	6081.78673	6037.4137	6978.12962	844.0737402			
1970.70351	5214.76355	5802.56175	6611.57071	799.409835			
1291.42832	3149.621	6292.314	7381.82737	893.284925			
958.502933	2206.94954	6498.52547	7712.76185	933.6426265			
1321.0553	3236.01235	6699.00885	8038.15005	973.3377798			
893.655053	2030.00511	8686.65833	11445.1499	1389.682481			
1450.64153	3618.20049	9863.78216	13604.4466	1654.118541			
1442.85864	3595.05383	3448.31409	3257.81958	391.8734491			
2160.6999	5819.9949	3070.25972	2781.88976	334.2530567			
2379.86505	6530.97999	4654.0784	4898.18951	590.9255006			
2251.73898	6113.71867	7068.47107	8646.97929	1047.645876			
2080.9174	5564.5423	4327.5769	4436.84931	534.8817349			
1024.92947	2390.61334	5466.42009	6096.18088	736.6455766			
1211.28028	2917.84545	5984.60235	6895.246	833.9729622			
2143.03434	5763.27276	4427.37069	4576.57057	551.8504711			
1506.33573	3784.53131	3321.19924	3095.58593	372.2242121			
2259.50359	6138.87761	5648.71019	6374.30439	770.5104004			
2407.03132	6620.01753	7714.77539	9739.60691	1181.108485			
1467.51264	3668.45817	3735.60004	3632.38493	437.2678802			
972.599965	2245.72711	17051.0993	28639.4716	3503.336399			
1899.2716	4990.06039	19349.426	34013.4242	4166.714891			
2417.64848	6654.86818	4332.65638	4443.93333	535.7419676			
2270.11193	6173.27775	7850.02795	9972.55857	1209.5797			
836.486897	1876.05333	5147.94118	5618.28056	678.4849287			
1048.69301	2456.88559	6429.10301	7600.92211	920.0020738			
1569.33825	3974.12276	3393.1377	3187.13	383.3107992			
1507.65043	3788.47219	11965.2751	17691.1723	2155.537334			
528.047776	1083.67878	8463.43271	11047.018	1340.969388			

2092.451	5601.3562	6483.45533	7688.4471	930.6769354		
2055.43831	5483.35579	8044.14338	10309.4214	1250.760638		
925.317364	2116.10027	4759.71032	5049.99873	609.3766802		
989.472702	2292.2827	3113.61148	2835.44593	340.7335141		
951.392192	2187.43121	6079.69524	7044.67594	852.1842476		
759.36808	1671.59499	7112.38932	8720.12817	1056.576717		
1367.44444	3372.03195	14232.5433	22399.8947	2734.510401		
1262.70401	3066.22615	8727.42969	11518.2687	1398.630434		
1070.6799	2518.46189	4262.77482	4346.73749	523.940143		
2618.51062	7319.6489	4441.36102	4596.24979	554.2407853		
2382.84467	6540.73616	6330.00477	7442.02715	900.6252512		
1518.73616	3821.72863	3590.51487	3441.87356	414.174601		
1885.32765	4946.38502	3266.31488	3026.22146	363.8253909		
1155.05413	2756.9952	46676.8688	112657.495	13946.16515		
983.396211	2275.49853	18812.9621	32737.3503	4009.101296		
2100.62741	5627.47809	21647.0963	39620.9862	4859.935644		
2246.10022	6095.45845	10574.6726	14954.9437	1819.690949		
2534.13639	7039.15608	3667.10547	3542.10625	426.3233401		
1451.81867	3621.70345	5790.16653	6592.37029	797.0708753		
2231.55294	6048.39025	82352.4212	243838.133	30401.69292		
794.281555	1763.68481	5070.73916	5504.00332	664.5831007		
395.686049	768.052405	5339.69807	5904.79021	713.3486965		
1085.22718	2559.33757	4355.16039	4475.35416	539.5576203		
2513.7702	6971.7183	7212.54203	8887.5473	1077.019452		
2432.30542	6703.02787	5056.63496	5483.19307	662.0517811		
942.663824	2163.51113	3767.74584	3674.96099	442.4301389		
2094.8085	5608.88588	3281.86665	3045.83396	366.1999723		
2985.10211	8558.1075	3193.56186	2934.91963	352.7725906		
2548.68367	7087.39005	5608.06818	6312.01215	762.9245294		
1425.63356	3543.91143	3186.01653	2925.49307	351.6315798		
1242.33782	3007.31826	3938.66342	3903.51832	470.150055		
497.517018	1009.35351	3680.66686	3559.93292	428.4843083		
610.985812	1289.69363	3770.91463	3679.16505	442.9398978		
1687.48462	4333.5971	3781.2083	3692.83054	444.5969222		
2506.93663	6949.11416	8660.40812	11398.1384	1383.929679		
1237.43448	2993.16339	8516.29673	11140.9654	1352.46294		
2782.34126	7869.24423	4525.7838	4715.47375	568.7239066		
514.968924	1051.73465	6189.9272	7218.95101	873.4274167		
468.984077	940.682432	38906.6433	87945.592	10862.78901		
1294.3041	3157.99006	20619.2485	37084.5329	4546.25532		
1841.7791	4810.38467	3677.35133	3555.57241	427.9557156		
1386.94304	3429.47279	3342.58556	3122.72689	375.5109013		
410.711986	802.974056	16291.9345	26919.35	3291.223881		
326.958954	611.704748	15176.0485	24443.1741	2986.093007		

703.400818	1525.68348	8678.54914	11430.6217	1387.904642			
1417.09531	3518.60487	8617.68264	11321.7311	1374.580129			
2000.4033	5308.6571	5036.70327	5453.82021	658.4790511			
435.765385	861.747292	3030.64472	2733.18731	328.3607386			
590.170442	1237.44962	15733.9915	25673.3581	3137.652			
727.419382	1588.03756	7192.39199	8853.79604	1072.897992			
2408.7189	6625.55499	6791.6875	8189.76469	991.8382724			
1671.00584	4283.15843	10742.9381	15279.4999	1859.501356			
850.943428	1914.7979	7476.43567	9332.6809	1131.38788			
748.006723	1641.80164	4874.39259	5216.19293	629.581288			
2834.19061	8044.50424	5209.15837	5709.33652	689.5635117			
1770.51133	4589.15982	10484.2555	14781.3087	1798.395718			
1204.35945	2897.96728	3185.34709	2924.6571	351.530394			
727.419382	1588.03756	4788.16505	5091.10145	614.3731455			
2508.22438	6953.37288	5249.73604	5769.90569	696.9336172			
2981.73322	8546.58629	4108.48907	4134.18079	498.1381309			
2404.22694	6610.81715	3957.35886	3928.73875	473.2096313			
2419.44357	6660.76344	3153.62204	2885.11329	346.7442464			
1803.17021	4690.32819	34972.5809	76076.655	9384.660427			
999.225121	2319.26189	23315.0539	43829.5787	5380.803628			
2169.63729	5848.72599	11918.5164	17597.2153	2143.997158			
512.293082	1045.21825	9091.13111	12175.9297	1479.133143			
1309.89791	3203.43338	13092.9687	19996.4829	2438.844533			
736.738319	1612.33819	7438.19818	9267.82645	1123.465225			
365.199029	697.990082	10483.0746	14779.0446	1798.118057			
448.890473	892.801488	13701.944	21271.8634	2595.704459			
833.110285	1867.02228	24576.5028	47085.7044	5784.106254			
602.324787	1267.9131	9308.62228	12573.7817	1527.851178			
1071.5041	2520.77491	8043.84875	10308.9079	1250.697851			
490.736195	992.963295	4419.68905	4565.77485	550.539215			
276.435376	500.694258	3197.91972	2940.36765	353.4320436			
243.466019	430.301024	4608.28431	4832.75924	582.9743724			
230.785497	403.7004	3804.70448	3724.07333	448.3854838			
235.857706	414.307654	6428.63861	7600.17541	919.9110088			
332.229672	623.487074	7092.82194	8687.51719	1052.595124			
745.614684	1635.53998	10110.3462	14069.0105	1711.059459			
369.003185	706.672758	12094.6964	17951.9206	2187.566251			
419.725273	824.040981	7273.21741	8989.3832	1089.455698			
238.39381	419.627873	19751.9289	34979.2666	4286.045773			
1118.42203	2653.0048	4117.09436	4145.96161	499.567923			
517.365291	1057.576	6810.07904	8219.94074	995.5207852			
224.445236	390.50457	3664.88892	3539.19481	425.9704202			
918.069781	2096.34194	4154.21571	4196.88296	505.7484164			
555.406856	1150.99285	4873.01989	5214.19524	629.3383937			

673.335709	1448.21079	7727.9886	9762.30028	1183.881809			
680.944023	1467.75422	12601.0085	18981.6068	2314.086134			
1214.79399	2927.94602	40509.5088	92909.3129	11481.53241			
488.200091	986.844362	38558.7651	86877.8766	10729.73741			
356.322664	677.798578	9162.20369	12305.5685	1495.006303			
1519.12652	3822.90053	16341.9423	27031.7867	3305.085153			
474.251517	953.300532	6000.97552	6920.91446	837.1009957			
755.759101	1662.12163	30022.7376	61817.8868	7611.804313			
2512.01138	6965.89932	3714.88961	3605.02433	433.9507131			
745.614684	1635.53998	10573.1473	14952.0102	1819.331163			
1351.74363	3325.89366	30362.0785	62770.0678	7730.083436			
871.15185	1969.17086	15002.4388	24063.6717	2939.351617			
2191.19417	5918.119	13805.8157	21491.4727	2622.722773			
1197.04126	2876.97179	12841.3732	19475.7125	2374.819347			
821.697815	1836.55101	9430.10438	12797.4723	1555.248446			
1410.07403	3497.81652	37595.3972	83939.2184	10363.62151			
2130.32767	5722.5288	15395.3598	24924.8151	3045.423492			
1029.65837	2403.778	24396.8231	46618.1479	5726.178804			
1952.80036	5158.29569	14180.8767	22289.3779	2720.908397			
2475.23786	6844.41652	5750.93487	6531.6973	789.6801683			
573.159587	1195.01723	13395.0346	20626.4925	2516.319284			
2662.90959	7467.95397	3071.92794	2783.94565	334.5018083			
1557.16808	3937.38303	12073.3912	17908.9272	2182.284896			
1146.31917	2732.14001	4982.95288	5374.81822	648.8705127			
968.791868	2235.24118	12948.5335	19697.0757	2402.03265			
1280.55665	3118.01489	3161.22817	2894.581	347.8901161			
1643.5805	4199.42723	5893.81524	6753.37744	816.68616			
1636.62109	4178.22244	3500.56905	3325.14296	400.0296792			
1435.44175	3573.01807	3009.41778	2707.185	325.2151468			
2992.92486	8584.87011	5795.58499	6600.76176	798.0931017			
2722.33166	7667.18734	5822.37506	6642.29257	803.1524284			
2755.13083	7777.51933	6568.03198	7825.16874	947.3539818			
1721.95677	4439.41719	4911.5127	5270.29018	636.159128			
2623.93413	7337.73909	19922.8815	35391.6399	4337.00401			
1738.35636	4489.90367	7197.59861	8862.51386	1073.962537			
1795.75492	4667.32636	10090.9261	14032.2707	1706.555719			
1984.3045	5257.72835	11975.161	17711.054	2157.979357			
2244.1539	6089.15763	7853.95531	9979.34456	1210.409164			
1569.22039	3973.76671	4415.89642	4560.44721	549.8921253			
2009.2552	5336.69394	9510.47459	12946.0338	1573.446255			
968.907505	2235.55948	17595.864	29890.981	3657.734695			
2964.76767	8488.60424	5196.5807	5690.59658	687.2833468			
2022.65023	5379.1657	4173.37799	4223.23317	508.9468595			
1058.20774	2483.50217	3383.81999	3175.23316	381.8698745			

1567.21905	3967.72129	3201.71332	2945.11245	354.0063826		
968.907505	2235.55948	3677.7489	3556.09521	428.0190899		
1468.9888	3672.86084	26190.5613	51340.3348	6311.465129		
2991.55774	8580.19202	39802.761	90711.7842	11207.56399		
1826.18972	4761.8496	58627.2269	153602.003	19068.71958		
616.171593	1302.76331	18221.3938	31345.3214	3837.228361		
1719.02944	4430.41509	18953.6957	33070.8584	4050.289149		
2147.67055	5778.15042	41181.2116	95010.7096	11743.57333		
2201.25069	5950.53676	9692.23076	13283.6692	1614.810928		
1426.03842	3545.11214	10639.9155	15080.5675	1835.099232		
2078.63228	5557.25313	9993.76682	13848.8429	1684.071804		
1152.11014	2748.61405	7753.7846	9806.64457	1189.301215		
2368.67399	6494.35815	12578.3617	18935.2266	2308.385988		
2553.97842	7104.9589	6461.48717	7653.03925	926.3583466		
2352.56057	6441.6869	19212.1552	33685.6736	4126.227808		
402.901373	784.790122	13741.4294	21355.2745	2605.966152		
721.372439	1572.30125	17618.3217	29942.8771	3664.138348		
279.12538	506.512327	11329.1408	16424.405	1999.993485		
490.961605	993.507445	9304.54153	12566.2857	1526.933159		
924.60282	2114.15095	10166.0731	14174.5785	1724.001014		
918.372343	2097.16618	12061.4427	17884.8273	2179.324486		
378.813015	729.142341	4953.80683	5332.10743	643.6763234		
675.383731	1453.46736	9132.2352	12250.8606	1488.307623		
327.723102	613.410691	17661.3983	30042.4864	3676.429813		
2087.20987	5584.62226	4135.35179	4170.98574	502.6051112		
1064.16551	2500.19207	3375.94289	3165.18488	380.6528763		
599.371909	1260.50104	3480.97222	3299.85242	396.9655791		
618.063341	1307.53636	5341.49186	5907.4881	713.6770533		
368.748551	706.091033	5851.63434	6687.72988	808.6879437		
801.963631	1784.0538	6271.75168	7349.03983	889.2872545		
495.102949	1003.51339	6766.88997	8149.12454	986.8789563		
183.084945	306.264045	5371.50024	5952.66955	719.1762102		
1490.46617	3737.01392	3570.99737	3416.45355	411.0939913		
1108.82431	2625.86667	3195.8926	2937.83309	353.1252484		
649.822621	1388.08368	4006.11889	3994.71843	481.214572		
941.211336	2159.53472	3826.06861	3752.5415	451.8378032		
1523.98877	3837.50247	5581.55891	6271.46866	757.9875184		
1080.45904	2545.92809	4936.37871	5306.61132	640.575822		
912.846063	2082.11971	3961.10632	3933.7993	473.8235617		
1343.48248	3301.65893	4501.25718	4680.75343	564.5058404		
2760.35564	7795.11838	5866.63853	6711.06196	811.5305505		
695.69939	1505.77616	5041.40805	5460.74976	659.3219041		
534.86566	1100.39186	3916.09375	3873.12888	466.4636139		
471.280232	946.179503	5011.39967	5416.59112	653.9509598		

1084.6926	2557.83359	8372.33836	10885.625	1321.226448			
1731.76784	4469.60967	7412.07016	9223.57978	1118.06029			
2072.13689	5536.54233	5386.50443	5975.29441	721.930069			
1170.71994	2801.66284	3586.00156	3435.99089	413.4616744			
590.97045	1239.45105	6121.70977	7110.96728	860.2642965			
1850.10993	4836.354	11958.3399	17677.2285	2153.824663			
1131.03887	2688.74809	15961.0258	26178.4813	3199.902042			
1153.50984	2752.59829	21201.4485	38515.7965	4723.23508			
928.800129	2125.60561	6491.27668	7701.06383	932.2157994			
2100.58669	5627.34793	5150.75871	5622.4629	678.9937542			
1830.51125	4775.29603	5207.97847	5707.57785	689.3495242			
2760.77107	7796.51797	18299.4301	31528.0308	3859.783544			
1755.4903	4542.74914	4308.20187	4409.85571	531.6039232			
1620.45259	4129.02547	4177.10859	4228.36822	509.5701809			
840.234674	1886.08533	4665.72899	4914.87291	592.9530124			
660.184387	1414.52968	3628.90033	3492.01269	420.2513955			
1380.38554	3410.13757	7007.53167	8545.75137	1035.287858			
2460.68726	6796.44383	3980.46867	3959.97355	476.9990417			
1260.35201	3059.41372	3533.55976	3367.83401	405.2023637			
2370.66212	6500.86169	3919.50652	3877.72005	467.0205389			
2225.3919	6028.47384	4015.28394	4007.15254	482.7232479			
1122.60126	2664.83591	19796.4549	35086.5493	4299.302514			
1730.12664	4464.55677	8273.69516	10711.5693	1299.937244			
2925.3668	8354.19397	3126.31933	2851.19614	342.6395148			
2972.20498	8514.0144	43836.1304	103436.44	12794.78941			
2222.62697	6019.53926	8515.77701	11140.0408	1352.349814			
2472.89595	6836.69167	8871.23812	11777.1531	1430.315354			
2562.27774	7132.51153	3801.31425	3719.56105	447.8383			
2913.84608	8314.95846	11244.3512	16257.4543	1979.501472			
2413.3081	6640.61738	3337.68036	3116.49626	374.7563718			
2449.06081	6758.15127	3880.70194	3825.60168	460.6987026			
2031.94583	5408.6713	19956.9075	35473.87	4347.166083			
566.084616	1177.44024	4268.08038	4354.09682	524.8336641			
715.054252	1555.88624	3252.16745	3008.40916	361.6688697			
715.054252	1555.88624	10089.0794	14028.7784	1706.127621			
1334.76794	3276.12528	70310.0611	196664.298	24470.64303			
2455.0196	6777.7726	3472.97681	3289.5487	395.7172733			
2127.2864	5712.78391	8123.60353	10448.1649	1267.725091			
1078.54016	2540.53484	4263.52883	4347.78317	524.0671021			
2949.59879	8436.81677	3631.08721	3494.87498	420.5983209			
1966.39919	5201.17834	4318.04966	4423.57036	533.2692627			
2446.00783	6748.10185	4105.41843	4129.97916	497.628203			
2026.0607	5389.98801	12487.9959	18750.4586	2285.679147			
2784.91253	7877.92084	4800.559	5109.032	616.5528918			

1060.91906	2491.09533	3220.71023	2968.90299	356.8862389			
601.305595	1265.35402	8289.64504	10739.6624	1303.373195			
1347.06546	3312.16637	3118.74537	2841.80614	341.5031802			
682.34092	1471.34701	4831.84717	5154.37126	622.0648791			
1183.82039	2839.10465	4042.09781	4043.58924	487.1444565			
366.420597	700.776318	3997.33735	3982.81422	479.7702242			
1834.44262	4787.53379	9619.0344	13147.421	1598.117682			
1968.3765	5207.41836	4026.99957	4023.06188	484.6536394			
678.698625	1461.98207	3669.64043	3545.4367	426.7270548			
2545.62979	7077.25997	4204.97289	4266.7747	514.2323645			
2722.02533	7666.15811	6436.70187	7613.14279	921.4924922			
868.142768	1961.05903	4382.94622	4514.2303	544.2789163			
2154.79261	5801.01724	7409.2361	9218.78378	1117.47445			
2346.13212	6420.69311	4244.45916	4321.35721	520.8587303			
657.641001	1408.03081	4434.01322	4585.91137	552.9850308			
1263.56711	3068.72669	8112.02719	10427.9212	1265.249735			
1211.85222	2919.48919	3947.38027	3915.27213	471.5759308			
2946.02482	8424.6225	3730.11362	3625.13148	436.3884617			
792.961626	1760.18884	3403.10515	3199.86945	384.8538253			
236.164658	414.95099	5470.21071	6101.93075	737.3455672			
790.375882	1753.34347	9878.09296	13631.2972	1657.409133			
1850.98215	4839.07421	6196.70844	7229.7088	874.7388686			
1093.14262	2581.62334	8309.72908	10775.0649	1307.703227			
1515.67905	3812.55281	3622.00498	3482.99185	419.1580311			
605.181202	1275.08971	7305.19617	9043.17883	1096.025717			
2562.47896	7133.1798	3355.40793	3139.02955	377.4852097			
1134.03324	2697.24244	11380.1207	16525.0011	2012.341808			
2766.93207	7817.27932	6479.35145	7681.82926	929.8697648			
1529.30925	3853.49088	8561.82681	11222.0478	1362.383233			
1548.39154	3910.92246	3591.21069	3442.78073	414.2845426			
1195.68586	2873.08593	5023.4669	5434.33717	656.1093419			
566.377514	1178.16708	3420.10628	3221.62968	387.4895752			
1877.81116	4922.86763	6595.77769	7870.15946	952.8423885			
2878.94449	8196.2791	13037.6275	19881.6219	2424.721896			
2912.54757	8310.53806	5614.73503	6322.21937	764.167511			
1759.95842	4556.54638	5639.17852	6359.68062	768.7294821			
2304.36717	6284.5701	21819.0827	40049.7108	4912.973784			
1382.30362	3415.79134	3730.32707	3625.4136	436.4226662			
2468.63686	6822.64655	3966.80316	3941.49558	474.7572637			
2079.5522	5560.18735	36900.2622	81835.5138	10101.60311			
1455.46484	3632.55729	3250.07174	3005.77292	361.3497108			
1239.3067	2998.56678	3706.40006	3593.82462	432.5929295			
2317.88041	6328.56166	3672.59797	3549.32337	427.1981971			
1553.01313	3924.85259	5410.02581	6010.80803	726.2528945			

2394.36714	6578.48637	6324.37257	7433.02316	899.5273385			
2681.19666	7529.17732	9403.26527	12747.9624	1549.184202			
2676.89659	7514.77383	3436.50194	3242.65188	390.0360592			
320.629467	597.60403	5081.1593	5519.39126	666.4549103			
2565.13357	7141.99649	3743.6753	3643.06799	438.563135			
2674.9558	7508.27443	5937.08506	6820.89569	824.9129506			
1325.91365	3250.21511	4709.99149	4978.39241	600.672955			
1241.79224	3005.74276	7192.97613	8854.774	1073.017413			
2684.81976	7541.3167	3854.06573	3789.93497	456.3727991			
1936.98119	5108.48389	3007.26009	2704.54559	324.8958605			
1556.45555	3935.23374	5815.87237	6632.20557	801.923596			
1042.48169	2439.53507	3698.1519	3582.95219	431.2748545			
891.856959	2025.13324	9127.88605	12242.9266	1487.336161			
2894.24099	8248.25934	9178.41171	12335.1833	1498.632581			
981.042655	2269.00303	12753.8335	19295.3726	2352.651216			
981.70329	2270.82599	3150.39596	2881.10013	346.2585455			
920.92489	2104.12194	8426.43084	10981.3855	1332.94038			
1594.77237	4051.08144	3663.1215	3536.87376	425.6890675			
2419.90522	6662.27968	5460.15601	6086.68223	735.4892212			
1056.25235	2478.02837	7477.35097	9334.2348	1131.57771			
835.817076	1874.26127	3199.94314	2942.89817	353.7383521			
676.041963	1455.15747	3033.31716	2736.46563	328.7573443			
1402.78707	3476.26268						
1270.95889	3090.1553						
2464.17296	6807.93104						
1030.31064	2405.59474						
1865.43821	4884.19502						
1803.04362	4689.93537						
2158.21281	5812.0037						
1292.87497	3153.83057						
768.044536	1694.40569						
2972.83168	8516.15612						
2185.5996	5900.09699						
2607.96744	7284.50265						
1564.78047	3960.3571						
2822.47371	8004.84448						
2150.70524	5787.89212						
2524.99567	7008.87577						
2824.00831	8010.03705						
1215.98476	2931.37029						
1046.54426	2450.88107						
1053.18898	2469.45688						
1036.57717	2423.06005						
578.091113	1207.29388						

1109.66915	2628.25368						
425.262428	837.026563						
1667.82608	4273.43676						

Table 14. Kokatahi volume–area scaling values for historical soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)
1388.77231	3434.86969	7504.86282	9380.97353	1137.28763
2142.00476	5759.96967	3110.19192	2831.21164	340.221118
1539.81111	3885.0811	4499.99101	4678.96287	564.288316
955.271352	2198.07567	6096.42517	7071.05306	855.3992
2883.46794	8211.64506	21535.9892	39344.6713	4825.75507
2895.2372	8251.64647	13963.1474	21825.2426	2663.79076
1043.54078	2442.49209	9410.25991	12760.8605	1550.76401
2182.57052	5890.34301	13016.1277	19837.0463	2419.24132
2275.70873	6191.43926	30354.9116	62749.9179	7727.58027
1906.95743	5014.16053	3302.38235	3071.75774	369.338872
1328.00718	3256.33836	10624.4717	15050.8056	1831.44871
2351.88027	6439.4647	6157.56986	7167.67776	867.177052
792.488564	1758.93615	7275.2606	8992.81777	1089.87515
263.469514	472.805754	11037.3592	15851.7933	1929.71701
544.965889	1125.22668	49313.9953	121400.707	15038.7095
477.018488	959.939648	13996.5496	21896.2781	2672.53188
1167.58595	2792.71767	3453.10162	3263.97248	392.618819
856.27592	1929.1216	25315.1393	49020.6422	6023.88881
374.404046	719.029449	8940.87736	11903.0634	1445.72767
424.324585	834.824852	16495.7889	27378.4692	3347.82759
484.645237	978.277789	17907.7166	30613.7442	3746.92811
414.617814	812.09238	5051.26338	5475.273	661.088418
169.868502	280.076107	11746.0722	17251.8544	2101.58312
564.207402	1172.7836	6223.49868	7272.25028	879.925142
188.069134	316.23667	24751.5555	47542.4058	5840.6939
2491.16488	6896.98962	5421.45624	6028.08629	728.356132
1223.72854	2953.65477	3455.79486	3267.43516	393.038298
1666.7529	4270.15646	9317.71703	12590.492	1529.8977
864.402588	1950.98387	4365.21456	4489.41103	541.264714
2780.80766	7864.06992	3180.87636	2919.07593	350.854853
1749.5936	4524.55093	10149.014	14142.2398	1720.03654
312.362945	579.268904	4733.43255	5012.11909	604.7723
1418.53225	3522.86179	3045.08248	2750.91063	330.504921
1146.31754	2732.13535	6523.91583	7753.77357	938.645034
351.484122	666.832772	6337.30208	7453.69737	902.048289
1102.26051	2607.33316	10585.2867	14975.362	1822.19527
591.670649	1241.20322	4275.7245	4364.70575	526.121752

2559.5234	7123.36558	11588.1932	16937.2598	2062.95389
2443.38064	6739.45595	8364.48514	10871.7408	1319.52813
2390.7877	6566.75555	3680.53305	3559.7569	428.462971
2344.76887	6416.24249	5660.62731	6392.60049	772.738591
1319.20641	3230.61002	11240.131	16249.1566	1978.48303
598.244767	1257.67364	4597.00625	4816.68103	581.020669
427.317691	841.854841	15830.3102	25887.337	3164.02089
2455.49031	6779.32295	19869.5384	35262.8278	4321.08573
1815.70135	4729.24063	5375.80623	5959.16025	719.966238
2763.90095	7807.06391	5273.40992	5805.32106	701.243276
1707.26363	4394.26279	8824.7001	11693.2086	1420.04072
2584.85262	7207.54421	7465.62182	9314.32742	1129.14576
1612.69585	4105.4571	6609.21632	7891.97524	955.503782
2304.07353	6283.61475	3211.52061	2957.38815	355.492341
1297.52531	3167.36868	4696.40025	4958.86515	598.299595
2299.29118	6268.05837	9032.93858	12070.0558	1466.17084
1001.76587	2326.299	13692.4206	21251.7587	2593.23111
868.197084	1961.2054	4720.86805	4994.03396	602.574096
331.536799	621.936131	28511.6829	57625.252	7091.20296
321.996171	600.64424	47567.3328	115590.389	14312.5697
369.563248	707.952518	11173.6842	16118.657	1962.46646
540.824265	1115.03226	24249.5475	46235.8352	5678.81648
1731.4285	4468.56485	7168.75629	8814.2497	1068.06904
2927.60357	8361.81505	18564.9186	32151.7271	3936.78663
2848.47915	8092.91137	3813.93232	3736.36259	449.87577
2964.83859	8488.8465	4116.87196	4145.65701	499.530955
2722.81095	7668.79778	16003.3688	26272.9767	3211.54859
1331.15202	3265.54005	3390.5936	3183.88053	382.917225
2777.54525	7853.06452	3784.80351	3697.60656	445.176056
2432.71874	6704.38676	3107.07317	2827.3513	339.75398
2244.86037	6091.44456	3041.04787	2745.95482	329.905353
1252.53887	3036.80104	3083.77013	2798.55132	336.269054
2075.91326	5548.58192	5495.63568	6140.53413	742.045289
644.717684	1375.08433	9709.89752	13316.6099	1618.84707
1427.31174	3548.88883	4032.21071	4030.14375	485.512952
1726.36753	4452.98673	5822.43223	6642.38128	803.163235
353.429574	671.238358	3792.11041	3707.31837	446.353714
722.394513	1574.95927	7463.1176	9310.07859	1128.62672
567.040854	1179.81345	4481.87223	4653.35987	561.178054
1538.00122	3879.63387	7584.42823	9516.49074	1153.84466
1755.49634	4542.76779	3356.64253	3140.60043	377.675454
1537.86816	3879.23345	12918.9109	19635.8176	2394.50161
1074.13291	2528.1547	3567.90556	3412.43129	410.606557
2187.22947	5905.34643	8167.6485	10525.2818	1277.15515

2286.21819	6225.56554			
985.674908	2281.79029			
1608.04015	4091.32153			
1247.88971	3023.35839			
967.772693	2232.43614			
1934.48778	5100.63972			
1224.47395	2955.80131			
1326.75885	3252.68698			
1808.60539	4707.19946			
1660.68971	4251.63131			

Table 15. Styx volume–area scaling values for recent soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)
1597.12337	4058.20714	4223.971464	4293.01381	517.417734
2001.66694	5312.658	3249.779933	3005.4059	361.305278
1396.6821	3458.2216	6457.242674	7646.20304	925.524574
2657.90617	7451.21713	4977.869848	5367.36302	647.963839
1686.81855	4331.55652	4296.216102	4393.17881	529.578939
2748.10066	7753.84931	8807.517586	11662.2553	1416.25226
2827.44667	8021.67327	33185.38633	70838.5255	8732.98613
2987.35368	8565.80903	18004.01937	30837.8606	3774.58928
2300.48039	6271.92614	4825.234616	5144.78026	620.89885
1048.67915	2456.84684	23846.32182	45193.3843	5549.69201
867.345047	1958.90945	7253.869752	8956.87727	1085.48593
2940.01547	8404.1252	4299.318015	4397.49319	530.102804
2937.07839	8394.11007	3205.408698	2949.73634	354.566092
1362.80437	3358.386	18378.40338	31713.2199	3882.64605
1818.05153	4736.5443	3336.241706	3114.66949	374.535152
2608.12561	7285.02972	20384.50951	36511.5374	4475.41996
1620.31067	4128.59406	3060.038689	2769.3023	332.730074
1764.83434	4571.61057	9741.792353	13376.1343	1626.14065
886.038579	2009.38156	21128.70012	38336.1715	4701.02064
971.745308	2243.37305	4920.053011	5282.75735	637.675123
2438.71987	6724.12206	6071.415328	7031.63115	850.59433
1885.58005	4947.17506	3425.685223	3228.77883	388.355559
1310.64163	3205.60335	4019.936741	4013.46885	483.489643
2798.39699	7923.44863	6790.197495	8187.32125	991.540094
2467.45013	6818.73393	7524.794844	9414.87378	1141.42926
2422.50958	6670.83452	8720.18575	11505.2685	1397.03949
1185.57448	2844.12402	25822.79323	50362.3672	6190.20996
1455.21777	3631.82165	4476.107471	4645.2217	560.189454
1568.63915	3972.01081	18867.7916	32867.1779	4025.1344

706.208624	1532.95184	3744.774951	3644.5234	438.739597
2501.69055	6931.76917	3715.404168	3605.70346	434.033047
1300.70194	3176.62187	4461.963165	4625.27003	557.765844
2772.32397	7835.45624	6915.610085	8393.65738	1016.72232
1208.40508	2909.58458	3659.444686	3532.04651	425.103924
2145.61695	5771.55961	12681.79486	19147.3001	2334.45088
1036.91525	2424.0029	13798.8885	21476.8084	2620.91856
1969.56925	5211.18305	4387.427697	4520.50883	545.04144
1636.27506	4177.16856	4084.846476	4101.85934	494.215566
2792.87269	7904.79173	3354.054637	3137.30788	377.276706
2769.1839	7824.86975	3037.746994	2741.90204	329.415041
2822.89652	8006.27507	4287.074262	4380.47018	528.03584
1549.80272	3915.1751	3203.762558	2947.67635	354.316735
1202.93768	2893.88637	4958.088733	5338.3765	644.438703
855.788469	1927.81153	4436.114524	4588.86729	553.344069
585.913267	1226.80797	4169.181398	4217.45867	508.245925
477.341055	960.714104	3751.509564	3653.44016	439.820725
1751.44243	4530.25544	4652.718331	4896.2429	590.688935
829.630624	1857.723	11078.34682	15931.9047	1939.54778
1515.13968	3810.9343	8201.258813	10584.2299	1284.36386
1397.20813	3459.7755			
479.588321	966.112409			
363.903082	695.036165			
437.469909	865.770149			
370.642028	710.41862			
624.562344	1323.95537			
311.381226	577.097622			
500.969795	1017.71598			
201.587846	343.540058			
719.910118	1568.49958			
1145.22786	2729.03725			
2087.92346	5586.90015			
1611.05205	4100.46531			
1623.94047	4139.6303			
892.983138	2028.18436			
795.399414	1766.64645			
1579.75161	4005.60269			
804.605426	1791.06724			
1266.74721	3077.94278			
975.837244	2254.64748			
779.28804	1724.03925			
738.238857	1616.25665			
383.664415	740.296311			
200.397588	341.121555			

201.044032	342.434731			
332.27213	623.582133			
80.8054791	115.432555			
1301.81161	3179.85528			
228.941599	399.855435			
1595.87736	4054.43034			
1009.76001	2348.46287			
461.911496	923.783149			
1062.12789	2494.48191			
997.67512	2314.97054			
486.081283	981.736953			
389.402133	753.523195			
601.559157	1265.99061			
992.304056	2300.11012			
633.645224	1346.95752			
823.738791	1841.99445			
297.813255	547.225898			
1591.24157	4040.38368			
1066.35912	2506.3417			
285.506537	520.356972			
2700.18435	7592.83159			
470.779695	944.980759			
1949.60518	5148.22833			
451.850764	899.830029			
489.159543	989.15854			
836.338456	1875.65616			
907.381481	2067.2585			
580.625875	1213.61186			
554.419912	1148.55325			
1357.89028	3343.94392			
1080.77588	2546.81879			
607.779111	1281.62251			

Table 16. Arahura volume–area scaling values for historical soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)
914.610291	2086.92129	3020.32678	2720.53996	326.830714
2526.52663	7013.94589	5856.78554	6695.73777	809.663556
2185.75625	5900.60148	5638.3394	6358.39364	768.572752
633.503737	1346.59871	3126.81638	2851.81266	342.714124
540.771545	1114.90259	7849.5179	9971.67734	1209.47199
505.703273	1029.19832	8681.0281	11435.0624	1388.44806
2469.67588	6826.07248	9631.65132	13170.8797	1600.99176
211.578676	363.948048	3915.78112	3872.70837	466.412605

2884.62915	8215.59039	3303.45418	3073.11371	369.503061
1795.55291	4666.69999	5032.10932	5447.05613	657.656333
2015.98975	5358.04045	3967.43903	3942.35487	474.861512
2374.89794	6514.72141	3860.02723	3797.90993	457.340029
602.804851	1269.11878	6606.09845	7886.91238	954.886139
564.296607	1173.00482	4175.15069	4225.67304	509.243024
2821.40761	8001.23749	9565.51666	13048.0384	1585.94218
1088.01556	2567.18463	3689.84128	3572.00626	429.947901
1842.50711	4812.65318	34612.8532	75014.3972	9252.47031
2446.31351	6749.10795	7108.93665	8714.3716	1055.87387
2567.41444	7149.57335	6348.34502	7471.36697	904.202913
1354.78386	3334.81963	12242.4525	18250.8385	2224.28866
1945.09276	5134.01603	9268.93127	12500.9235	1518.92852
2292.95366	6247.45288	7957.27587	10158.3077	1232.28582
1081.18408	2547.96638	3094.11469	2811.32642	337.814864
1654.09911	4231.50955	3369.66731	3157.1856	379.684064
1661.69081	4254.68913	3140.70251	2869.05061	344.800258
1274.61513	3100.76355	3563.46821	3406.66076	409.907267
1807.30371	4703.15803	4400.04255	4538.19456	547.189405
628.461087	1333.82098	9699.3176	13296.8805	1616.42966
1507.35787	3787.59517	3956.86995	3928.07865	473.12955
1054.00808	2471.74829	3146.19615	2875.87788	345.626518
2562.0094	7131.62041	4168.14995	4216.03972	508.073688
1278.54681	3112.17756	4274.50266	4363.00955	525.915805
972.019648	2244.12865	15488.6145	25130.3683	3070.74744
1390.16175	3438.96983	3051.32634	2758.58477	331.433379
1256.61568	3048.59669	3436.91999	3243.18836	390.101047
991.508398	2297.91005	4170.71791	4219.57266	508.50253
1422.97849	3536.03893	7467.54778	9317.5955	1129.545
1777.34032	4610.28465	3994.98902	3979.63243	479.384181
1288.42276	3140.87809	13342.2608	20516.0514	2502.73641
643.686621	1372.46121	4134.60304	4169.95871	502.480457
372.571422	714.832686	8490.18352	11094.532	1346.78216
741.016736	1623.51478	7846.38889	9966.2718	1208.81126
1031.24595	2408.20022	3080.71837	2794.78547	335.81339
731.96963	1599.8956	8835.12314	11711.9957	1422.34017
1018.13009	2371.7053	11621.6344	17003.7677	2071.11992
1542.84026	3894.20071	6287.35466	7373.91596	892.320301
700.301842	1517.6679	5136.96311	5601.99254	676.503352
1872.43172	4906.04771	4673.02781	4925.33229	594.224156
1658.39172	4244.61357	4059.48654	4067.26494	490.017421
2020.72049	5373.04369	3980.60332	3960.15573	477.021146
251.054344	446.348763	7415.08978	9228.69051	1118.68458
288.559398	527.001733	3124.61309	2849.08008	342.383436

319.560784	595.228506	3310.61872	3082.18162	370.601072
739.424865	1619.35484	7311.55659	9053.88866	1097.33374
304.642011	562.228226	5866.65247	6711.08364	811.533192
570.875577	1189.33825	12936.14	19671.4403	2398.88103
1546.98178	3906.67483	18982.8828	33140.1374	4058.84549
350.086175	663.669944	5271.56637	5802.56109	700.907411
889.883646	2019.7888	3095.34242	2812.84364	337.998454
669.704546	1438.89842	14354.5643	22661.4753	2766.70714
1028.77376	2401.31446	4756.84702	5045.8676	608.874515
1004.39983	2333.59795	3299.64637	3068.29718	368.91985
1337.44059	3283.95275	4793.10054	5098.23972	615.24091
1280.40917	3117.58648	18791.2056	32685.8719	4002.74411
2223.41101	6022.07259	3941.45158	3907.27687	470.606007
1806.31443	4700.08693	3085.5806	2800.78606	336.539458
1139.58121	2712.99218	4574.00671	4783.93646	577.041952
1068.7736	2513.11338	3903.21119	3855.81108	464.362951
2295.84043	6256.83742	3230.08688	2980.66437	358.310026
2484.99462	6876.61465	3958.93837	3930.8715	473.468369
2047.29921	5457.46219	3348.53597	3130.28958	376.42676
632.335868	1343.63766	5670.69479	6408.06769	774.622303
2171.6659	5855.25057	4718.17698	4990.16273	602.103566
2348.69824	6429.07214	3487.91708	3308.80921	398.050726
364.44593	696.273259	5480.89486	6118.14492	739.319512
1101.54345	2605.30978	9821.97367	13526.084	1644.51536
2300.02147	6270.4335	3379.68704	3169.95999	381.23121
1847.1669	4827.17723	4385.15097	4517.31886	544.654018
346.41367	655.372604	3090.69093	2807.09651	337.303032
592.070674	1242.20441			
1514.71097	3809.64791			
816.146015	1821.75716			
1081.55534	2549.01022			
2014.02803	5351.82097			
548.320932	1133.49594			
849.387924	1910.6229			
827.984647	1853.3268			
223.401744	388.339607			
876.216164	1982.83537			
319.830099	595.827008			
1092.55448	2579.96635			
2003.19918	5317.50999			
846.328819	1902.4165			
1194.00705	2868.27404			
320.157011	596.55364			
1420.71451	3529.32826			

1127.7985	2679.5608			
556.417401	1153.49167			
607.505627	1280.93454			
1574.3932	3989.39902			
398.326046	774.16975			
2942.90448	8413.9783			
231.654676	405.514903			
353.835039	672.157147			
1985.42315	5261.26466			
706.685142	1534.18592			
1902.87375	5001.35316			
2007.50124	5331.1367			
1755.87606	4543.94008			
371.73798	712.925396			
770.591584	1701.11144			
647.31576	1381.69767			
1431.68129	3561.85402			
2431.77085	6701.27039			
2027.77818	5395.43934			
727.736806	1588.86431			
701.443214	1520.61929			
289.105612	528.192041			
1079.52138	2543.29244			
1790.12777	4649.88346			
816.062301	1821.53423			
988.648575	2290.00517			
702.409559	1523.11882			
1399.11788	3465.41787			
1533.7373	3866.80563			
1155.24967	2757.55203			
1155.61452	2758.59102			
1439.34332	3584.60699			
523.171057	1071.74969			
428.027419	843.523194			
502.987162	1022.6071			
476.367053	958.375914			
526.497357	1079.88394			
358.277768	682.237702			
261.478092	468.545469			
1091.83706	2577.94541			
2350.45971	6434.82479			
1236.74158	2991.164			
2681.47017	7530.09361			
1061.40529	2492.45743			

152.927983	247.083853			
246.932248	437.619585			
416.277473	815.971958			
428.180125	843.882231			
2373.31165	6509.53047			
2229.72634	6042.4844			
649.21857	1386.54448			
577.20307	1205.08167			
254.979594	454.686851			
1997.84746	5300.56637			
1316.19704	3221.81996			
1755.37142	4542.38214			
800.235214	1779.46761			
1745.92189	4513.22537			
445.905789	885.724082			
476.653641	959.063802			
1597.66474	4059.84827			
2057.28139	5489.22211			
2269.44503	6171.11424			
798.94336	1776.04105			
1900.4818	4993.85391			
1405.96665	3485.6648			
403.997099	787.337017			
187.220864	314.535767			
702.533105	1523.43843			
669.637543	1438.72668			
2321.01282	6338.7661			
144.234773	230.420918			
1797.11061	4671.53029			
796.530618	1769.64427			
237.436772	417.618916			
651.889608	1393.35273			
2103.01601	5635.11285			
1413.26708	3507.26791			
610.073978	1287.39775			
2268.66338	6168.57863			
952.690335	2190.9924			
2077.21952	5552.74743			
1575.24231	3991.966			
1761.97574	4562.77794			
901.921675	2052.42752			
2627.9025	7350.98025			
2329.37195	6366.01064			
1592.65017	4044.65097			

2516.43466	6980.53505			
1245.7691	3017.23005			
229.85119	401.751403			
199.761729	339.830678			
447.317042	889.069368			
607.967778	1282.09714			
222.938234	387.378577			
893.225152	2028.84014			
1024.17935	2388.5262			
1475.31097	3691.72655			
1430.22946	3557.54534			
2772.88575	7837.35049			
1123.53105	2667.46925			
2104.86645	5641.02864			
2593.33523	7235.77082			
1156.55607	2761.27261			
671.132479	1442.55929			
1780.20644	4619.15536			
755.821346	1662.28495			
504.002555	1025.07037			
370.668581	710.479338			
766.390026	1690.05207			
306.020099	565.263721			
172.945142	286.138374			
361.896388	690.466211			
273.064826	493.419689			
1758.51419	4552.08596			
1364.76506	3364.15109			
1029.80603	2404.18926			
1191.03252	2859.75152			
1398.53717	3463.702			
1151.09951	2745.73788			
1328.63431	3258.17301			
235.70296	413.983384			
147.246118	236.171637			
334.671575	628.958055			
236.71164	416.097801			
830.731262	1860.66361			
272.409197	492.006668			
204.307772	349.077047			
830.866824	1861.02584			
647.345939	1381.77452			
1546.0282	3903.80209			
392.738587	761.23191			

764.409948	1684.84415			
557.812477	1156.94276			
2260.06292	6140.69058			
1889.54306	4959.58201			
874.203311	1977.40247			
1788.42386	4644.60381			
215.301192	371.600096			
732.420528	1601.07143			
766.11002	1689.31545			
607.289225	1280.39021			
895.888691	2036.0597			
334.931048	629.539847			
137.39396	217.443693			
198.978842	338.242404			
108.617179	164.277601			
818.642816	1828.40798			
2232.08637	6050.11516			
2014.74831	5354.10443			
797.598286	1772.47447			
283.632597	516.284991			
268.071332	482.674233			
79.8166411	113.749345			
607.059941	1279.81351			
1391.55699	3443.08793			
355.977803	677.016048			
835.431681	1873.2303			
535.291819	1101.4379			
551.768727	1142.00399			
467.311797	936.682203			
355.040401	674.889712			
278.037804	504.158756			
280.286706	509.027452			
207.797781	356.202549			
1591.55191	4041.32378			
1308.8277	3200.31124			
2550.75529	7094.26319			
2597.65004	7250.13558			
1561.31328	3949.89051			
908.836841	2071.21474			
453.08781	902.769751			
2013.17771	5349.12545			
845.255757	1899.53924			
1083.8501	2555.46364			
1688.17651	4335.71694			

2847.92887	8091.04626			
847.148615	1904.61514			
2776.16874	7848.42174			
1903.19533	5002.36153			
867.639706	1959.70341			
179.321579	298.768673			
210.268836	361.261675			
375.577666	721.719159			
728.957102	1592.0433			
544.45649	1123.97201			
1772.81047	4596.27026			
969.589545	2237.437			
961.569771	2215.37637			
611.555092	1291.12734			

Table 17. Taipo volume–area scaling values for historical soil and mixed landslides and the associated error propagation values (below)

Soil Area (m ²)	Soil Volume (m ³)	Mixed Area (m ²)	Mixed Volume (m ³)	V-A Error (Mixed)
1328.93882	3259.06389	5447.98306	6068.2348	733.243491
2076.8675	5551.62482	3348.0948	3129.72872	376.358838
455.216585	907.832209	7053.30692	8621.76027	1044.567
1191.47994	2861.03318	5870.55608	6717.15744	812.273192
2006.2435	5327.15224	3145.90978	2875.52188	345.583435
348.778597	660.713781	4333.88272	4445.64408	535.94971
949.530542	2182.32579	3851.93359	3787.0838	456.027003
1752.65976	4534.01214	3029.12849	2731.32781	328.13578
117.402603	180.250695	3259.1325	3017.17501	362.730132
1748.63703	4521.59989	3319.58225	3093.53639	371.976028
136.317835	215.413426	7918.63486	10091.2785	1224.09171
744.175412	1631.77425	5368.05972	5947.48479	718.545142
2710.90375	7628.8055	4842.03165	5169.15229	623.861923
1029.24913	2402.63826	3905.92994	3859.46414	464.806063
1030.75475	2406.83185	5629.93954	6345.51439	767.004316
261.887344	469.420481	23248.1915	43658.7237	5359.64895
427.298551	841.809857	7308.34351	9048.47799	1096.67292
759.057457	1670.77928	5261.9867	5788.22512	699.162863
215.15893	371.307189	6223.19196	7271.76287	879.865719
268.132938	482.806569	3393.59278	3187.71135	383.381212
1132.52415	2692.96096	4510.88329	4694.37222	566.160317
334.504254	628.582932	11059.3493	15894.7603	1934.9896
1376.32881	3398.1849	8002.44428	10236.8084	1241.88286
827.244428	1851.35032	11407.3648	16578.8273	2018.94932
1989.09104	5272.86234	4422.4754	4569.68999	551.014748

2448.4104	6756.01012	4294.8317	4391.25364	529.34518
1495.48479	3752.03041	5323.87662	5881.0086	710.454319
858.079642	1933.97051			
590.534644	1238.3607			
1413.58153	3508.19891			
436.568411	863.642142			
344.096079	650.145159			
825.939354	1847.86644			
618.197901	1307.87597			
661.330647	1417.46019			
680.011085	1465.35551			
347.522381	657.875749			
1274.60074	3100.72178			
654.577037	1400.2082			
1369.82543	3379.03769			
1272.14436	3093.5942			
2281.34111	6209.72492			
718.434249	1564.6642			
543.691891	1122.08919			
866.922078	1957.76986			
1071.47893	2520.70429			
600.580057	1263.53278			
1605.53504	4083.71881			
459.451584	917.917073			
967.64767	2232.09209			
1583.66003	4017.42831			
823.761653	1842.05544			
216.400613	373.864989			
168.082136	276.565906			
862.530373	1945.94373			
509.907919	1039.41526			
1251.04037	3032.46719			
2128.59135	5716.96491			
978.086974	2260.85			
2248.5517	6103.39609			
155.700935	252.438069			
287.577029	524.862052			
419.078788	822.527009			
2133.82774	5733.74712			
2242.84402	6084.91778			
1049.41701	2458.90928			
2645.44193	7409.54966			
1992.78528	5284.5475			
1837.19891	4796.11672			

924.667153	2114.32644			
2454.86724	6777.27078			
532.069203	1093.53174			
1406.85994	3488.30703			
569.153367	1185.05904			
1394.09066	3450.56812			
1571.71344	3981.29951			
1016.36431	2366.79893			
419.114596	822.610854			
601.475735	1265.78117			
1283.13144	3125.49565			
875.305894	1980.37815			
1620.7918	4130.05664			
1414.98365	3512.35067			
1695.45273	4358.02027			
266.772555	479.885704			
548.456559	1133.83043			
2904.23101	8282.23586			
1042.93817	2440.80952			
325.137943	607.642487			
217.709512	376.564318			
1241.21045	3004.06283			
712.298759	1548.73607			
2288.77183	6233.86229			
1274.3422	3099.97148			
670.276547	1440.36471			
610.65409	1288.85832			
528.809057	1085.5429			
1436.66419	3576.64846			
588.573421	1233.45581			
2766.58658	7816.11483			

Appendix III — Volume–Area Scaling Plots

V-A scaling values plotted for each catchment, split into their respective categories. Starting with recent landslides, followed by historical landslides. The individual plots for soil, mixed, and bedrock landslide volumes can be requested.

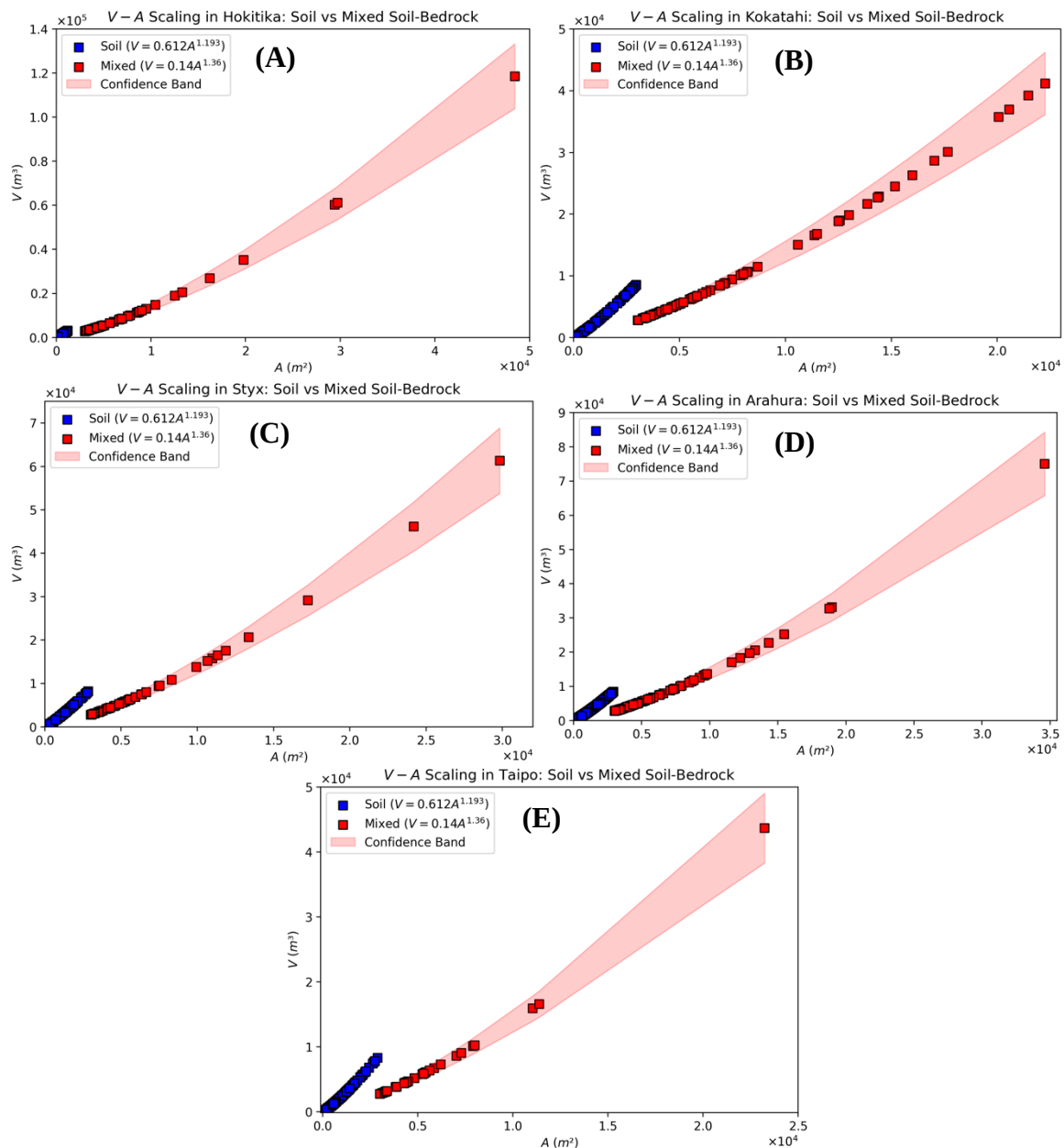


Figure 21. Recent landslide data for each catchment showing the volume-area scaling estimations for both soil and bedrock landslides. (a) Hokitika, (b) Kokatahi, (c) Styx, (d) Arahura, (e) Taipo.

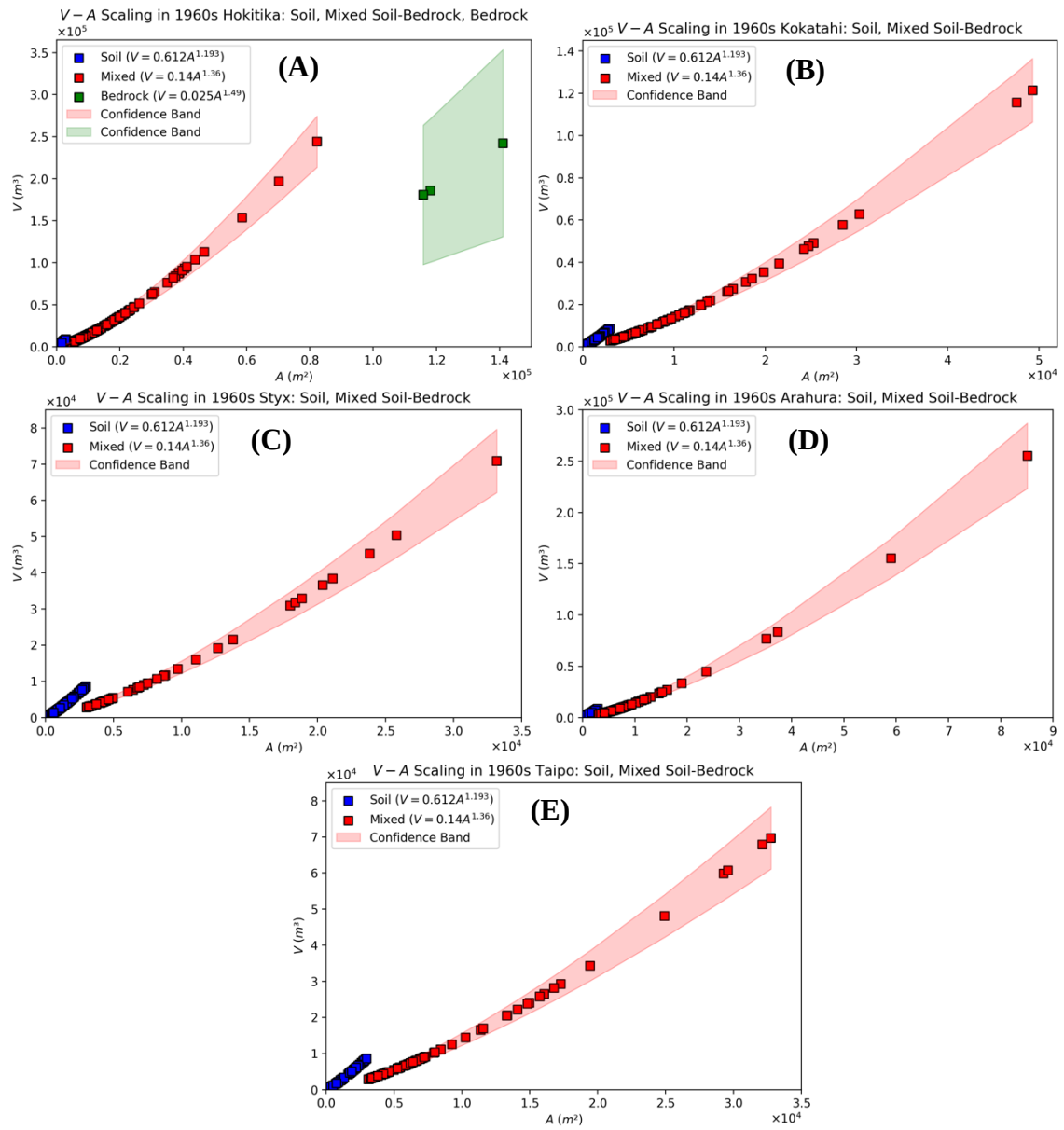


Figure 22. Historical landslide data for each catchment showing the volume-area scaling estimations for both soil and bedrock landslides. (a) Hokitika, (b) Kokatahi, (c) Styx, (d) Arahura, (e) Taipo.

Appendix IV — Landslide Elevation density

Landslide density as a function of elevation across all this study's catchment selection. Four elevation bands are used and computed using QGIS. The plots are displayed with recent landslide data first, then followed by historical landslides.

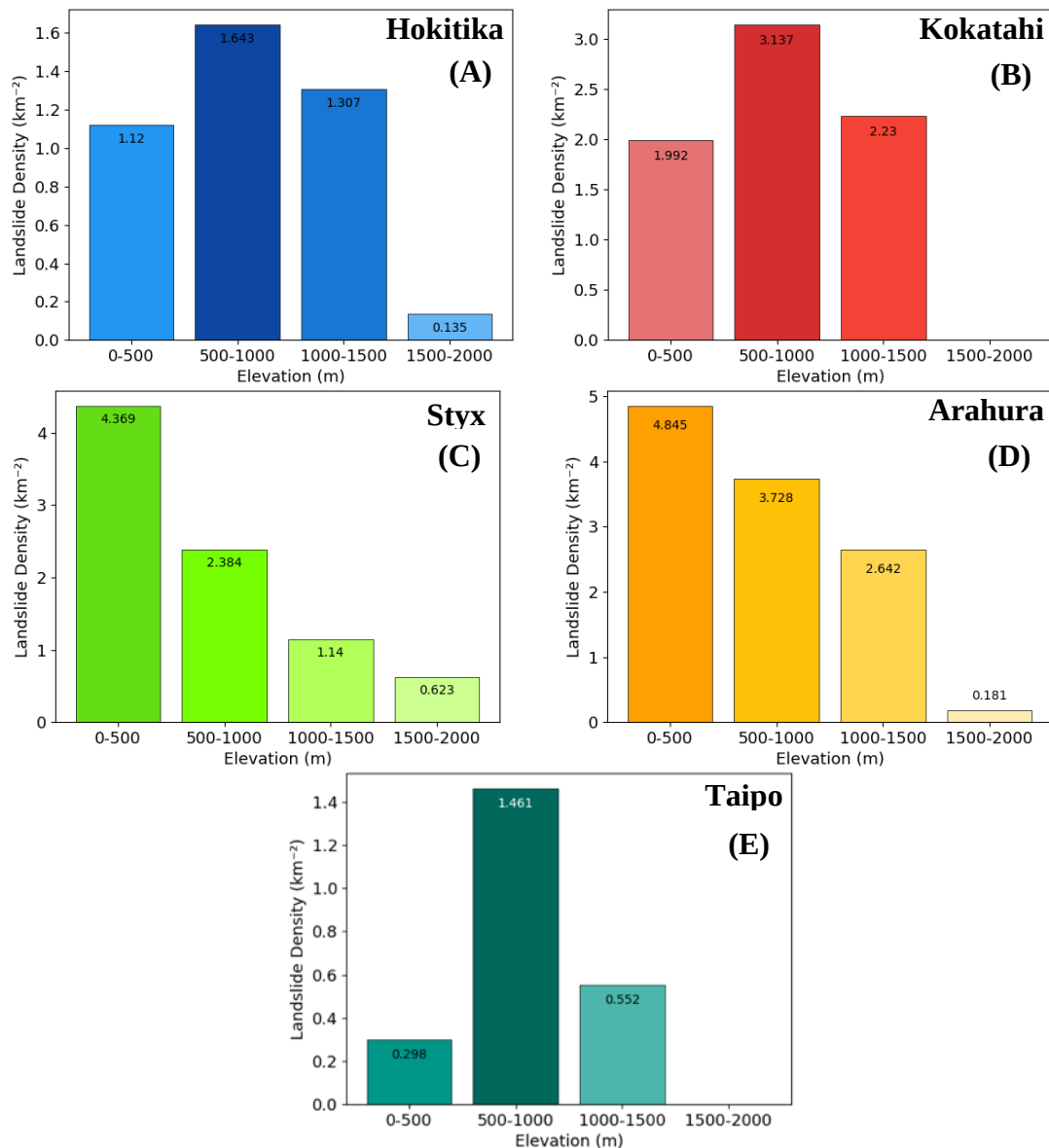


Figure 23. Recent landslide elevation density plots per km² (a) Hokitika, (b) Kokatahi, (c) Styx, (d) Arahura, (e) Taipo.

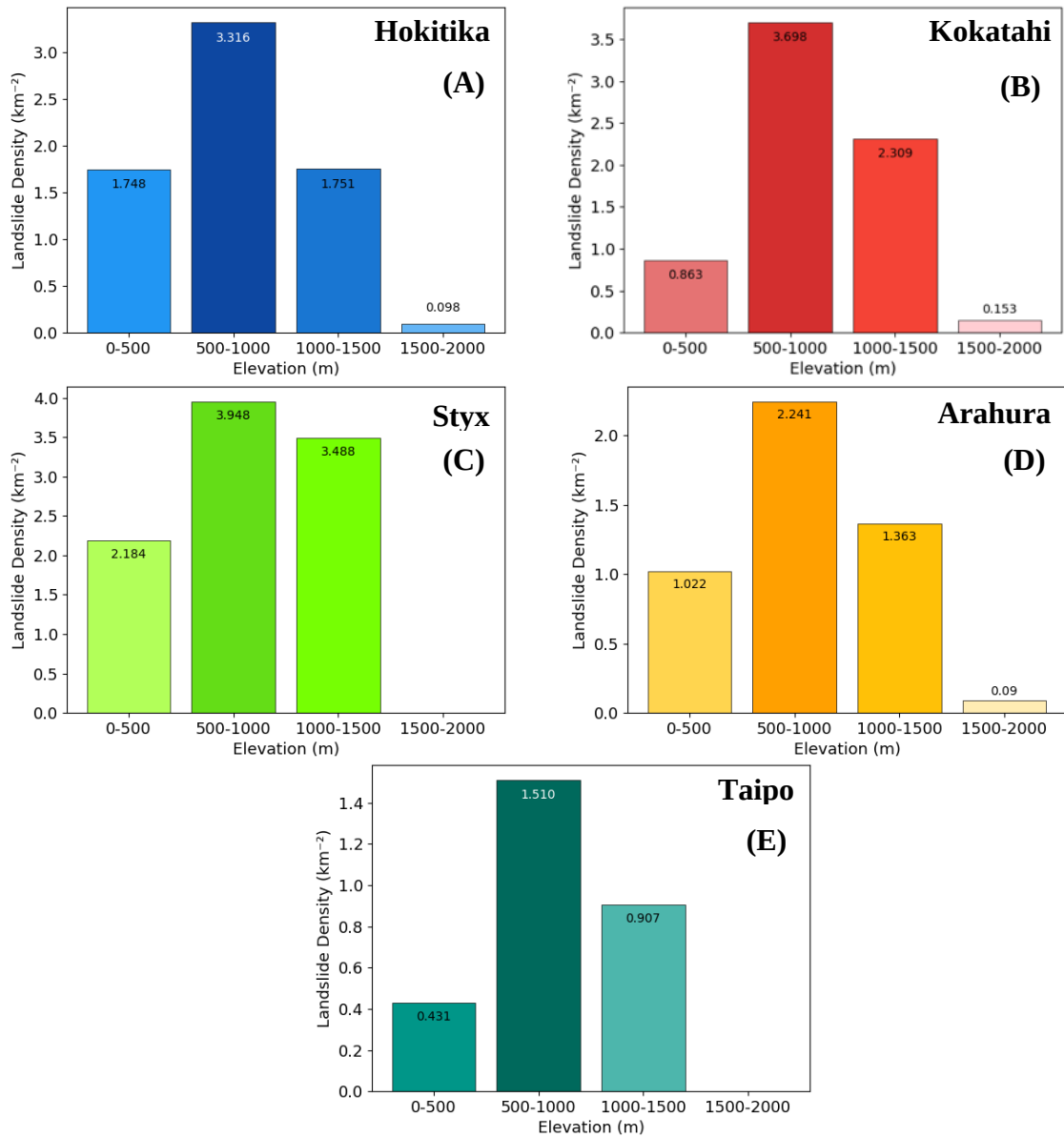


Figure 24. Historical landslide elevation density plots per km^2 (a) Hokitika, (b) Kokatahi, (c) Styx, (d) Arahura, (e) Taipo.

Appendix V — Landslide Median Area at Alpine Fault

All median area values for each Distance from Fault interval, computed in QGIS.

<i>RECENT</i>	Hokitika	Kokatahi	Styx	Arahura	Taipo
0–5km	668	0	3461	1762	2007
5–10km	386	1598	1958	841	964
10–15km	711	2286	1843	1662	1370
15–20km	2159	2623	0	0	0
20–25km	2160	0	0	0	0

Table 18. Recent landslide median area values for each distance from fault interval in m²

<i>1960s</i>	Hokitika	Kokatahi	Styx	Arahura	Taipo
0–5km	4333	0	3883	4161	3470
5–10km	1732	1750	1611	2953	2247
10–15km	4741	3286	1062	2296	3709
15–20km	3127	3089	0	0	0
20–25km	2674	0	0	0	0

Table 19. Historical landslide median area values for each distance from fault interval in m²